

Image quality in modern CT imaging

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Image quality in modern CT imaging: Optimization and Quantification

Eva Janna Ina Hoeijmakers

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Image quality in modern CT imaging: Optimization and Quantification

PROEFSCHRIFT

voor het behalen van de graad van Doctor aan de Universiteit Maastricht,
onder gezag van de Rector Magnificus, Prof. dr. Jan Smits,
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door

Eva Janna Ina Hoeijmakers

Promotor:

Prof. dr. J.E. Wildberger (Universiteit Maastricht/Maastricht UMC+)

Copromotoren:

Dr. C.R.L.P.N. Jeukens (Maastricht UMC+)

Dr. B. Martens (Universiteit Maastricht/Maastricht UMC+)

Beoordelingscommissie:**Voorzitter:**

Prof. dr. M.E. Kooi (Universiteit Maastricht/Maastricht UMC+)

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Prof. dr. R.G.H. Beets-Tan (Universiteit Maastricht/Maastricht)

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Prof. dr. ir. C. van Pul (Eindhoven University of Technology/Máxima MC Veldhoven)

Dr. A.A. Jacobi-Postma (Maastricht UMC+)

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CHAPTER

1



General Introduction

Over the years, computed tomography (CT) has become an essential imaging modality in modern clinical practice. By using X-rays to generate cross-sectional images of the body, CT plays a central role in the diagnostic work-up of patients as well as in treatment planning [1]. Worldwide, hundreds of millions of CT scans are performed each year, and this number keeps rising rapidly [2; 3]. The widespread adoption of CT is driven by its high diagnostic accuracy, fast acquisition times and general availability. It guides clinical decision-making, reduces additional downstream testing, and contributes to improved survival [4]. The adoption of CT has been fueled by continuous innovation. Advances in (detector) technology, and increasingly advanced reconstruction algorithms keep improving diagnostic image quality, acquisition speed, and clinical applicability [5]. These advances, however, ask for careful re-evaluation of standard CT acquisition protocols to fully use their clinical potential.

The quality of a CT image is not straightforward: It is not an absolute property of an image, but rather a combination of physical image characteristics, e.g. noise, contrast, temporal and spatial resolution, as well as the ability of a radiologist to perform a specific diagnostic task. Accordingly, acceptable image quality may differ for different clinical questions and is context dependent. For instance, a CT angiography of the aorta may tolerate higher image noise, whereas low image noise is a prerequisite for the detection of metastases in a portal-venous CT of the abdomen. As a result, image quality can vary for every individual acquisition and this is influenced by different parameters throughout the entire imaging procedure (Figure 1). The parameters include [1; 6-9]:

- Scan- and reconstruction protocol. This includes acquisition settings, such as radiation dose, tube voltage (kV), tube current-time product (mAs), and rotational speed. They determine the number and energy of the emitted X-ray photons and therefore directly affect image noise and signal. Furthermore, it also includes reconstruction parameters, e.g. the choice of reconstruction algorithm, kernel, and slice thickness which further define the amount of noise and structural detail.

- Patient-related factors, e.g. body habitus, anatomical variability, and motion during the scan, may influence image quality.
- Contrast medium parameters, including injection rate, contrast volume and timing, influence the level of enhancement between different tissues.

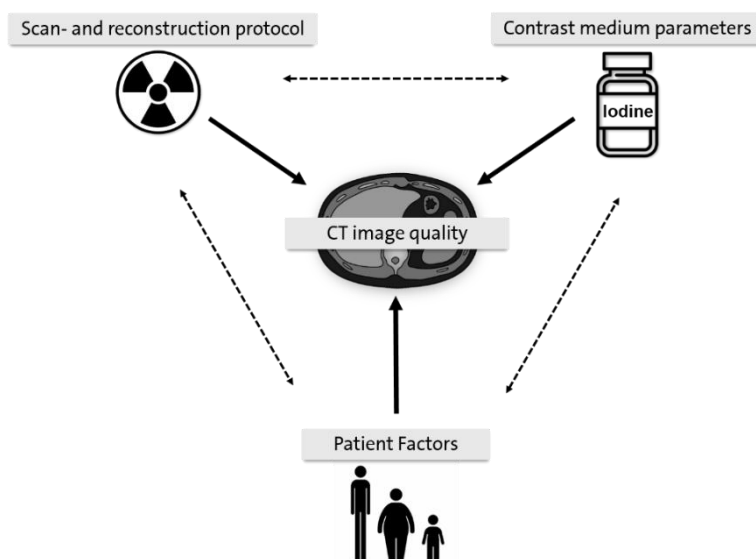


Figure 1 Image quality in CT imaging is influenced by scan- and reconstruction protocol, contrast medium parameters and patient factors.

Radiation dose

In CT imaging, radiation dose and image quality are intrinsically linked. Within the framework of radiation protection, justification of the examination is the first fundamental principle, followed by optimization of radiation exposure, which aims to achieve adequate diagnostic image quality at the lowest reasonable dose [10]. The ALARA principle (As Low As Reasonably Achievable) provides guidance in this context: radiation exposure should be minimized as much as possible without compromising the diagnostic value of the CT acquisition [11; 12].

Higher radiation generally reduces image noise and, therefore, improves image quality. However, increasing radiation dose also increases the risk of radiation-induced biological effects. Ionizing radiation might damage living cells and tissues at molecular level, particularly by breaking DNA strands. Radiation effects are

classified as deterministic or stochastic. Deterministic effects occur only above a threshold dose (therefore rare in CT) and increase in severity with higher exposure. These may include skin burns, radiation sickness, and organ dysfunction. In contrast, stochastic effects have no threshold. Their probability increases with dose, but their severity is dose-independent. The main stochastic risks are development of cancer and genetic mutations [10]. Conversely, very low doses reduce patient risk but might compromise image quality to an extent where it may become insufficient for reliable diagnosis. Therefore, consistent technological innovation in CT hardware and reconstruction algorithms seeks to improve diagnostic image quality while lowering patient dose [5; 13].

In CT imaging, radiation dose is well defined and routinely quantified, most commonly using the volume CT dose index ($CTDI_{vol}$), often expressed in milligray (mGy). By regulation, modern CT scanners are required to calculate and display this dose metric for each examination, enabling standardized dose monitoring [14].

Contrast media

A similar balance exists for contrast media, which is commonly used in CT to increase the visibility of specific tissues and pathological processes, by enhancing the differences in X-ray attenuation between structures [15]. Iodine is the most used contrast medium and commonly administered intravenously. It leads to strong X-ray attenuation and therefore it appears very bright on CT images. Contrast media dose is most commonly defined in terms of total iodine load (g iodine) or patient-weight-normalized iodine dose (g iodine / kg body weight).

Reducing contrast media dose may have a positive effect on the risk of contrast-induced acute kidney injury although there is not yet a consensus on the true risk [16-19]. However, particularly in patients with preexisting severe renal impairment (chronic kidney disease stage IV / V) caution is recommended.

In addition, reduced contrast usage supports more sustainable imaging practices, by decreasing the amount of iodinated contrast media that pollutes sewage, surface, and drinking water [20]. Unfortunately, current purification systems cannot completely remove these substances [21]. Beyond clinical and environmental considerations, contrast dose optimization also offers financial and logistical

advantages. Reduced contrast media usage lowers materials costs for the hospital and may mitigate the impact of supply shortage.

Yet, insufficient contrast dose can hinder the visualization of vascular structures and subtle abnormalities, thus indicating that contrast media also requires the ALARA balance.

The past 10-20 years already significant steps have been taken in our center to reduce contrast media. This includes protocol individualization and the introduction of the 10-to-10 rule: a contrast media dose reduction of 10% for every 10 kV decrease in tube voltage (Figure 2) [9; 22; 23].

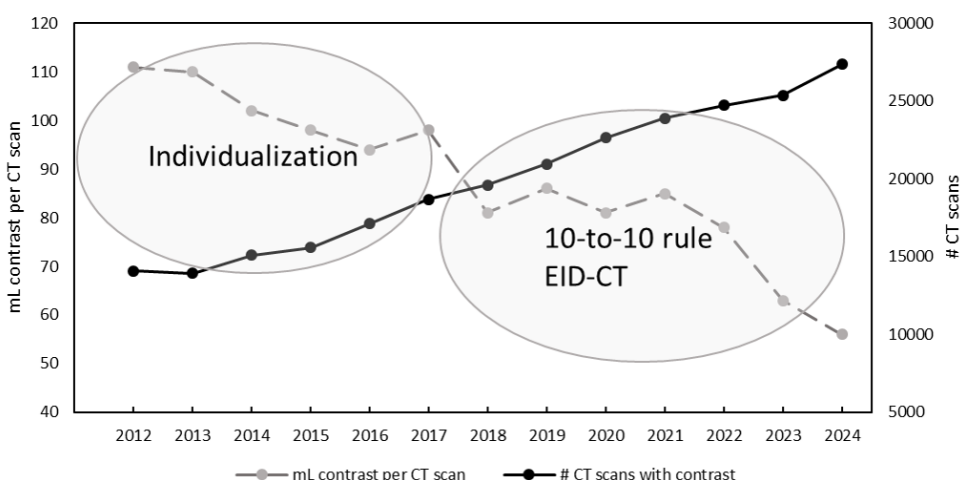


Figure 2 While the number of CT scans (with contrast media) increases over time, the contrast media volume has decreased due to protocol individualization and the 10-to-10 rule (Numbers from Maastricht University Medical Centre+).

Developments in Computed Tomography

Since its introduction in the 1970s, CT technology has undergone continuous development [24]. Early systems were limited by long acquisition times, which led to patient discomfort and susceptibility to motion artifacts [1]. A major technological breakthrough was the introduction of slip-ring technology, enabling continuous gantry rotation and thereby spiral CT acquisition. This allowed volumetric data acquisition within a single breath-hold, significantly decreasing scan times [25]. Later, the focus shifted towards enhancing image quality with each successive generation, while simultaneously improving dose efficiency [26]. The transition from single-slice

to multi-slice and multi-detector systems enabled faster volumetric imaging and reduced motion artifacts [27]. The development of dual-energy CT introduced spectral imaging capabilities by acquiring two energy spectra (e.g. dual-source systems, rapid kV-switching) or by energy separation on the detector domain (dual-layer detectors). This allowed material decomposition based on energy-dependent attenuation differences, enabling improved tissue characterization. Moreover, it enabled reconstruction of virtual monoenergetic images (VMIs): images simulated as if made with a single energy level [28].

Most recently, photon-counting detector CT (PCD-CT) represents another major technological advancement [29-31]. In PCD-CT, conventional energy-integrating detectors are replaced with photon-counting detectors that register individual X-ray photons and can discriminate their individual energies, typically expressed in kilo-electronvolts (keV). This design provides intrinsic spectral resolution, higher spatial resolution and improved dose efficiency, thereby offering potential for enhanced image quality [32]. The PCD-CT provides VMIs as default output, which altered the way radiologist view images. Instead of multi-energy images originating from the poly-energetic energy spectra of an energy-integrating detector, PCD-CT offers the possibility to view VMIs by default. In addition, PCD-CT facilitates material decomposition applications and therefore the possibility to reconstruct virtual non contrast images and iodine maps [33].

Optimization challenge

From both a clinical and research perspective, CT protocol optimization aims to achieve an appropriate balance between diagnostic image quality and patient burden, the latter mainly consisting of radiation exposure and iodinated contrast medium administration. This task has become increasingly complex because CT technology itself is constantly evolving, offering new capabilities and imaging techniques with each generation of scanners. Therefore, CT protocol optimization is an ongoing process rather than a one-time adjustment.

In recent years, increasing attention has been given to personalized CT acquisition protocols, aiming to tailor scan parameters to individual patient characteristics rather than relying on fixed, one-size-fits-all settings [9; 22; 34]. This seeks to deliver the

lowest radiation dose required, and the lowest contrast medium dose, that still achieves adequate diagnostic image quality. The introduction of PCD-CT adds a new level of complexity to this process, as VMIs are now viewed by default. Rather than simply replacing conventional polyenergetic images, VMIs introduce new degrees of freedom in energy selection. This will fundamentally change the optimization landscape and requires protocol optimization strategies that explicitly account for the energy-dependent trade-offs between contrast enhancement, noise, and overall image quality.

Such optimization is inherently challenging. Radiation dose and contrast media dose are relatively straightforward to quantify, whereas assessing diagnostic image quality is far more complex [7; 12; 15; 35-39]. Image quality is influenced by multiple interacting factors and ultimately depends on clinical perception and the diagnostic task, which cannot be easily captured by a single physical metric.

Image quality

Image quality is generally divided in four main categories: noise, contrast, spatial resolution, and artifacts [7; 12; 40]. They are illustrated in Figure 3.

- Noise refers to random variations in pixel values, that do not represent true anatomical variation. Excessive noise reduces the signal-to-noise ratio and can therefore obscure subtle findings. Noise tends to decrease when radiation dose is increased.
- Contrast is the difference in grayscale values between tissues, and determines how well structures can be distinguished. Contrast generally improves at lower photon energies and can be further enhanced using iodinated contrast media.
- Spatial resolution reflects the level of detail an imaging system can visualize.
- Artifacts are unwanted distortions or errors in the image that do not originate from the patient. This includes motion artifacts, beam-hardening, and metal artifacts and it may degrade diagnostic accuracy.

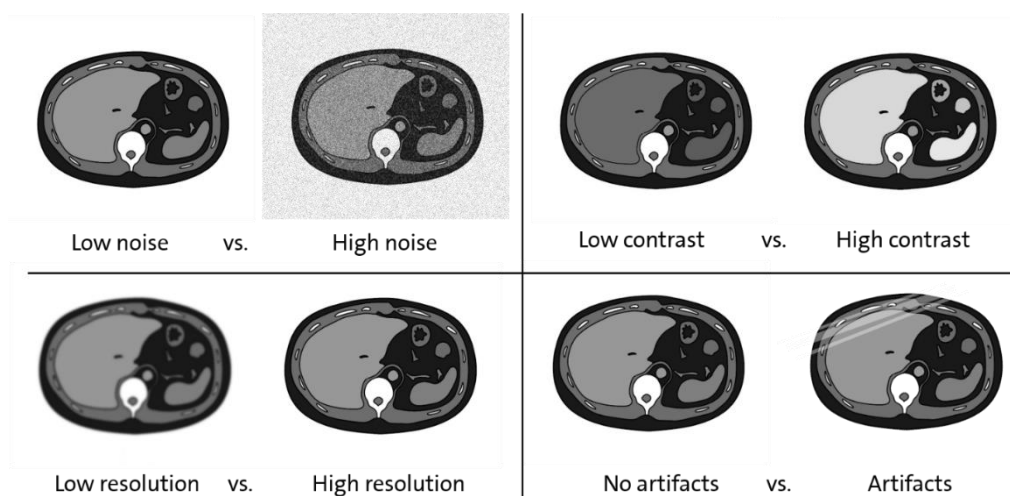


Figure 3 Schematic visualization of the main components of CT image quality: noise, contrast, spatial resolution, and artifacts.

The relation between these physical parameters and the image quality perceived by the radiologist is not straightforward, as sufficient diagnostic image quality also depends on the diagnostic task at hand as well as preferences of the radiologist [12]. Diagnostic image quality is often assessed subjectively by observers (i.e. radiologists). Typically, a Likert scale method is used in which observers need to assess diagnostic image quality on a scale of 1 to 5 (e.g. 1-very poor, 2-poor, 3-moderate, 4-good, 5-very good image quality). However, aside from the fact that subjective image quality assessment in general is very time-consuming, this form of assessment can lead to high inter- and intraobserver variation [41-43]. In addition, observers tend to fear assessing the highest and lowest values, called the central tendency bias [44]. This results in the majority of the scans receiving the same moderate Likert score, and thus not being very informative.

Objective parameters for diagnostic image quality assessment are used as well. The most common ones require manually indicating region-of-interests (ROIs) and measuring the CT signal (i.e. Hounsfield Unit (HU)) distribution. Common parameters to measure noise and contrast are the mean HU value $\pm$ standard deviation, the signal-to-noise ratio (SNR) and the contrast-to-noise ratio (CNR). However, similar to the subjective Likert scale method, manually assigning ROIs is time-consuming

and leads to interobserver variation. In addition, size of the ROI might influence the results and might not be representative for the complete CT-series [45].

An alternative choice would be image quality assessment with phantoms, as phantom images are reproducible and the physical properties and geometry of the images object are a priori known [46; 47]. However, phantoms cannot reproduce the anatomical complexity and variability seen in real patients. They lack realistic tissue heterogeneity, pathological structures, physiological motion, and clinically relevant artefacts, all of which influence diagnostic image quality. As a result, phantom-based measurements do not fully reflect how an imaging technique performs in routine clinical practice, making clinical validation essential.

Ideal scenario

In an ideal scenario, image quality assessment would be performed on clinical scans in a fully automated manner, preferably immediately after the acquisition on the CT console. It would enable immediate identification of suboptimal acquisitions. Scans of insufficient quality could be repeated in the same session, preventing patient recall and reducing workflow delays. Continuous on-scanner monitoring would also support early detection of gradual image-quality degradation, ensuring timely technical intervention and improving overall system reliability. Moreover, large-scale, automated assessment at the scanner level would facilitate systematic protocol optimization, as image quality trends could be evaluated across institutions, scanners, and patient populations in a consistent and data-driven manner.

To achieve this, automated and standardized image quality assessment methods for clinical CT data are warranted, which are currently lacking.

Aim of this thesis

The aim of this thesis is 1) to assess how PCD-CT improves CT image quality and enables optimized protocols with reduced contrast medium dose and 2) to improve both subjective and objective image quality assessment methods to facilitate protocol optimization and large scale IQ evaluation.

Outline of this thesis

Part I - Exploiting the full potential of photon-counting detector CT

Chapter 2 presents an educational review aimed at helping radiologists understand the fundamentals of conventional EID-CT and PCD-CT. It highlights the differences between kVp-based imaging in conventional CT and virtual monoenergetic images (VMI) with specific keV levels in photon-counting CT, and it discusses the potential benefits of VMIs for image quality (i.e. for contrast, noise, and contrast-to-noise ratio). The review is supported by controlled phantom experiments. Building on this, **Chapter 3 and 4** evaluate the clinical impact of PCD-CT on contrast-medium usage compared to EID-CT. These chapters investigate to what extent PCD-CT may enable reduced contrast doses while maintaining sufficient diagnostic image quality, in both abdominal CT and pulmonary embolism CTPA scans.

Part II – Diagnostic image quality assessment

Studies on CT protocol optimization as the one from chapter 3 and 4 usually require image quality assessment at some point. **Chapter 5** addresses the limitations of the standard subjective image quality assessment method by developing and investigating a pairwise comparison method as an alternative to the Likert scale scoring. It aims to improve reliability in observed-based image quality assessment. Recognizing the need for objective image quality assessment methods, **Chapter 6** provides a systematic review of advanced objective image quality metrics, that go beyond the standard manual ROI-based measurements. Finally, **Chapter 7** introduces an approach for fully automated assessment of contrast and noise, aiming to provide scalable solutions for large datasets.

Chapter 8 provides a discussion of the findings in this thesis and it outlines future directions and the potential role of AI-based image quality evaluation.

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CHAPTER

2



It is all about energy

The changing role of X-ray photons in photon-counting detector CT: What the radiologist needs to know about kV and keV

E.J.I. Hoeijmakers

B. Martens

J.E. Wildberger

T.G. Flohr

C.R.L.P.N. Jeukens

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Abstract

CT image quality is strongly influenced by the energy of the X-ray beam. The tube voltage (kV) determines a polyenergetic X-ray spectrum, which affects both image contrast and noise in conventional energy-integrating detector CT (EID-CT)-imaging. In photon-counting detector CT (PCD-CT), spectral datasets are always available by default, and virtual monoenergetic images (VMI) can be reconstructed at any specific photon energy (keV). This enables radiologists to tailor energy selection to dedicated clinical tasks, achieving an optimal balance between contrast, noise, and radiation dose, and ultimately enhancing diagnostic confidence. This article educates on the differences between polyenergetic and monoenergetic images to provide radiologists with practical insights into optimizing spectral imaging protocols and leveraging keV differences in routine. Using an anthropomorphic phantom with iodine rods, differences in image quality, specifically image contrast and noise, are demonstrated between polyenergetic EID-CT and PCD-CT VMIs.

Keywords

Computed tomography; Image quality; Photon-counting CT; kV; Monoenergetic CT

Introduction

Computed Tomography (CT) has revolutionized diagnostic imaging by providing detailed cross-sectional images of the body. Image quality (IQ) depends heavily on scanner technology, acquisition parameters, patient characteristics, and post-processing algorithms. Importantly, optimal IQ for the given task must be balanced against radiation exposure to minimize the risks associated with ionizing radiation [1].

CT systems use a broad polyenergetic X-ray spectrum generated by the X-ray tube, with its shape influenced by the tube voltage (kV). In conventional energy-integrating detector CT (EID-CT), the detector sums the energy of all incoming photons, so image contrast and noise are determined by the combined effect of the entire spectrum. EID-CT systems are also capable of spectral imaging through dual-energy CT (DECT) techniques, by acquiring data with two different X-ray spectra using techniques such as dual-source systems and rapid kV-switching, or by energy separation on the detector domain with dual-layer detectors [2]. This leads to the ability to generate virtual monoenergetic images (VMIs), simulating tissue attenuation at a single (monoenergetic) energy level (measured in kiloelectronvolts, keV) [3].

More recently, photon-counting detector CT (PCD-CT) has emerged as a promising next-generation technology that enables spectral imaging as well. This system directly detects individual X-ray photons including their energy and assigns them to discrete energy bins. PCD-CT offers potential benefits including improved spatial resolution, enhanced contrast-to-noise ratio (CNR), electronic noise removal, and reduced radiation dose [4]. PCD-CT provides VMIs for default viewing as a standard acquisition mode. Given these advantages, PCD-CT has the potential to become the new standard in clinical CT imaging over the coming decades [5].

This review educates on the impact of energy on CT IQ, both at the acquisition stage (kV) and during post-acquisition reconstruction using VMIs (keV level). Using an anthropomorphic phantom with iodine rods, polyenergetic EID-CT images and PCD-CT VMIs are compared across various kV and keV levels. Key differences in IQ are demonstrated via noise and contrast graphs. The aim is to provide radiologists with

practical insights into optimizing spectral imaging protocols to balance diagnostic IQ with radiation dose.

CT technology – the basics

Table 1 Essential definitions

keV (kiloelectronvolt): A unit of energy. In CT, keV is used to describe the energy level of individual photons or the reference energy in virtual monoenergetic image reconstructions. kV (kilovoltage): The X-ray tube voltage applied during CT data acquisition. It determines the maximum kinetic energy than an electron can achieve on its way from the cathode to the anode plate of the X-ray tube. For example, at a tube voltage of 120 kV, the maximum electron energy is 120 keV. This is then also the upper energy limit of the polyenergetic X-ray spectrum generated by the incident electrons. At a tube voltage of 120 kV, it includes photons with energies ranging from a few keV up to, but not exceeding, 120 keV. VMI (virtual monochromatic image): A reconstruction generated from spectral data of conventional dual-energy CT or PCD-CT that simulates images as if acquired at a single, specific photon energy level (measured in keV).

X-ray tube

An X-ray tube generates X-rays by accelerating electrons from a heated cathode toward a rotating metal anode (typically tungsten) using a high tube voltage (kV). The kV determines the electron acceleration, and thus their energy, e.g. at 120 kV, electrons hitting the anode have an energy of 120 keV (Table 1). As the accelerated electrons collide with the anode, they rapidly decelerate, converting most of their energy (>99%) into heat. A small fraction is emitted as X-ray photons, forming a polyenergetic X-ray beam, i.e. it contains photons with energies up to 120 keV, called a spectrum. The spectrum is composed of:

- 1) A continuous Bremsstrahlung spectrum, with photon energies ranging from very low up to the energy determined by the kV value.

- 2) Discrete characteristic peaks which are sharp increases in intensity at specific energies, caused by electron interactions with the atoms of the anode material.

Although kV sets the maximum photon energy, most X-ray photons have lower energies [6]. Therefore, a spectrum is usually characterized not only by its maximum energy, but also by its effective photon energy, which depends on several factors, including beam filtration. Filtration is used to selectively remove low-energy photons at the tube aperture. These photons would be fully absorbed by the patient without contributing to image formation, leading to an unnecessary increase in radiation dose. This pre-patient filtration effectively increases the effective energy of the X-ray beam, producing a 'harder' spectrum. Figure 1 shows a typical filtered polyenergetic X-ray spectrum [7].

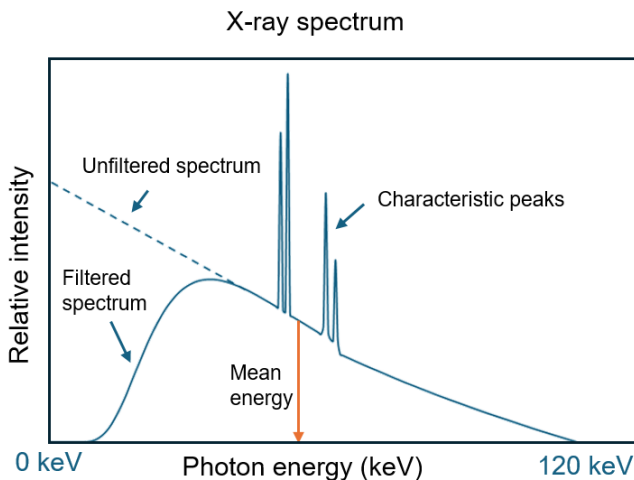


Figure 1 Typical X-ray energy spectrum at 120 kV tube voltage (solid line). The dashed line shows the spectrum without filtration of low-energy photons. The orange arrow indicates the mean energy of the spectrum which depends on the amount of filtration [7].

Image contrast and noise in CT

The number of X-ray photons reaching the CT detector depends on total attenuation along their path through the body. Variations in tissue attenuation result in differences in detected photon number and energy, determining image contrast. In

CT, attenuation is quantitatively expressed in Hounsfield Units (HU), a standardized scale where water is set at 0 HU and air at -1000 HU. Good contrast enables visualization of subtle density variations, essential for detecting lesions, and abnormalities [8].

X-ray attenuation is primarily caused by two physical interactions:

- 1) Photoelectric effect: The process of photons transferring their full energy to an inner-shell electron, ejecting it from the atom. This process increases with atomic number (Z) of the absorbing material ($\propto Z^3$), and decreases sharply with increasing photon energy E ($\propto 1/E^3$). As a result, it dominates at lower energies (lower kV-spectra) and in high- Z materials such as bone and iodine (Figure 2a). Hence, iodinated contrast agents are most effective at low kV, where vascular and lesion contrast is maximized [9].
- 2) Compton scattering: The process of a photon transferring part of its energy to an outer-shell electron. The photon is deflected with reduced energy, while the electron is ejected from the atom. Because Compton scattering is relatively independent of Z , it produces more uniform attenuation, resulting in less attenuation differences between tissues. It is the dominant interaction at higher energies and in low- Z tissues such as fat and soft tissue (Figure 2a).

Thus, image contrast depends on both the X-ray spectrum energy and tissue composition [6]. In general, attenuation decreases as photon energy increases, because higher-energy photons are less likely to interact with matter. Figure 2b illustrates this for bone and soft tissue: while bone shows much stronger attenuation than soft tissue at low energies, the curves converge at higher energies where Compton scattering dominates [10]. Consequently, bone-soft tissue contrast diminishes at higher energies, as their attenuation becomes more similar.

Noise in CT refers to random pixel intensity fluctuations that can obscure fine detail and reduce diagnostic confidence [8]. It arises from quantum noise (due to statistical variation in the number of photons reaching the detector) and electronic noise (unwanted electrical signals from the CT electronics) [6]. Electronic noise becomes particularly noticeable when photon counts are low, typically due to high attenuation (e.g. obese patients or dense tissues) or low tube current (mAs). Increasing photon

energy (higher kV) improves penetration and thus reduces noise, at the cost of image contrast. Conversely, lowering kV increases contrast but also increases noise at the same radiation dose. The contrast increase is particularly pronounced for iodine and can outweigh the noise increase. Therefore, modern CT scanners incorporate tube voltage selection algorithms that propose an appropriate kV based on patient size and clinical indication to optimize contrast, noise, and radiation dose.

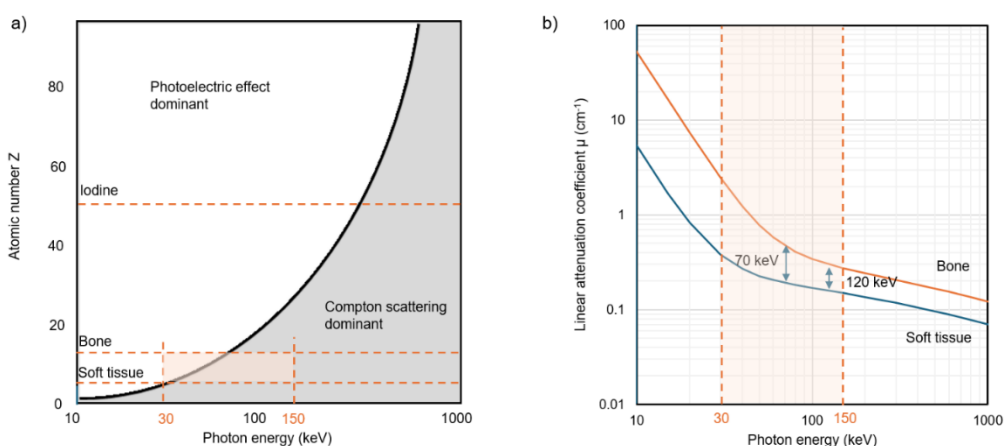


Figure 2 a) Graph illustrating the relative importance of the photoelectric effect and Compton scattering for different atomic numbers (linear scale) and photon energies (logarithmic scale). b) Graph showing the linear attenuation coefficient of bone and soft tissue (logarithmic scale) plotted against the photon energy (logarithmic scale). The orange areas in both graphs indicate the relevant CT energy range. The blue arrows denote the difference between bone and soft tissue at 70 and 120 keV which is related to the contrast between the two tissues [10].

Conventional CT technology

Conventional CT scanners use EIDs that convert absorbed X-ray photons into visible light via a scintillator, subsequently detected by photodiodes (so-called indirect detection). EIDs measure total energy over time and cannot distinguish individual photon energies, causing loss of (spectral) information. Consequently, in conventional CT using a single kV, materials with similar attenuation for that kV value yield similar CT values in a CT-image and cannot be differentiated [11-13].

Dual energy CT technology

Because photoelectric effect and Compton scattering depend differently on photon energy, material attenuation varies more at low energies and converges at higher energies (Figure 2b). DECT uses this characteristic by acquiring data at two distinct effective energies, allowing comparison of attenuation behavior between materials [2].

By analyzing energy-dependent attenuation, DECT can distinguish two materials (see Table 2 for the underlying mathematics of material decomposition). For instance, soft tissue and iodine can both have similar low CT values at a high mean energy. At low effective energy, iodine—due to its higher atomic number and thus more photoelectric effect—shows significantly increased attenuation, enabling effective material separation [3; 14]. This allows generation of material specific maps and virtual non-contrast images by subtracting iodine contributions.

Another impactful post-processing application of DECT is VMIs [15-18]. VMIs are reconstructed from spectral CT data to simulate how each image voxel would appear if scanned with a single photon energy (e.g. 70 keV), rather than with a polyenergetic spectrum (e.g. 120 kV). Users can select a specific VMI energy during image reconstruction, optimized for the clinical task. In this way, the prospective, kV-based determination of image characteristics is replaced by a retrospective keV-adjustment. Still, the optimal acquisition kV must be determined dependent on patient size and tube power requirements [19]. However, depending on the CT-system, the selection may be limited to a fewer options (e.g., 120 kV or 140 kV for PCD-CT in spectral standard mode).

VMIs (particularly at higher keV, approx. ≥ 70 keV) can mitigate beam-hardening artifacts and provide more predictable attenuation values [20-22]. Moreover, VMIs allow for the reconstruction of low keV images, which increase iodine contrast but also increase noise. In addition, high keV images can be reconstructed, which reduce noise and can partially reduce metal artifacts, although they cannot fully replace dedicated metal artifact reduction techniques [23]. Importantly, in commercial implementations, the displayed VMI may include additional vendor-specific post-processing steps, such as energy-dependent noise reduction, which modify the noise characteristics compared with the idealized equation [16].

Table 2 Technical note (optional read)**Mathematics behind material decomposition and VMI**

X-ray attenuation through a material follows an exponential decay described by the Beer-Lambert law,

$$I = I_0 \cdot e^{-\mu(E) \cdot x}$$

with I_0 the incoming X-ray intensity, I the transmitted intensity, $\mu(E)$ the linear X-ray attenuation coefficient at energy E , and x the thickness of the material.

Material decomposition:

DECT is based on the fact that the attenuation of each voxel can be modeled as a linear combination of two basis materials,

$$\mu(E) = w_1 \cdot \mu_1(E) + w_2 \cdot \mu_2(E)$$

with $\mu(E)$ the effective attenuation in the voxel measured, $\mu_1(E), \mu_2(E)$ the known energy-dependent attenuation coefficients of the base materials (e.g., iodine and soft tissue) and w_1, w_2 the respective contributions of the base materials to the voxel. The latter are unknown and estimated by solving the two equations for two energy levels. Because X-ray attenuation is caused by only two physical interactions with different energy dependence (photoelectric effect and Compton scattering), this approach allows differentiation of only two materials simultaneously, even when more energy levels are available. Three materials can be differentiated if an additional boundary condition is considered, such as conservation of volume. K-edge imaging, i.e., exploiting a K-edge within the X-ray spectrum relevant to CT, is beyond scope here [3].

VMI calculation:

Once the base material contributions are known, a VMI at any monochromatic level E_{mono} can be reconstructed:

$$\mu_{VMI}(E_{mono}) = w_1 \cdot \mu_1(E_{mono}) + w_2 \cdot \mu_2(E_{mono})$$

Instead of iodine and soft tissue images, photoelectric effect images and Compton effect images may be used as a basis. Alternatively, a VMI is simply a weighted addition of the original low energy and high energy images μ_{low} and μ_{high} , calibrated to correctly represent the iodine attenuation at the desired energy [18].

$$\mu_{VMI}(E_{mono}) = w(E_{mono}) \cdot \mu_{low} + (1 - w(E_{mono})) \cdot \mu_{high}$$

It can be shown that these approaches are mathematically equivalent. In commercial implementations, the displayed VMI may include additional vendor-specific post-processing steps, such as energy-dependent noise reduction, which modify the noise characteristics compared with the idealized equation [16].

Photon-counting CT Technology

PCDs convert photons directly into electrical signals, enabling measurement of each individual photon's energy (Figure 3). Each photon produces a voltage peak proportional to its energy. By applying an energy threshold, (low-energy) electronic noise can be eliminated, improving IQ. Detected photons are read-out into energy bins, inherently enabling multi-energy data acquisition from a single X-ray spectrum. This enables reconstruction of VMIs from a single acquisition [12; 24]. Moreover, PCD-CT enhances the contribution of low-energy photons because each photon is counted equally regardless of its energy. In contrast, in EID-CT, high energy photons contribute more to the summed (energy-integrated) signal than the low energy photons. Since low energy photons carry valuable contrast information, their enhanced contribution in PCD-CT improves the quality of VMIs with respect to VMIs calculated with DE.

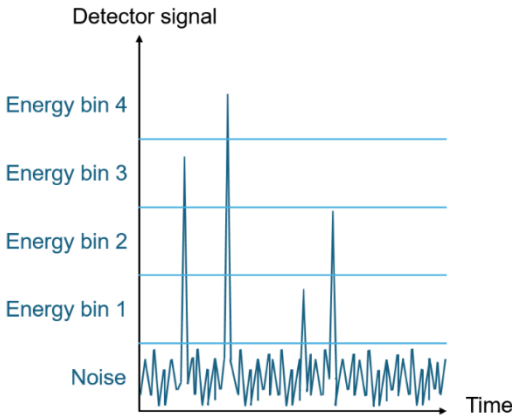


Figure 3 Graph illustrating the signal detected by a photon-counting detector, which is able to register individual photon energy peaks and to sort them in their corresponding energy bin.

It is important to note that in PCD-CT, spectral datasets are available by default. This shifts the focus from polyenergetic CT images to standard viewing of VMIs at a chosen keV-level (Table 3). Nevertheless, kV selection remains relevant, as it defines the upper energy limit of the X-ray beam and affects penetration as well as the available tube power reserves. Whereas for polyenergetic images, often lower kV is used to optimize contrast, in PCD-CT the focus is more on ensuring sufficient spectral information across the energy range, which favors higher kV. Lower kV, often used for dose reduction in pediatrics, narrows the available energy spectrum for VMI reconstruction.

Table 3 Key concept

VMIs at a certain keV-level (either with DECT or PCD-CT) are NOT equivalent to conventional kV images. VMIs SIMULATE images as if acquired with monoenergetic photon beams based on mathematical modeling of attenuation and material decomposition. Conversely, conventional kV images are acquired using a polychromatic spectrum, where the detector integrates the total deposited energy. As a result, attenuation values, contrast behavior, and noise characteristics in VMIs differ from those in kV images, and they cannot be directly matched or substituted one-to-one.

However, for each kV setting on EID-CT, the corresponding mean photon energy of the polychromatic X-ray spectrum can be calculated using a validated spectrum simulation tool (e.g. SpekPy). Although the mean energy of a kV spectrum is not identical to the respective VMI keV-level, this approach provides a reasonable approximation for matching the energy content of the polyenergetic spectra to the VMI keV-level.

Phantom set-up and methodology

To assess IQ for polyenergetic images and VMIs across varying kV and keV settings, an anthropomorphic phantom (Kyoto PBU-60, Kyoto Kagaku Co, Ltd, Kyoto, Japan) was scanned with nine material Gammex rods (Sun Nuclear – a Mirion Medical Company, Middleton, WI, USA) (Figure 4a). The rods were placed on the anterior side of the Kyoto phantom, directly above its liver region. These included seven iodine concentrations (2.0-20.0 mg/mL), covering a clinically relevant range, as well as liver and muscle-equivalent rods, as defined by the International Commission on Radiation Units and Measurements (ICRU) [25]. Scans were performed on a third generation dual-source EID-CT (SOMATOM Force, Siemens) and on a first generation dual-source PCD-CT (NAEOTOM Alpha, Siemens Healthineers, Forchheim, Germany), with the phantom centered at the isocenter of both scanners.

Detailed acquisition and reconstruction parameters are provided in Table 4a. Although a dual-source CT system was used, polyenergetic images on the EID-CT were acquired in conventional single-kV mode, at tube voltages from 70 to 140 kV, in 10 kV steps. On the PCD-CT, data were acquired at a fixed 120 kV tube voltage (clinical abdominal protocol). VMIs were reconstructed in a range of 53 – 90 keV, at keV levels matching with the effective energies of the EID-CT spectra according to Table 4b, which were obtained from measurements of the attenuation of a 15 mg/ml iodine rod in a 20 cm water phantom and comparison to NIST-tables of the expected keV-dependent attenuation. The effective energy as received by the detector depends on the size of the phantom. As the Kyoto phantom is larger than the 20 cm water phantom used for Table 4b, the measured kV-dependent CT-values of the EID-CT were eventually matched to the corresponding CT-values of the VMIs at the respective keVs to get an approximation of the effective energies of the EID-CT spectra with the Kyoto phantom. This enables a direct comparison of the corresponding noise and CNR results. In addition, a polychromatic image (T3D image) was made, reconstructed from the low-energy threshold PCD-CT data, for which the CT values were also matched to the CT-values of the EID-CT to obtain the effective energy. Moreover, for both scanners, images were acquired across a range of CT Dose Index volume ($CTDI_{vol}$) values (2-25 mGy). Scans were reconstructed with

iterative reconstruction (ADMIRE 3 on the EID-CT, QIR 3 on the PCD-CT) with equivalent soft tissue kernels.

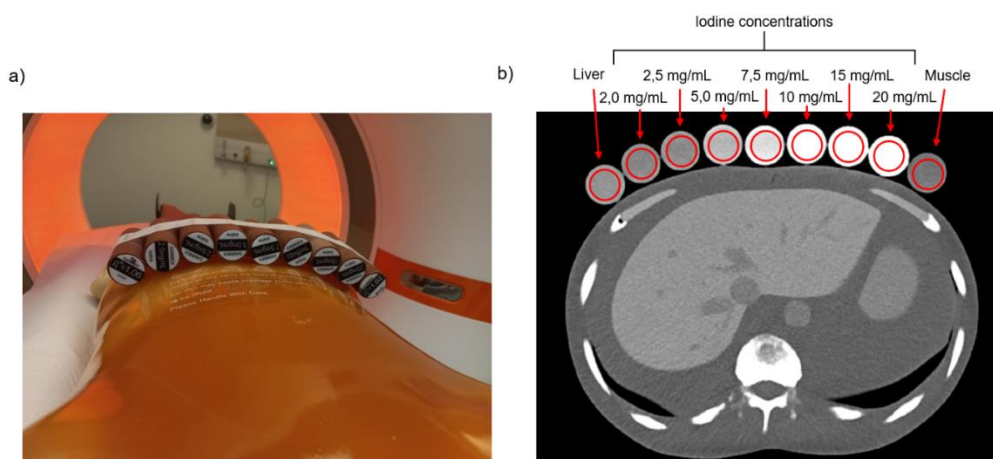


Figure 4 a) Photo of the set-up of the Kyoto phantom with the Gammex rods. b) Example scan (window width 400, window level 40) of the phantom set-up with circular regions of interest (ROIs) drawn in the iodine rods, liver rod and muscle rod of the Kyoto phantom. These ROIs were used for measurements of mean attenuation, contrast, contrast-to-noise ratio.

Image quality parameters

Image quality measurements were conducted on a single axial slice corresponding to the largest visible liver cross-section. Mean attenuation was measured in all iodine, liver, and muscle rods of the phantom by placing circular ROIs with 1 cm radius centered in the rods (Figure 4b). Contrast was calculated as the signal difference between the ROIs and the muscle ROI. In addition, the global noise level (GNL) was determined by calculating the standard deviation of pixel values using a 7x7 mask centered on pixels within the soft tissue attenuation range [26]. A histogram of these standard deviations was generated, and the GNL was defined as the median standard deviation, providing a measure of image noise in homogeneous soft tissue regions. Lastly, CNR was calculated as contrast divided by the GNL.

Table 4 a) Overview of acquisition and reconstruction parameters.

**Due to tube power limitations, some high $CTDI_{vol}$ values cannot be achieved at low kV settings on the EID-CT.*

†See Table 4b for the actual kV and keV values used. b) Approximation of the effective energy of the kV spectra on the 3rd generation dual-source EID-CT in keV, which were obtained from measurements of the attenuation of a 15 mg/ml iodine rod in a 20 cm water phantom and comparison to NIST-tables of the expected keV-dependent attenuation.

a)

CT parameters	3 rd generation dual-source EID-CT	1 st generation dual-source PCD-CT
Peak Tube voltage	70 – 140 kV [†]	120 kV (reconstructed for 53 – 90 keV) [†]
$CTDI_{vol}$ (mGy)	2 – 25*	2 – 25
Slice thickness (mm)	3	3
Increment (mm)	2	2
Pitch	0.9	0.9
Acquisition mode	Axial	Axial
Gantry rotation time (s)	0.5	0.5
FoV (mm)	320	320
Iterative reconstruction	ADMIRE 3	QIR 3
Kernel	Br40d	Br40

b)

Polyenergetic X-ray spectrum (kV)	Approximate effective energy (keV) for a 15 mg/ml iodine rod in a 20 cm water phantom
70	53
80	57
90	61
100	65
110	66
120	67
130	69
140	70

EID-CT, energy-integrating detector CT; PCD-CT, Photon-counting detector CT; $CTDI_{vol}$, Computed Tomography Dose Index volume; FoV, Field of View;

Influence of energy selection in CT on image quality

Effect of kV / keV on Contrast

Figure 5a compares mean attenuation values of iodine (5 and 10 mg I/mL), liver, and muscle rods obtained with polyenergetic images from EID-CT and PCD-CT VMIs. Results for the other concentrations are provided in the Supplementary Material. For the polyenergetic images, attenuation is shown versus polyenergetic tube voltage (kV), and for the VMIs from the PCD-CT as a function of the reconstructed energies (keV). For 10 mg I/mL, attenuation decreases with increasing energy for both imaging types, from ~420 HU at 70 kV to ~180 HU at 140 kV in polyenergetic EID-CT images, and from ~460 HU at 53 keV to ~150 HU at 90 keV in the VMIs. The measured kV-dependent CT-values of the EID-CT are matched to the corresponding CT-values of the VMIs at the respective keVs to get an approximation of the effective energies of the spectra (Table 1 in the supplementary material).

The mean CT values for the liver and muscle rods remain relatively constant across this energy range, ~80 HU for liver and ~40 HU for muscle, because attenuation in low-Z tissues is dominated by the Compton effect, which changes only slightly with photon energy. Liver exhibits slightly higher CT values than muscle due to its higher electron density and, consequently, a higher probability of Compton scattering. Furthermore, as PCD-CT is less affected by beam hardening, the effective energy of the T3D is lower compared to the 120 kV polyenergetic EID-CT scan.

Since iodine attenuation decreases with increasing photon energy (due to diminishing photoelectric effect), whereas muscle attenuation remains nearly constant (Compton-dominated), image contrast (Section 'Image contrast and noise in Computer Tomography') decreases at higher energies. As expected, the 10 mg I/mL rod demonstrates higher attenuation, and therefore higher contrast, than the 5 mg I/mL rod for both scanner types, with clear separation between the concentration levels across the full energy range. Higher iodine concentrations yield steeper attenuation curves and thus more contrast, because a larger amount of iodine attenuated more X-rays, particularly at lower energies.

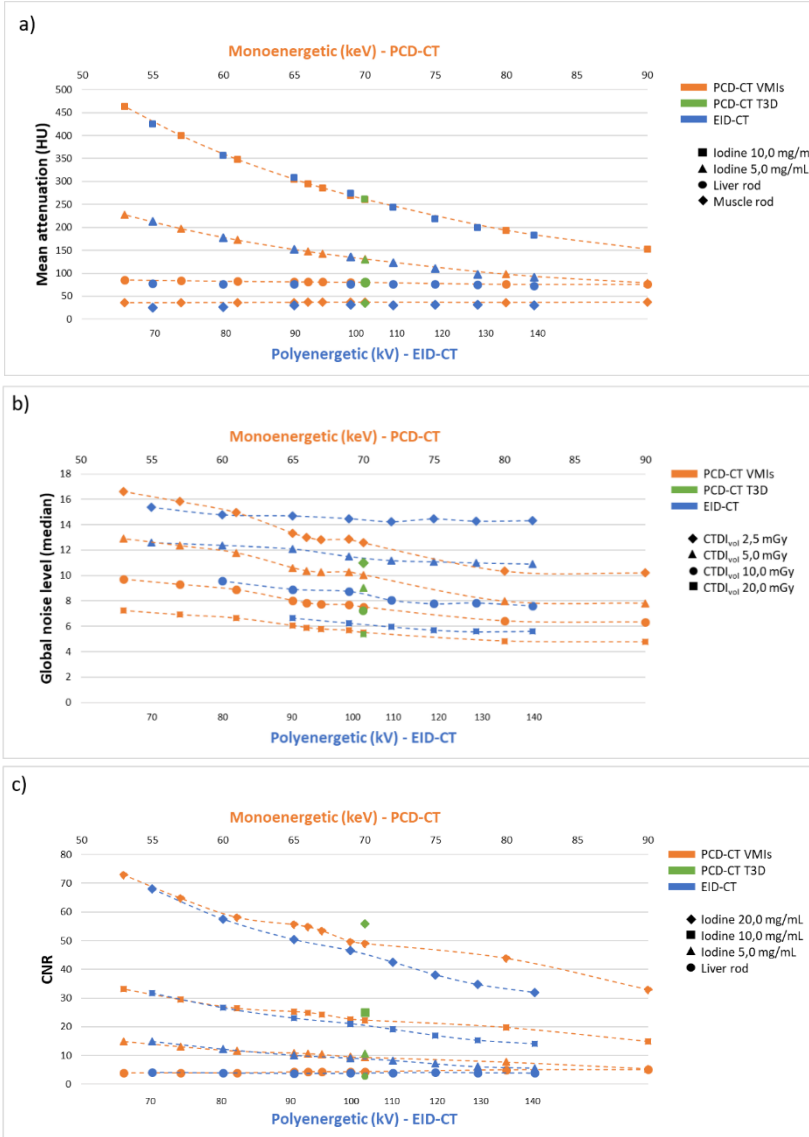


Figure 5 a) Mean attenuation values (in Hounsfield Units; HU) measured in the iodine (5.0 mg/mL and 10.0 mg/mL), liver, and muscle rods across different energy levels for CTDI_{vol} = 5 mGy. b) Global noise level (GNL) measured for different CTDI_{vol} levels across different energy levels. c) Contrast-to-noise ratio (CNR) measured in different iodine rods and a liver rod across different energy levels for CTDI_{vol} = 5 mGy. Results are shown for both polyenergetic images from an energy-integrating detector CT (EID-CT, lower x-axis), and virtual monoenergetic images and a T3D reconstruction from a photon-counting detector CT (PCD-CT, upper x-axis). Note that the effective energy of the kV values on the lower x-axis approximately corresponds to keV levels on the upper x-axis; as a result, the lower x-axis does not follow a linear scale.

Figure 6 shows clinical example scans of the same oncology patient, acquired with EID-CT at 90 kV (top) and with a PCD-CT (bottom). The VMIs were reconstructed at 65 keV, corresponding to the effective energy of the polyenergetic acquisition. Additionally, the intrinsically available spectral information of the PCD-CT, allowed for reconstruction at lower and higher keV. Reconstructions at 55 and 82 keV are shown, demonstrating the enhanced and reduced contrast, respectively.

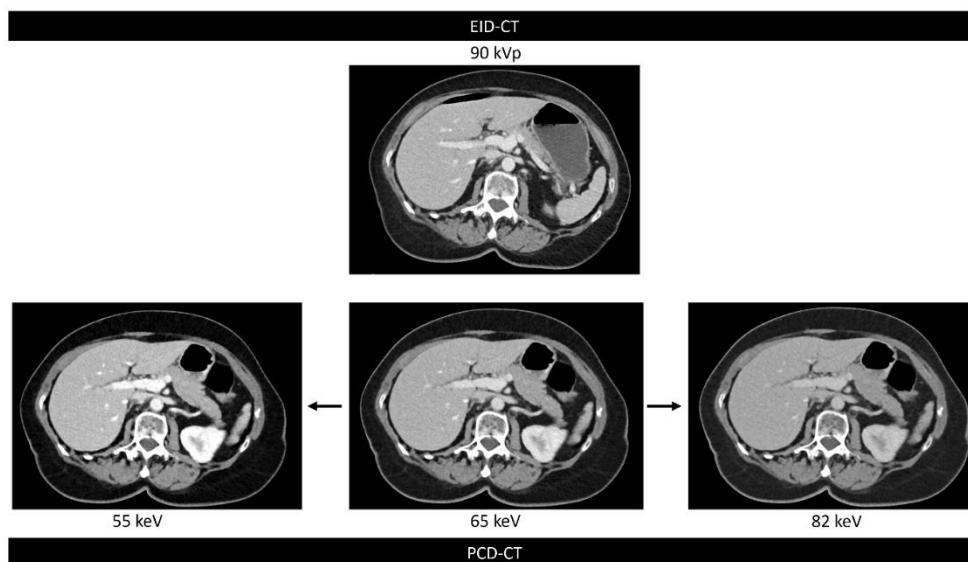


Figure 6 Clinical example scan of an oncology patient (Female, 77y, BMI 21.9 kg/m²) made with an EID-CT (upper row, 90 kV) and a follow-up scan five months later of the same patient (BMI 22.4 kg/m²) with a PCD-CT (lower row). The PCD-CT scan was acquired with 120 kV and reconstructed at 65 keV, corresponding to the effective energy of the 90 kV X-ray spectrum of the used EID-CT. Additionally, the PCD-CT images are reconstructed at 55 keV (corresponding to ~70 kV) and 82 keV (corresponding to ~140 kV) to demonstrate the intrinsically available spectral information, showing enhanced and reduced contrast, respectively. (The clinical images shown in this manuscript are included for illustrative purpose only. They were acquired as part of routine clinical care and were retrospectively selected. Ethics committee approval was waived by the institutional review board.)

Effect of kV / keV on noise

Figure 5b shows the GNL as a function of energy for both polyenergetic EID-CT images and PCD-CT VMIs at different CTDI_{vol} levels. For both detector types, noise was lower for higher radiation dose, as expected due to the larger number of detected photons resulting in reduced statistical variation (i.e. improved photon

statistics). At the lower CTDI_{vol} level of 2.5 mGy, noise would theoretically double compared to 10 mGy scans (as noise $\propto \sqrt{\text{dose}}$). However, the actual noise increase observed is only a factor of 1.6 – 1.8. This may be caused by an increased post-processing data filtration at low radiation dose levels to reduce potential noise streaks.

For the polyenergetic images, the GNL at a given CTDI_{vol} remained relatively stable across the energy range, with a slight noise reduction at higher kV values due to improved photon penetration, resulting in more photons reaching the detector. In the VMIs, the GNL showed a more pronounced decrease at higher energy levels, especially for lower radiation doses. In VMIs of PCD-CT, noise should theoretically increase significantly the further the keV values deviate from the effective energy of the spectrum of the CT scan (here 120 kV, corresponding to 75 keV). Theoretically, the VMI curves shown in Figure 5b should have a minimum at about 75 keV, and the noise should rise substantially especially for lower keVs. This is a consequence of the mathematical combination of low- and high-energy data when reconstructing VMIs. To obtain low-energy VMIs from a higher-energy CT scan (120 kV), high weighting factors (see Table 2) must be applied, which amplify image noise. However, modern image processing in VMI reconstruction prevents excessive noise at both ends of the energy spectrum [16]. It is fine-tuned resulting in the noise curves shown in Figure 5b. While noise in polyenergetic images is a natural consequence of the underlying physical principles, the energy dependence of noise in VMIs is strongly determined by image processing [16].

The noise in the polychromatic (T3D) image of the PCD-CT at 120 kV (effective energy 70 keV) is lower than the noise in the 120 kV image of the EID-CT (effective energy 75 keV), especially at lower CTDI_{vol}. One reason for this may be the absence of electronic noise in PCD-CT.

Figure 5c shows the CNR values for multiple iodine concentrations at a fixed radiation dose across different energy levels for both polyenergetic images and VMIs. For both imaging types, CNR decreases with increasing energy due to the reduction in iodine contrast at higher energies. However, for polyenergetic images, the decline in CNR at higher energies is more pronounced than for the VMIs. This is a result of the noise

pattern observed in Figure 5b. Iodine CNR in VMIs is similar to polyenergetic EID-CT images for energies up to 60 keV, and higher for higher energies.

Summary and conclusions

This review illustrates the fundamental difference between conventional tube potential (kV) in polyenergetic CT images from an EID-CT and VMIs (keV) from a PCD-CT. While both parameters describe the 'energy' of the X-ray beam, they are conceptually different. A kV setting defines a polyenergetic X-ray spectrum with a broad range of photon energies, whereas keV levels represent VMIs, simulating images as if acquired with a single photon energy.

This review does not aim to provide a head-to-head performance comparison between EID-CT and PCD-CT, but rather to illustrate how fundamental differences in image formation between polyenergetic and monoenergetic imaging with PCD-CT affect contrast and noise characteristics, even at comparable energy levels. In this work, VMIs were reconstructed from PCD-CT data, but it should be emphasized that VMIs can also be generated with other spectral CT technologies. Likewise, polyenergetic images can be generated from PCD-CT data.

Understanding the differences is essential for interpreting attenuation and noise behavior. In polyenergetic images from an EID-CT, increasing the kV hardens the X-ray beam, shifts the effective energy upward, and thereby reduces iodine contrast. Increasing the keV level in PCD-CT VMIs reduces contrast as well. Regarding noise, both systems show reduced noise at higher radiation doses, but PCD-CT demonstrates a stronger energy dependence with a reduction in noise for higher energy levels. CNR is similar for energies up to 60 keV in the VMIs from the PCD-CT and in the polyenergetic EID-CT images and higher in the VMIs at higher energies because it decreases less rapidly. Clinically, this allows for retrospective optimization of IQ without needing multiple acquisitions at different tube voltages [27]. This may improve diagnostic confidence, particularly when assessing iodine-enhanced lesions. Moreover, the ability to select optimal VMI levels post-acquisition may facilitate the

optimization of acquisition protocols in future examinations which could contribute to radiation dose reduction, while maintaining diagnostic IQ. In clinical practice, automated VMI optimization tools such as CARE keV already prospectively link radiation dose to a task-specific VMI energy [19]. In that context, dose optimization has already been performed prior to acquisition, and the benefit of retrospective VMI adjustment may be limited to fine-tuning image impression, enhancing contrast in the event of a missed contrast bolus, or reducing metal artifacts.

In the presented phantom experiment, kV spectra were linked to VMI keV levels by matching the effective photon energy of the polyenergetic beam. Although this is not identical to a true monoenergetic acquisition, it provides a reasonable and intuitive framework for comparison. Although the quantitative results of this study are specific to the scanners used, the trends observed are likely generalizable to other scanners. Absolute values may differ due to variations in detector technology, reconstruction algorithms, and filtration.

This review is limited by the fact that both scanners are from a single vendor, as only one commercial PCD-CT system was available at the time. However, the educational insights regarding polyenergetic versus monoenergetic imaging are physics-driven and the essence will be the same for different vendors. Also, the present study focused on image quality in terms of contrast and noise, other factors such as spatial resolution were not evaluated. Reported improvements in spatial resolution for PCD-CT are primarily related to intrinsic detector properties rather than to virtual monoenergetic reconstruction [4]. Furthermore, while this review focused on iodine-based soft-tissue contrast, other applications such as calcium discrimination or lung nodule imaging may exhibit different spectral characteristics and optimal energy ranges, which should be explored in future studies.

In conclusion, kV and keV should not be seen as interchangeable terms. kV defines the X-ray spectrum at acquisition whereas keV in VMIs is a reconstruction parameter and not a direct modulation of beam energy. Both determine the trade-off between image contrast and noise, but PCD-CT and DECT uniquely offer spectral data that allow radiologists to optimize IQ retrospectively by selecting the most suitable keV levels.

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Supplementary material

Table 1 Approximation of the effective energy of the kV spectra on the dual-source EID-CT in keV experimentally obtained for the Kyoto phantom.

Polyenergetic X-ray spectrum (kV)	Approximate effective energy (keV) for the Kyoto phantom
70	55
80	60
90	65
100	69
110	72
120	75
130	78
140	82

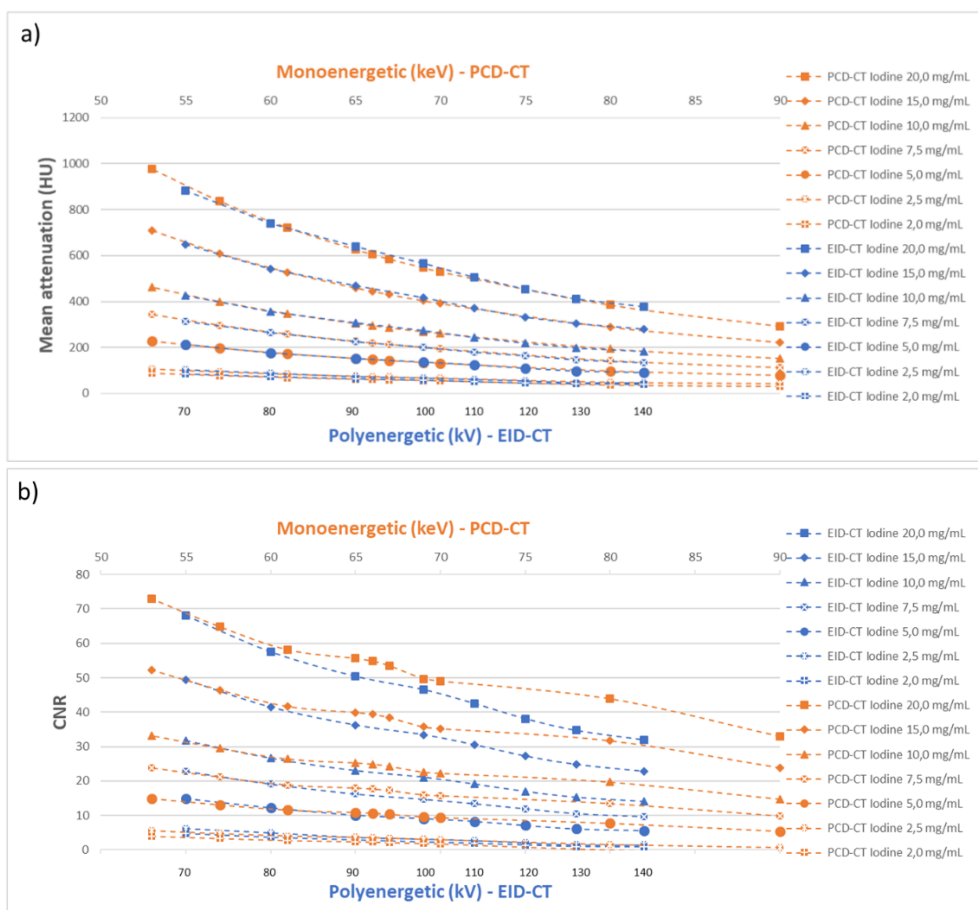


Figure 1 a) Mean attenuation values (in Hounsfield Units; HU) and b) Contrast-to-noise ratio (CNR). Results are measured in rods with varying iodine concentrations across different energy levels for CTDIvol = 5 mGy and shown for both polyenergetic images from an energy-integrating detector CT (EID-CT, lower x-axis), and virtual monoenergetic images from a photon-counting detector CT (PCD-CT, upper x-axis). Note that the effective energy of the kVp values on the lower x-axis approximately corresponds to keV levels on the upper x-axis; as a result, the lower x-axis does not follow a linear scale.

CHAPTER

3



PCD-CT enables contrast media reduction in abdominal imaging compared to an individualized kV-adapted contrast media injection protocol on EID-CT

*E.J.I. Hoeijmakers**

*L. Stammen**

J.E. Wildberger

N.G. Eijssvoogel

J. M. Hersbach

J.C.J.G. Pernot

T.G. Flohr

B. Martens[†]

C.R.L.P.N. Jeukens[†]

*E.J.I. Hoeijmakers and L. Stammen have equal contributions.

[†]B. Martens and C. Jeukens have equal contributions

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Abstract

Objectives: This study aims to demonstrate reduced iodine contrast media (CM) in routine abdominal CT scans in portal venous phase (PVP) using a photon-counting detector CT (PCD-CT) compared to total body weight (TBW) and kV-adapted CM injection protocols on a state-of-the-art energy-integrating detector CT (EID-CT) while maintaining sufficient image quality (IQ).

Materials and Methods: Consecutive contrast-enhanced abdominal PVP CT scans from an EID-CT (Nov 2022-March 2024) and a PCD-CT (Sep 2023-Dec 2023) were compared. CM parameters (total iodine load (TIL), iodine delivery rate (IDR) and dosing factor (DF)) were reported. An individualized acquisition and CM injection protocol based on TBW and kV was applied for the EID-CT and a TBW adapted CM injection protocol was used for the PCD-CT. Objective IQ was evaluated with mean attenuation (Hounsfield Units, HU), signal-to-noise ratio (SNR) and contrast-to-noise ratio (CNR)). Subjective IQ was assessed via a 5-point Likert scale by 2 expert readers based on diagnostic confidence.

Results: Based on 91 EID-CT scans and 102 PCD-CT scans a TIL reduction of 20.1% was observed for PCD-CT. PCD-CT demonstrated significantly higher SNR (9.9 ± 1.7 vs. 9.1 ± 1.8 , $p < 0.001$) and CNR (5.1 ± 1.7 vs. 4.3 ± 1.3 , $p < 0.001$) compared to EID-CT. Subjective IQ assessment showed that all scans had sufficient diagnostic IQ.

Conclusions: PCD-CT allows for CM reduction while providing higher SNR and CNR compared to EID-CT, using clinical individualized scan and CM injection protocols.

Key words

Computed tomography; Diagnostic imaging; Abdomen; Contrast media; Image quality; Photon-counting CT; Injection protocol

Introduction

About 40% of computed tomography (CT) examinations use iodinated contrast media (CM) [1]. Its risks, particularly to patients with impaired kidney function, are debated [2-4]. While there is no uniform opinion about the incidence and frequency of contrast-induced acute kidney injury, there is a consensus that CM should be minimized.

To reduce CM dose, patient-specific adjustments like weight-based CM protocols for abdominal CT are used [5; 6]. Moreover, technical advancements in modern CT help minimize CM dose by using lower tube voltages than the conventional 120 kV, which improves iodine attenuation and contrast-to-noise ratio (CNR) [5-8]. In practice, the simple 10-to-10 rule effectively reduces CM dose by 10% for every 10 kV reduction in tube voltage [7; 8]. Dual-energy CT systems can also reconstruct virtual monoenergetic images (VMI) at lower energy levels enhancing iodine CNR and reducing CM dose [9-14].

Recently introduced photon-counting detector CTs (PCD-CTs) directly convert absorbed X-ray photons into individual electrical signals enabling spectral data acquisition, image noise reduction, and improved iodine attenuation compared to conventional energy-integrating detector CTs (EID-CTs), thereby improving iodine CNR [15; 16]. Previous phantom studies have shown an 11-38% improvement of iodine CNR with PCD-CT [17; 18]. The inherently higher iodine CNR of PCD-CT can contribute to lower CM dose, even in combination with other CM dose-reducing techniques like low-energy VMIs.

CT angiography studies have shown PCD-CT enables a possible CM reduction [19-23]. Additionally, existing studies on CM reduction in abdominal portal venous phase scans did not compare PCD-CT with optimized individualized kV-adapted CM protocols in EID-CT [24-26].

This retrospective study aims to show that a reduction in CM dose can be achieved for abdominal PVP CT scans in everyday clinical practice, performed on a PCD-CT compared to scans performed on a state-of-the-art EID-CT scanner with an individualized kV-adapted CM injection protocol without loss of image quality (IQ).

Material and Methods

Ethics

This study received a waiver of written informed consent from the local ethical committee and institutional review board (METC 2020-1535-A-1), as retrospective data were analyzed anonymously.

Study population

Consecutive contrast-enhanced abdominal PVP CT scans were collected retrospectively from patients ≥ 18 years undergoing a routine clinical scan protocol for various clinical indications (e.g. trauma, oncological, infection), that did not involve any other contrast-enhanced phase beforehand. Scans performed on a third-generation dual source EID-CT (SOMATOM Force; Siemens Healthineers, Forchheim, Germany) were collected (Nov 2022-March 2024) as well as scans performed on a first-generation dual source PCD-CT (NAEOTOM Alpha; Siemens) (Sep 2023-Dec 2023). Exclusion criteria included large motion artifacts, non-adherence to the kV-tailored CM protocol on the EID-CT, and adjusted CM dosing due to practical considerations like fragile veins or small catheter size. Repeated patient inclusion was permitted as it was not expected to impact the outcome.

Scan protocol

The EID-CT settings were: gantry rotation time of 0.5 s, 96 x 0.6 mm collimation, 120 kV reference tube voltage with automated adaptation to the patients' size (CARE kV, Siemens) ranging from 70-120 kV in 10 kV increments, and 150 mAs reference tube current with automated modulation (CARE Dose 4D, Siemens). Reconstructions used 3-mm slice thickness, 2-mm increment and a smooth reconstruction kernel (Br40, Siemens) with an iterative reconstruction level 3. The PCD-CT settings were: gantry rotation time of 0.25 s for combined thorax/abdominal CT and 0.5 s for only abdominal CT, 144 x 0.40 mm collimation, 120 kV fixed tube voltage and reference IQ level of 145 optimized for soft tissue with contrast (CARE keV, Siemens) with automated tube current modulation (CARE Dose 4D, Siemens). Following the vendor-recommended standard protocol available on the scanner, VMIs were

reconstructed at 60 keV, again with 3-mm slice thickness, 2-mm increment and a Br40 kernel with iterative reconstruction level 3. For secondary analysis, based on recent studies, VMIs at 55 keV were reconstructed from spectral post processing (SPP) images to analyze the attenuation at a lower keV level [21; 27]. The SPP reconstructions were performed with a 2-mm slice thickness, 1.5-mm increment, and a quantitative reconstruction kernel (Qr40, Siemens) with iterative reconstruction level 3.

Radiation dose information (CT dose index ($CTDI_{vol}$), tube voltage and tube current) were collected from the DICOM-header. Patient characteristics (age, gender, weight, height) were provided by a PACS workstation (IDS7 version 24.2; Sectra AB, Linköping, Sweden) and the CM data acquisition program (Certegra Informatics Solution; Bayer Healthcare, Berlin, Germany).

Contrast media injection protocol

Iodinated CM (Ultravist 300mgI/ml or Ultravist 370mgI/ml; Iopromide, Bayer) was administered through an 18-, 20-, or 22-gauge needle. The scan was initiated approximately 70 s after injection. The exact timing was optimized by using bolus tracking, considering potential differences in cardiac output. On both scanners CM was injected with a programmable CM power injector (Stellant and Centargo; Bayer). On the EID-CT scanner an individualized CM protocol, based on total body weight (TBW) and tube voltage was used, with a maximum dosing factor of 0.521 g I/kg at 120 kV, reduced by 10% for every 10 kV decrease, following the 10-to-10 rule [7]. The PCD-CT used a TBW-adapted CM protocol with a fixed 120 kV tube voltage. The dosing factor was tailored to VMIs at 60 keV (corresponding to data acquisition with ~ 90 kV) by applying the 10-to-10 rule and reduced by 20%, resulting in 0.3 g I/kg. The CM injection duration at both scanners was standardized at 30 s, regulated by a dedicated CM injection software (P3T; Bayer Healthcare, Berlin, Germany). Consequently, flow rate was based on patient's weight, iodine concentration, and scanner type. Information on total iodine load (TIL), iodine delivery rate (IDR) and dosing factor (DF) were acquired by a dedicated CM data acquisition program (Certegra Informatics Solution; Bayer).

Image quality

Prior to IQ analysis, one observer (B.M.) removed all scans with motion artifacts. Objective IQ assessment included the objective IQ using region of interest (ROI) analysis quantifying the attenuation (in Hounsfield Units, HU), signal-to-noise ratio (SNR) and CNR. ROIs were placed in the right and left liver lobes and in the left paraspinal muscle. They were maximized in size, carefully avoiding any large vessels, bile ducts or liver lesions. Mean and standard deviation (SD) of HU values were measured for each ROI. Attenuation was defined as the average of the mean HU values from the liver ROIs. SNR was calculated as the mean HU divided by the mean SD of the liver ROIs. CNR was calculated by subtracting the mean HU of the paraspinal muscle from the mean HU of the liver ROIs, and dividing by the SD of the paraspinal muscle.

Subjective IQ was assessed by two expert readers (B.M. and N.E.) with respectively 8- and 5-years' experience in abdominal imaging. IQ was assessed on diagnostic screens with freely adjustable window settings, while being blinded to all scan, CM and patient information. The observers reviewed a training set of 5 abdominal PVP CT scans for each scanner in consensus, not included in the study dataset, to standardize their assessments. The final dataset was rated for overall diagnostic confidence (confidence of accurately diagnose presence or absence of lesions) by both observers independently using a 5-point Likert scale (1 = very poor/2 = poor/3 = moderate/4 = good/5 = very good). For the analysis, scores 1-2, and 4-5 were grouped in poor and good IQ respectively, resulting in a poor, moderate, good category.

Statistical analysis

Continuous variables were reported as mean $\pm$ SD. Analysis included two groups: EID-CT and PCD-CT with a sub-analysis of EID-CT scans for the different kV-levels. Categorical values were presented as absolute numbers with percentages and analyzed using the Chi-square test. Normal data distribution was assessed for continuous variables by applying the Shapiro-Wilk test. The Mann-Whitney U test was employed when the data were not normally distributed. Otherwise, the independent sample T -test was conducted. Outcomes were presented as mean $\pm$ SD.

Examining subjective IQ, the percentage of scans assessed as moderate or good was computed, indicating sufficient diagnostic confidence. Moreover, the interobserver agreement between the two observers was assessed using the intraclass correlation coefficient (ICC).

Analysis was performed using SPSS software (IBM SPSS Statistics, Version 28; IBM Corp, Armonk, NY).

Results

Baseline characteristics

Figure 1 shows the data collection flowchart. The EID-CT group is further categorized based on the acquisition tube voltage.

The final dataset included 91 EID-CT scans and 102 PCD-CT scans. Detailed baseline characteristics of the study population are given in Table 1. No statistically significant differences between the EID-CT and PCD-CT group were observed.

Detailed CM injection and radiation dose parameters are summarized in Table 2. The mean TIL for the EID-CT was 28.3 ± 7.2 gI, significantly higher than for the PCD-CT, which was 22.6 ± 5.0 gI ($p < .001$). This corresponds to a TIL reduction of 20.1% (11.4% for the 90 kV scans and 42.6% for the 100 kV scans). Conversely, radiation dose values were higher for the PCD-CT compared to the EID-CT ($CTDI_{vol} = 7.4 \pm 2.2$ mGy and 6.5 ± 2.5 mGy, respectively; $p < .001$).

Table 1 Baseline Characteristics

	EID-CT (n = 91)	PCD-CT (n = 102)	p
Age [years]	64.9 ± 14.9	61.4 ± 15.8	.115*
Sex [% female]	46.2%	55.9%	.177†
TBW [kg]	74.8 ± 15.2	75.3 ± 16.5	.828*
Height [cm]	170.8 ± 8.8	169.8 ± 9.1	.377*
BMI [kg/m ²]	25.6 ± 4.8	26.0 ± 4.9	.382*

*EID, energy-integrating detector; PCD, photon-counting detector; TBW, total body weight; BMI, body mass index. *Mann-Whitney U test, † Chi-square test*

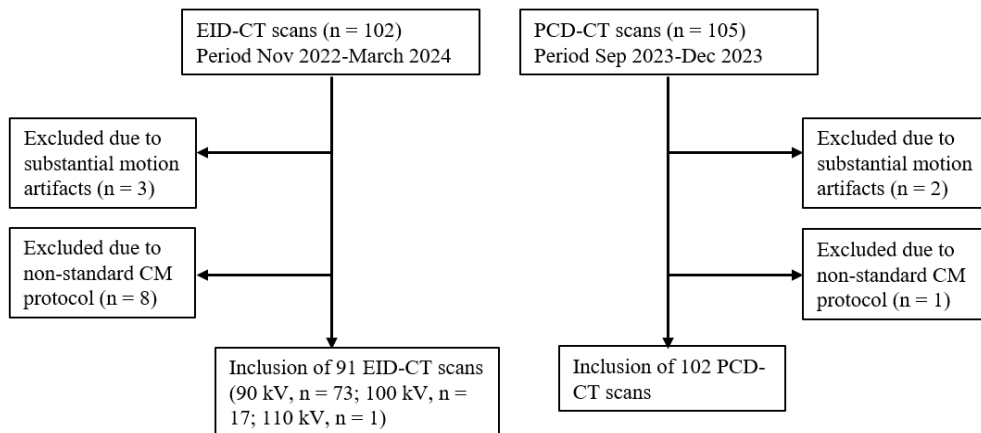


Figure 1 Flowchart of data collection (PCD-CT, photon-counting detector CT; EID-CT, energy-integrating detector CT; CM, contrast media).

Table 2 Contrast media injection and radiation dose parameters.

	EID-CT (n = 91)				PCD-CT (n = 102)	p
	90 kV (n = 73)	100 kV (n = 17)	110 kV (n = 1)	All (n = 91)		
TIL [g I]	25.5 ± 4.2	39.4 ± 5.0	47.0	28.3 ± 7.2	22.6 ± 5.0	<.001*
IDR [g I/s]	0.8 ± 0.1	1.3 ± 0.2	1.5	0.9 ± 0.2	0.7 ± 0.2	<.001*
DF [g I/kg]	0.36 ± 0.01	0.42 ± 0.00	0.47	0.38 ± 0.02	0.30 ± 0.00	.000*
CTDI _{vol} [mGy]	5.5 ± 1.3	10.1 ± 2.6	12.9	6.5 ± 2.5	7.4 ± 2.2	<.001*

*EID, energy-integrating detector; PCD, photon-counting detector; TIL, total iodine load; IDR, iodine delivery rate; DF, dosing factor; CTDI_{vol}, CT dose index; *Mann-Whitney U test*

Objective image quality

PCD-CT showed lower mean CT attenuation in the liver compared to EID-CT (111 ± 15 HU vs. 120 ± 14 HU, $p < .001$). However, both SNR (9.9 ± 1.7 vs 9.1 ± 1.8) and CNR (5.1 ± 1.7 vs 4.3 ± 1.3) were significantly higher for PCD-CT ($p < .001$), indicating an average SNR-increase of 8.7% and a CNR increase of 18.6% (see Table 3).

Secondary analysis of the PCD-CT scans reconstructed at 55 keV showed a mean CT attenuation in the liver of 120 ± 17 HU, which was not significantly different from the EID-CT scans ($p = 0.959$).

Table 3 Objective image quality outcomes.

	EID-CT (n = 91)	PCD-CT (n = 102)	p
Mean HU Liver ($\pm$ SD)	120 ± 14	111 ± 15	$<.001^*$
SNR Liver ($\pm$ SD)	9.1 ± 1.8	9.9 ± 1.7	$<.001^*$
CNR Liver ($\pm$ SD)	4.3 ± 1.3	5.1 ± 1.7	$<.001^*$

*EID, energy-integrating detector; PCD, photon-counting detector; HU, Hounsfield unit; SD, standard deviation; SNR, signal-to-noise ratio; CNR, contrast-to-noise ratio, *Mann-Whitney U test*

Subjective image quality

Figure 2 shows the results of the subjective IQ assessment. Both observers rated all scans moderate or good, indicating sufficient diagnostic IQ for all scans (ICC = 0.605, 95%CI 0.476-0.702). Representative images are shown in Figure 3, illustrating that the IQ of all scans is diagnostic sufficient and comparable.

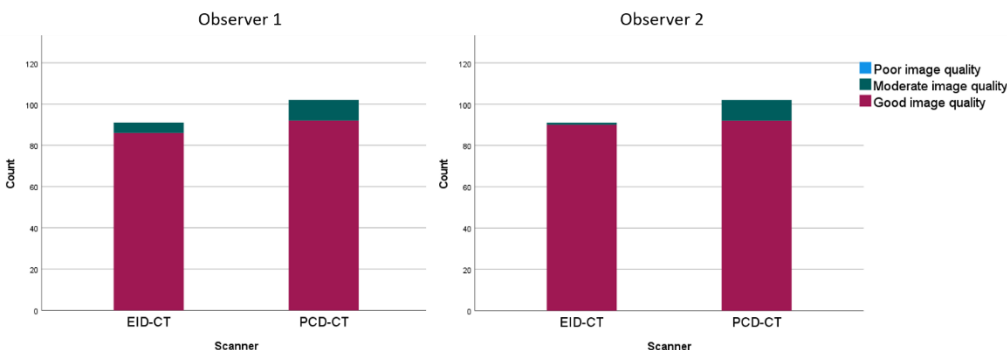


Figure 2 Subjective image quality assessment by 2 observers. PCD, Photon-counting detector; EID, Energy-integrating detector.

Discussion

This study shows that a PCD-CT allows CM dose reduction in abdominal PVP CT scans, compared to an EID-CT, both with an optimized CM protocol, while maintaining sufficient diagnostic IQ. EID-CT employed a CM protocol tailored to TBW and kV, while PCD-CT used a TBW adapted protocol, allowing a fair comparison of optimized clinical CM protocols.

A TIL reduction of 20.1% was found for the PCD-CT, along with a significant increase in SNR (8.7%) and CNR (18.6%) compared to the EID-CT. Moreover, overall diagnostic IQ on both CT scanners was rated as moderate or good in all cases. The initial lower liver attenuation on the PCD-CT can be attributed to the vendor-recommended reconstruction VMI energy level of 60 keV. Reconstruction at 55 keV, in line with recent literature, resulted in attenuation of the liver parenchyma comparable to EID-CT as visible in Figure 3 [28].

Prior studies showed potential CM reduction of similar magnitude with PCD-CT, while maintaining sufficient IQ [24-26]. Dane et al. [24] showed that PCD-CT scans

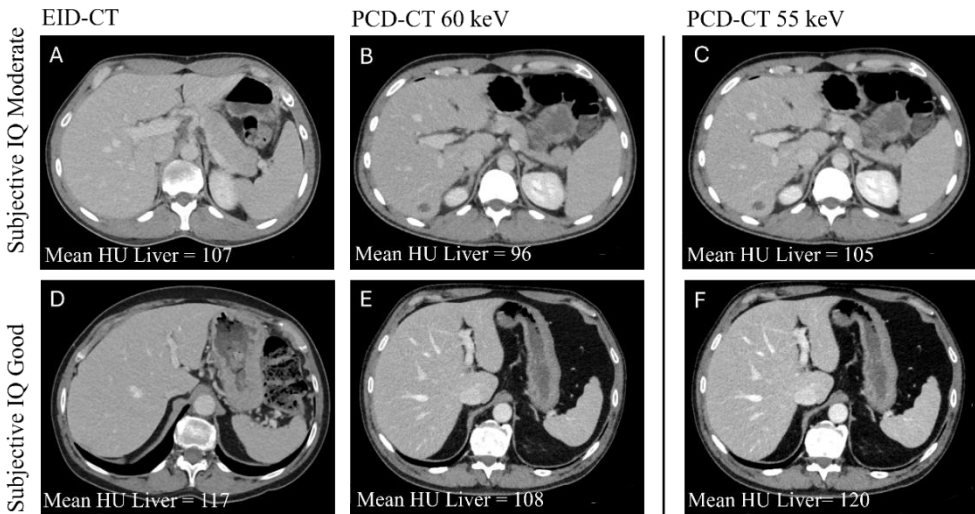


Figure 3 Representative abdominal PVP scans from both a photon-counting detector CT (PCD-CT) and an energy-integrating detector CT (EID-CT), illustrating comparable image quality (IQ). Image A and B were rated as moderate and image D and E as good by both observers. Image C and F are the 55 keV reconstructions of image B and E respectively but were not assessed subjectively.

with 20 ml less CM than weight-based dosing with EID-CT achieved comparable attenuation and IQ. Layer et al. [25] used 50% CM reduction on both EID-CT and PCD-CT scans and found a smaller difference in SNR and CNR with the PCD-CT. However, both studies only considered 120 kV scans for EID-CT without individualized kV-adapted CM protocols. Hagen et al. [26] reported 27% CM reduction and 31% radiation dose reduction using PCD-CT, achieving higher SNR and CNR levels compared to EID-CT. The difference in radiation dose reduction to the present study likely results from their exclusive inclusion of overweight or obese patients. The present study is unique by using individualized kV-adapted CM protocols for EID-CT as basis for comparison with PCD-CT.

This was a retrospective single-center study with a rather small patient sample, with a few other limitations. First, two different CM concentrations were used due to a shortage of CM at the time, though CM dose was adjusted by dosing factor. Therefore, total dose is comparable, and it has not impacted the results. Second, SNR and CNR calculations could not be performed on the 55 keV reconstructions due to different slice thicknesses, which would have affected the noise value. However, HU-values are unaffected by the slice thickness and are representative for demonstrating the effect of keV-level on the attenuation. Third, the radiation dose for PCD-CT images was higher than for EID-CT images (CTDI_{vol} of 7.4 mGy compared to 6.5 mGy). This can be explained by using a safe approach for a new scanner: initially using a higher dose and then gradually reducing it through an iterative process. Therefore, EID-CTs at this institution have already achieved a very low radiation dose compared to others globally [29]. Moreover, radiation dose for the PCD-CT was comparable to radiation dose reported in other studies (ranging from 6.57 mGy to 9.4 mGy) [24; 25; 30]. While a reduction in radiation dose may result in less noise for PCD-CT, it cannot account for the observed increase in CNR despite reduced CM dose, (18.6% increase in CNR versus 13.8% increase in CTDI_{vol}, while this increase in CTDI_{vol} could only explain an increase in CNR of ~ 7%). Lastly, subjective IQ assessment using Likert scores is always prone to variability because of different interpretations of the scale and hesitance to select the extreme Likert scores [31]. This is addressed by grouping the lowest scores (1-2) and highest scores (4-5) into a poor and good IQ rating, respectively.

In conclusion, the PCD-CT allows for reduced CM dose while providing higher SNR and CNR compared to an EID-CT even when an individualized scan and CM injection protocol is used in daily clinical practice.

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CHAPTER

4



Reducing contrast media dosage for pulmonary embolism CTPA in PCD-CT:

a comparative study of EID-CT and PCD-CT in the era of individualized protocolling

*L. Stammen**

*E.J.I. Hoeijmakers**

T.G. Flohr

H.A. Gietema

J. Vandewall

J.E. Wildberger

C.R.L.P.N. Jeukens†

B. Martens†

** L. Stammen and E.J.I. Hoeijmakers have equal contributions*

† B. Martens and C.R.L.P.N. Jeukens have equal contributions

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Abstract

Objectives: To evaluate diagnostic image quality (IQ) for pulmonary embolism detection in photon-counting detector CT (PCD-CT) with significantly reduced contrast media (CM) dose vs conventional energy-integrating detector CT (EID-CT), using optimized and individualized CM protocols.

Methods: Consecutive CT pulmonary angiography (CTPA) scans performed on the EID-CT (Jan 2024–Mar 2024) with individualized kilovoltage (kV) and total body weight (TBW) adapted CM protocol, and on the PCD-CT (Aug 2023–Feb 2024) with a TBW adapted CM protocol matching the 70 kV EID-CT protocol, were retrospectively collected. EID-CT scans were performed at 70–120 kV, based on patients' size, while PCD-CT scans were performed at 120 kV with 55keV virtual mono-energetic images. Objective IQ assessment included mean attenuation (in Hounsfield Units), signal-to-noise ratio and contrast-to-noise ratio (CNR). Two board-certified radiologists assessed diagnostic IQ subjectively using a five-point Likert scale.

Results: In 140 EID-CT and 118 PCD-CT scans, PCD-CT reduced total iodine load by 26.7% and CT dose index volume by 24.4%. Objective IQ parameters showed no significant differences, except for a decrease in CNR in the proximal pulmonary arteries in PCD-CT scans ($p = 0.02$). Subjective IQ was rated as moderate/good by observers 1 and 2 in 94.9% and 97.9% of the EID-CT scans, respectively, and 96.5% and 100% of the PCD-CT scans, respectively.

Conclusion: PCD-CT with a TBW-adapted CM protocol can achieve substantial reductions in both CM and radiation dose compared to EID-CT with individualized kV and TBW adapted CM protocols, while maintaining diagnostic IQ in CTPA scans.

Key words

Computed tomography; diagnostic imaging; pulmonary artery; contrast media; photon-counting detector

Introduction

Computed tomography pulmonary angiography (CTPA) has become the cornerstone for diagnosing pulmonary embolism (PE), providing high-resolution imaging of the pulmonary vasculature [1-3]. Previous studies demonstrated a high sensitivity (96-100%) and specificity (89-98%), establishing CTPA as the gold standard technique for diagnosing PE [4]. The use of iodinated contrast media (CM) is integral to CTPA imaging, facilitating the visualization of pulmonary arteries and the identification of emboli as areas without CM. However, the administration of CM is not without concerns or side effects, including potential life-threatening contrast-induced acute kidney injury (CI-AKI) in patients with renal insufficiency. Though the severity of the incidence and risks associated with CI-AKI is still a subject of ongoing debate, caution is recommended in patients with severe pre-existing renal insufficiency [5-10]. Additionally, excess CM can end up in wastewater, so minimizing its use can help reduce the ecological footprint of medical practices [11; 12]. This approach also contributes to addressing the ongoing CM shortage and offers financial benefits [13]. Previous studies focused on optimizing CM protocols for CTPA to reduce CM dose while preserving diagnostic image quality (IQ) [14-17]. A previous study has demonstrated that total body weight (TBW) adapted CM protocols lead to more homogenous attenuation and reduced CM dose for CTPA imaging [17]. Furthermore, adapting the CM dose to the optimal kilovoltage (kV) setting, following the simple 10-to-10 rule (reduction of the CM dose by 10% when the kV setting is reduced by 10 kV) facilitates the implementation of highly individualized CM protocols, tailored to both tube voltage and TBW [18; 19].

In recent years, advancements in CT technology have resulted in the introduction of photon-counting detector CTs (PCD-CTs) [20-22]. As opposed to energy-integrating detector CT (EID-CT), PCD-CT enables the counting of individual x-ray photons, which allows detection of their energies and sorting them in several energy bins. Lower energy photons contribute more strongly to the image than in EID-CT, leading to increased image contrast particularly in contrast-enhanced CT scans. In addition, the electronic noise is eliminated, as it is below the threshold energy for counting [23]. Therefore, image noise is reduced with PCD-CTs [22], and in combination with

increased image contrast, this leads to an inherently higher iodine contrast-to-noise ratio (CNR). The available spectral data provided when using standard scan mode enables reconstruction of virtual monoenergetic images (VMI) at various energy (keV) levels. For CTPA, the ability to reconstruct VMIs at lower keV levels, along with higher CNR, provides the opportunity to adapt CM and scan parameters while maintaining sufficient IQ [24].

Previous studies suggest that reduced CM dose or reduced radiation dose using PCD-CT still results in sufficient IQ in CTPA [25-28], in CT angiography [29; 30], and in abdominal CT [31; 32]. However, for CTPA, the CM reduction potential of PCD-CT with TBW adapted CM protocols, compared to highly individualized, kV- and TBW-adapted CM protocols in EID-CT, has not yet been evaluated.

This study aims to investigate the feasibility of reducing CM doses while maintaining diagnostic IQ for PE detection with PCD-CT compared to conventional EID-CT, using optimized routine protocols for both scanners. In EID-CT, the kV is automatically adapted to the patient's morphology, and the CM protocol is adapted to both the kV and the TBW. For PCD-CT, a standardized scan protocol at 120 kV with standardized reconstruction of VMIs at 55 keV is used for all patients, and the CM protocol is only adapted to TBW.

Material and Methods

Ethics

This study received a waiver of written informed consent from the local ethical committee and institutional review board (METC 2020-1535-A-1), due to its retrospective nature.

Study population

Consecutive CTPA scans used in daily clinical practice were collected retrospectively from a third-generation dual-source EID-CT system (SOMATOM Force; Siemens

Healthineers, Forchheim, Germany, between Jan 2024–Mar 2024) and a first-generation dual-source PCD-CT (NAEOTOM Alpha; Siemens, between Aug 2023–Feb 2024). Scans of patients ≥ 18 years were included that were solely conducted to rule out PE. Scans with adjustments to the CM protocol for diagnosis of other potential pathologies such as aortic dissection or coronary assessment were excluded from the analysis. Additionally, scans with obvious breathing artifacts were excluded, as were scans with modified CM dose or kV settings (PCD-CT). The picture archiving and communication system (PACS) workstation (IDS7 version 24.2; Sectra AB, Linköping, Sweden) and the CM data acquisition program (Certegra Informatics Solution; Bayer Healthcare, Berlin, Germany) were consulted to collect patient characteristics (age, gender, weight, height).

Scan protocol

The EID-CT scans were conducted with a gantry rotation time of 0.25 s and a pitch of 1.3 using single-energy (SE) mode. Other scanner settings were: slice collimation of 192×0.6 mm, a quality reference tube current of 105 mAs and a reference tube voltage of 100 kV_{ref}, with automatic tube current and tube voltage adaptation (CARE Dose 4D and CARE kV, Siemens) resulting in tube voltages ranging from 70 kV to 120 kV with increments of 10 kV. PCD-CT scans had a gantry rotation time of 0.25 s, a pitch of 2.4, and a slice collimation of 144×0.4 mm. The scans were made in the QuantumPlus mode, providing spectral information in four energy bins, which are combined to two bins for all established dual-energy (DE) applications. In PCD-CT, a fixed tube voltage is utilized to provide the full spectral data. For this study, a tube voltage of 120 kV was applied, with reconstruction of VMIs at low keV as a substitute for low kV scanning. The reference IQ level (CARE keV IQ level) was set at 70 with automatic adaptation of the tube current and the VMI keV-level (CARE Dose 4D and CARE keV, Siemens), optimized for vascular studies. Reconstructions consisted of 1.0 mm slices with 0.7 mm (EID-CT) and 0.8 mm (PCD-CT) increments, using a soft tissue kernel (Bv40) and (quantum) iterative reconstruction strength 3. The PCD-CT scans were reconstructed using a VMI level at 55 keV as proposed by CARE keV. Radiation dose information, including CT dose index volume (CTDI_{vol}), in mGy and tube voltage,

was obtained from the digital imaging and communications in medicine (DICOM) header.

Contrast media injection protocol

All patients were injected with iodinated CM (Ultravist 300 mgI/ml or Ultravist 370 mgI/ml; Iopromide, Bayer) at room temperature, followed by a saline chaser (40cc) [33]. A programmable CM power injector was used for CM injections (Centargo; Bayer) [34]. The CM was administered with a fixed injection duration of 8 s, regulated by dedicated CM injection software (P3T; Bayer). Prior to the CTPA, a dynamic test bolus with an injection duration of 2-3 seconds, followed by a saline chaser (40cc), was obtained to determine the optimal scan delay (time to peak enhancement in the pulmonary trunk + 7 seconds), accounting for patient specific factors like cardiac output [35; 36]. On the EID-CT scanner, a CM injection protocol individualized to tube voltage and TBW was used, according to the 10-to-10 rule, as described elsewhere [18; 19]. The TBW adjustment followed a non-linear relationship, as detailed and visualized by Hendriks et al [18]. For example, at 120 kV, an iodine dose of 0.257 g I/kg was used for a 70 kg patient, the dose was adapted to 0.240 g I/kg for a 90 kg patient, and 0.215 g I/kg for a 110 kg patient. Injection duration is kept constant at approximately 7 s, therefore influencing the IDR. On the PCD-CT, a VMI level of 55 keV was used, which is an estimate of the mean energy of a 70 kV spectrum. While this is only an approximation and does not imply the equivalence of 70 kV images and 55 keV VMIs, it provides a reasonable assumption for determining the CM protocol for PCD-CT. Therefore, a CM protocol identical to the 70 kV CM protocol used for the EID-CT scanner was used [33]. CM protocols were defined by a dosing factor (DF, g I/kg) that prescribes the amount of iodine in grams per kilogram. This ensures that the total iodine load (TIL, g I) was independent of the CM concentration used (300 mg I/ml or 370 mg I/mL).

A dedicated data acquisition program (Certegra Informatics Solution; Bayer) recorded the CM parameters for each patient, including the iodine concentration (g/mL), the total CM dose (mL) and the flow rate (mL/s). The TIL, iodine delivery rate (IDR), and DF were calculated as follows: $TIL = [\text{total CM dose} \times \text{iodine concentration}]$

in grams iodine (g I); $IDR = [\text{flow rate} \times \text{iodine concentration}]$ in grams iodine per second (g I/s); $DF = [TIL / TBW]$ in grams iodine per kg body weight (g I/kg).

Image quality assessment

Objective assessment of IQ included the quantification of the mean attenuation (in Hounsfield Units, HU) as the primary outcome, and signal-to-noise ratio (SNR), and CNR as a secondary outcomes. Regions of interests (ROIs), maximized in size, were drawn to measure the mean attenuation in HU and the standard deviation (SD). ROIs were placed in four specific anatomical locations: proximal in the left and right pulmonary artery, and subsegmental in a branch of the upper left pulmonary artery and of the lower right pulmonary artery. If not possible, due to the presence of PE or post-operative status for example, an adjacent vessel at the same side was chosen (e.g., if the upper left could not be used, the lower left pulmonary artery was chosen, and vice versa for the right side). In addition, an ROI was placed in the left paraspinal muscle. The proximal SNR was calculated by dividing the mean attenuation of left and right proximal pulmonary arteries by the mean SD. The distal SNR was calculated similarly, by dividing the mean attenuation of the distal pulmonary arteries by its mean SD. The CNR for the proximal pulmonary arteries was calculated by subtracting the attenuation of the paraspinal muscle from the mean attenuation of the proximal pulmonary arteries and dividing this difference by the SD of the paraspinal muscle. The distal CNR for the subsegmental pulmonary arteries was calculated accordingly. Subjective IQ was assessed independently by two board-certified radiologists, H.G. and B.M., with respectively 18 years' and 8 years' experience in thoracic and cardiovascular imaging. Prior to the evaluation, the radiologists jointly reviewed ten additional scans (5 performed on the EID-CT and 5 on the PCD-CT), not included in the dataset, to familiarize themselves with the areas of focus and to systematically align for the subjective assessments. Scans were reviewed under clinical viewing conditions on diagnostic screens with standard window level settings for CTPA scans (with the flexibility to adjust settings). Both radiologists were completely blinded to all scan-, reconstruction-, and patient information. Scans were rated for diagnostic confidence (confidence to accurately diagnose presence or absence of PEs) using a

5-point Likert scale (1 = very poor, 2 = poor, 3 = moderate, 4 = high, 5 = excellent) in which a Likert score of 3 or higher indicated sufficient diagnostic confidence. For the analysis, Likert scores 1 and 2, and 4 and 5 were grouped as poor and good IQ, respectively.

Statistical analysis

Statistical analysis was performed using SPSS statistical software package version 28 (IBM Corp, Armonk, NY, USA). Continuous data were reported as mean $\pm$ SD and a p -value < 0.05 was considered statistically significant. The normality of continuous data was assessed using the Shapiro-Wilk test. If data followed a normal distribution, an independent sample T-test was conducted; if not, the Mann-Whitney U test was used. Categorical variables were expressed as percentages and proportions were evaluated using a Chi-squared test.

In order to evaluate interobserver agreement in subjective IQ assessment between the two observers, the intraclass correlation coefficient (ICC) was calculated. Moreover, the percentage of scans that were rated as moderate or good was assessed, reflecting sufficient diagnostic confidence.

Results

A total of 143 EID-CT CTPAs scans and 133 PCD-CT CTPAs were performed during the inclusion periods. After exclusions, the final dataset comprised of 140 EID-CT (mean age, 64.3 ± 15.6 , 66 women) and 118 PCD-CT scans (mean age, 64.3 ± 15.3 , 72 women) (Figure 1). The majority of exclusions for the PCD-CT were due to the use of a tube voltage of 140 kV rather than the specified 120 kV ($n = 9$). Characteristics of the study population per CT scanner are summarized in Table 1 (and Supplementary material Table 1). A significant difference in sex distribution ($p = 0.03$) was demonstrated between the two groups, while other characteristics showed no statistically significant differences.

Table 2 summarizes CM and radiation dose parameters, showing that TIL, IDR and DF were significantly lower for the PCD-CT. A 26.7% reduction in TIL was found in the PCD-CT group compared to the EID-CT group (10.2 ± 1.4 gI vs. 13.9 ± 4.1 gI respectively, $p = 0.000$). Furthermore, $CTDI_{vol}$ was higher for the EID-CT group than for the PCD-CT group (4.5 ± 2.7 mGy vs. 3.4 ± 1.1 mGy, with a reduction of 24.4%, $p < 0.001$).

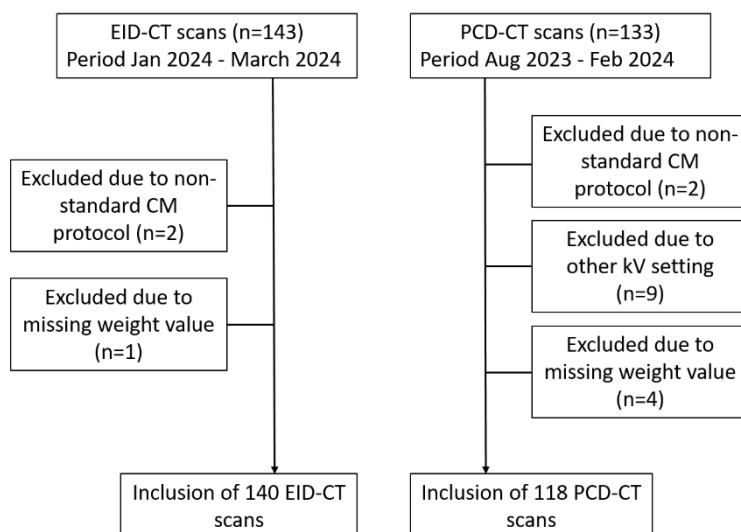


Figure 1 Flowchart showing patient selection process. EID, energy-integrating detector; PCD, photon-counting detector; CM, contrast media

Table 1 Baseline characteristics

Parameters	EID-CT scans (n = 140)	PCD-CT scans (n = 118)	p
Age [years]	64.3 ± 15.6	64.3 ± 15.3	0.90*
Gender [% female]	47.1%	61.0%	0.03 [†]
TBW [kg]	80.1 ± 18.8	78.3 ± 18.6	0.47*
Height [cm]	172 ± 10	170 ± 9	0.09°
BMI [kg/m ²]	27.1 ± 5.5	26.7 ± 5.3	0.64*

*Mann-Whitney U test; [†]Chi-square test; °Independent sample T-test
 EID, energy integrating detector; PCD, photon counting detector; TBW, total body weight; BMI, body mass index.

Table 2 Contrast media injection and radiation dose parameters

Parameters	EID-CT scans (n = 140)					PCD-CT scans (n = 118)	p*
	70 kV (n = 26)	80 kV (n = 38)	90 kV (n = 40)	100 kV (n = 23)	110 kV (n = 6)	120 kV (n = 7)	All (n = 140)
TIL [gI]	9.6 ± 2.3	11.9 ± 1.5	14.3 ± 2.1	17.6 ± 2.5	18.7 ± 3.2	22.9 ± 4.3	13.9 ± 4.1
IDR [gI/s]	0.8 ± 0.2	1.0 ± 0.1	1.2 ± 0.2	1.5 ± 0.2	1.5 ± 0.2	1.9 ± 0.3	1.2 ± 0.4
DF [gI/kg]	0.15 ± 0.03	0.16 ± 0.02	0.18 ± 0.02	0.19 ± 0.02	0.22 ± 0.03	0.22 ± 0.03	0.17 ± 0.03
CTDI _{vol} [mGy]	1.8 ± 0.5	3.3 ± 0.9	4.7 ± 1.1	6.4 ± 1.3	7.2 ± 1.5	12.0 ± 3.7	4.5 ± 2.7
							10.2 ± 1.4
							0.8 ± 0.1
							0.13 ± 0.02
							3.4 ± 1.1

*Mann-Whitney U test
 EID, energy integrating detector; PCD, photon counting detector; TIL, total iodine load; IDR, iodine delivery rate; DF, dosing factor; CTDI_{vol}, CT dose index volume

Objective image quality

Detailed results of the objective IQ assessment can be found in Table 3. For the primary outcome, both proximal and distal mean attenuation of the pulmonary arteries were not significantly different between the two groups ($p = 0.66$ and $p = 0.71$, respectively). Results for the secondary outcomes showed comparable SNR values between both groups, in both proximal and distal segments ($p = 0.32$ and $p = 0.13$, respectively). CNR was slightly higher for the proximal pulmonary arteries at EID-CT (18.7 ± 9.3) vs PCD-CT (15.9 ± 5.9 ; $p = 0.02$), whereas the CNR for the distal pulmonary arteries did not demonstrate a significant difference (17.8 ± 9.2 and 15.6 ± 5.9 for the EID-CT and PCD-CT, respectively, $p = 0.09$). Stratified analyses were conducted to account for the significant gender difference between the two groups (see Supplementary material Table 2). Results showed no significant differences except for the CNR among females in both the proximal and distal pulmonary arteries ($p = 0.003$ and $p = 0.02$, respectively), as well as in SNR within the distal pulmonary arteries ($p = 0.01$).

Table 3 Objective image quality outcomes

	EID-CT scans (n = 140)	PCD-CT scans (n = 118)	p*
Proximal			
Mean attenuation ($\pm$ SD) [HU]	384.7 $\pm$ 105.8	378.9 $\pm$ 100.6	0.66
SNR ($\pm$ SD)	14.9 $\pm$ 4.2	14.4 $\pm$ 4.1	0.32
CNR ($\pm$ SD)	18.7 $\pm$ 9.3	15.9 $\pm$ 5.9	0.02
Distal			
Mean attenuation ($\pm$ SD) [HU]	368.7 $\pm$ 107.4	373.6 $\pm$ 101.4	0.71
SNR ($\pm$ SD)	20.5 $\pm$ 6.7	19.2 $\pm$ 5.2	0.13
CNR ($\pm$ SD)	17.8 $\pm$ 9.2	15.6 $\pm$ 5.9	0.09

*Mann-Whitney U test

EID, energy integrating detector; PCD, photon counting detector; HU, Hounsfield unit; SD, standard deviation; SNR, signal-to-noise ratio; CNR, contrast-to-noise ratio.

Subjective image quality

Subjective IQ assessment showed a rating of moderate or good IQ in 94.9% of the EID-CT scans and 96.5% of the PCD-CT scans for observer 1. Observer 2 rated 97.7% of EID-CT and 100% of PCD-CT scans as having moderate or good IQ (Figure 2). There was a good interobserver agreement, with an ICC of 0.667 (95% CI: 0.574–0.740). Representative images are shown in Figure 3 and illustrate comparable IQ, and including an example of PE findings on both scanners.

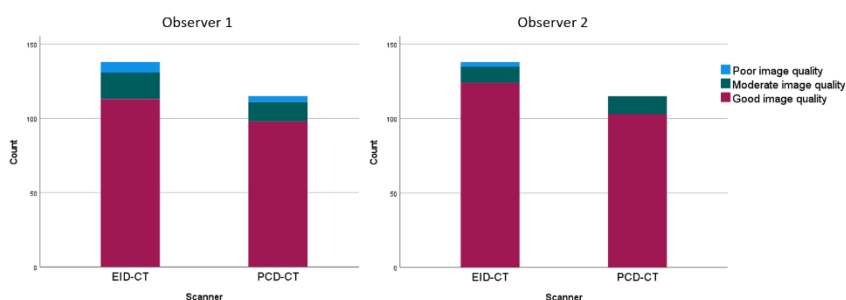


Figure 2 Results of subjective image quality assessment by 2 observers. PCD, photon-counting detector; EID, energy-integrating detector

Discussion

This study evaluated the potential for reducing CM in CTPA scans using PCD-CT compared to EID-CT, utilizing tailored CM protocols based on TBW for the PCD-CT and highly individualized CM protocols adapted for both tube voltage and TBW for the EID-CT. The results indicate that a substantial and additional CM reduction is achievable with PCD-CT in clinical setting, while maintaining IQ and decreasing radiation dose compared to an individualized and optimized EID-CT protocol on a 3rd-generation dual-source CT. While the TBW limits the use of low kV with correspondingly increased iodine contrast in EID-CT because of limited X-ray tube power reserves, patients can be examined on the PCD-CT with the standard protocol at 120 kV regardless of their body weight, and the reconstruction of VMIs at 55 keV is also always possible. Specifically, using PCD-CT resulted in a 26.7% reduction in TIL and a 24.4% reduction in CTDI_{vol}. Although the potential reduction in patient risk

from lowering CM dosage may be subject to debate, this approach offers environmental and financial benefits, while minimizing radiation dose remains consistent priority.

Objective IQ analysis showed no significant differences in attenuation, SNR and CNR values between PCD-CT and EID-CT except for CNR measured in the proximal pulmonary arteries, which exhibited a significantly lower CNR on the PCD-CT compared to EID-CT. However, this lower CNR is not considered clinically relevant, as it is likely due to increased noise in the paraspinal muscle, which served as the reference region for noise calculation but is not of primary interest in the study. This increase in noise may be attributed to cross-scatter correction effects, caused by dual-source scanning used for the PCD-CT [37]. Therefore, in future studies, it may be advised to use an alternative reference region in CNR calculations, e.g. the cardiac septum [35]. Furthermore, subjective IQ analysis showed that almost all CT scans, and more PCD-CT than EID-CT scans, were rated as having moderate or good IQ, indicating sufficient diagnostic IQ.

Previous studies have demonstrated that using a PCD-CT for CTPA can enable a 20-50% CM reduction compared to EID-CT, while still maintaining diagnostic IQ, which is consistent with the findings of the present study [25; 27]. However, these studies employed fixed CM protocols not adjusted for TBW and, for the EID-CT, not adjusted for tube voltage or TBW. Another difference is the comparison of spectral (DE) PCD-CT with SE EID-CT in the present study, as opposed to the DE EID-CT mode utilized in the previous study [25]. With EID-CT, both SE and DE CTPA are used in daily clinical practice. Vlahos et al (2022), and Yadav et al (2023), demonstrated that DE CTPA offers superior diagnostic sensitivity compared to SE CTPA [38; 39]. In addition, it may provide additional diagnostic capabilities by providing deeper physiological insights and hemodynamic correlates, which could inform and optimize the management of patients with pulmonary hypertension [38]. On the other hand, in a study by Chen et al (2019) DE CTPA did not outperform SE CTPA with regard to diagnostic IQ [40], and previous studies demonstrated a lower radiation dose and significantly reduced scan time for SE CTPA compared to DE CTPA, the latter being particularly advantageous for patients experiencing shortness of breath [26]. Additionally, limitations regarding DE CTPA on EID-CT were described, depending

on the DE configuration of the scanner (e.g. limited field of view and slight temporal misregistration) [38].

These limitations are overcome by PCD-CT for PE assessment. The availability of spectral data enables reconstruction of VMIs at lower keV levels for scans with insufficient contrast attenuation. This increases the iodine contrast in the image, potentially reducing the need for repeat scans and thus avoiding increased CM and radiation doses [22; 41]. However, further research is required to investigate this potential improvement.

This study has several limitations. First, it is a retrospective single-center study with a relatively small sample size. A prospective study design would have allowed for

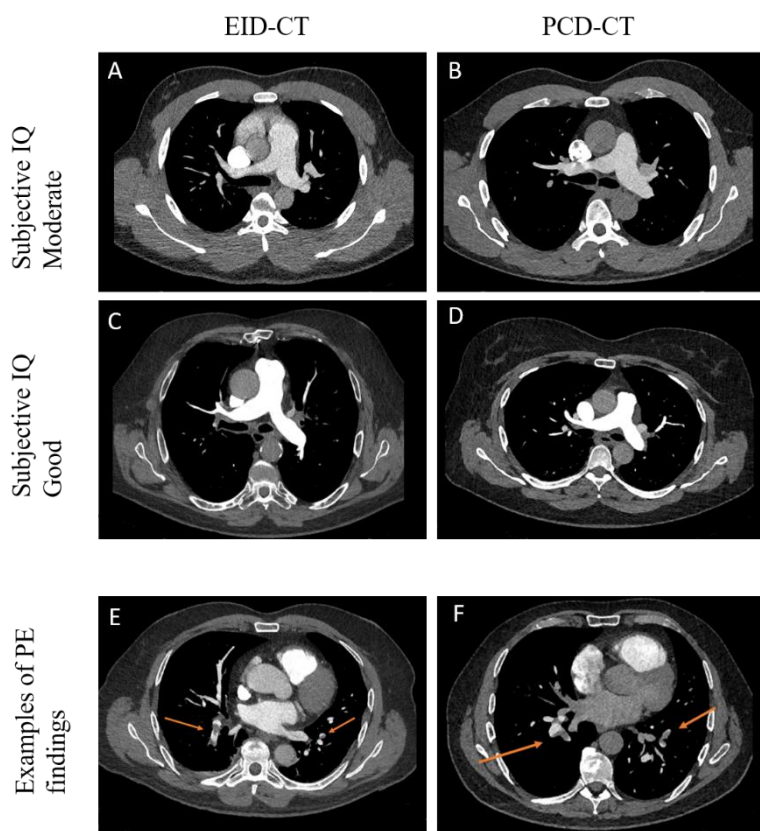


Figure 3 Representative CT pulmonary artery scans from both a photon-counting detector CT (PCD-CT) and an energy-integrating detector CT (EID-CT). Images A and B were rated by both observers as moderate, images C and D were rated as good by both observers. Images E and F are examples of PE findings for both scanners.

adjustments to improve consistency between the two scanners, although differences in settings will always exist due to the differences in scanner technique between EID-CT and PCD-CT. However, the retrospective study design provided the opportunity to examine their performance in routine daily clinical practice. Second, the use of Likert scores is inherently subjective, as observers tend to avoid selecting extreme scores and may interpret the scale differently [42]. To mitigate this issue, we grouped the lowest scores and highest scores into categories of poor and good IQ, respectively. Moreover, since the primary aim of the subjective IQ assessment is to determine whether both scanners provide sufficient diagnostic IQ (moderate/good), small variations in the specific Likert scores between observers (e.g. due to difference in observer experience, or due to familiarity to specific scanner types) are not of major concern, even though they affect the ICC for subjective IQ assessment. As shown in Figure 2, both observers classified the majority of scans as having sufficient diagnostic IQ, which aligns with the study's primary objective. A third limitation was the lower pitch of 1.3 used for the EID-CT in our study, resulting from a standardization of EID-CT scan protocols in our institution. The minimal difference in scan time (± 1 s) and the consistent delay used for both scanners, suggest this pitch variation does not impact contrast enhancement. Lastly, a difference in the percentage of female and male participants was observed between the EID-CT and PCD-CT (47.1% vs 61.0%, respectively). Females generally have a lower blood volume compared to males [43]. A lower blood volume might lead to increased iodine concentration in the blood and, as a result, increased attenuation [44]. Stratified analysis showed only significant differences in CNR in females within both the distal and proximal pulmonary arteries, along with a significant difference in SNR in the distal pulmonary arteries. This indicates that the unequal gender distribution between the two groups, may have introduced bias into these results. However, the mean outcome measures did not show clinically relevant differences. More importantly, attenuation, the primary outcome of this study, showed no significant differences in analyses stratified by gender. Nevertheless, evidence for sex-based differences in attenuation remains limited, and further studies are needed to examine this phenomenon [44].

In conclusion, this study compared TBW-adapted CM protocols for CTPA scans on a PCD-CT with highly individualized kV- and TBW-adapted protocols on EID-CT and demonstrated that PCD-CT can achieve substantial reductions in CM and radiation dose, while maintaining sufficient diagnostic IQ.

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Supplementary material

Table 1 Baseline characteristics

Parameters	EID-CT scans (n = 140)					PCD-CT scans (n = 118)	p*
	70 kV (n = 26)	80 kV (n = 38)	90 kV (n = 40)	100 kV (n = 23)	110 kV (n = 6)	120 kV (n = 7)	All (n = 140)
Age [years]	64.1 ± 18.3	62.5 ± 16.9	67.1 ± 14.3	64.8 ± 12.9	70.0 ± 11.1	52.9 ± 12.1	64.3 ± 15.6
Gender	80.8%	39.5%	50.0%	26.1%	33.3%	28.6%	47.1%
[% female]							
TBW [kg]	62.7 ± 13.1	75.4 ± 11.5	82.1 ± 13.4	94.7 ± 18.9	85.3 ± 3.6	106.6 ± 31.5	80.1 ± 18.8
Height [cm]	166 ± 7.5	172 ± 9	171 ± 10	176 ± 9	174 ± 7	179 ± 12	172 ± 10
BMI [kg/m ²]	22.8 ± 4.6	25.6 ± 3.3	28.1 ± 4.3	30.6 ± 6.8	28.2 ± 2.4	32.9 ± 8.2	27.1 ± 5.5

*Mann-Whitney U test; † Chi-square test; ‡Independent sample T-test
EID, energy integrating detector; PCD, photon counting detector; TBW, total body weight; BMI, body mass index.

Table 2 Objective image quality outcomes stratified by gender

	EID-CT scans (Male, n = 74; Female, n = 66)	PCD-CT scans (Male, n = 46; Female, n = 72)	p*
Proximal			
<i>Mean attenuation ($\pm$SD) [HU]</i>			
Male	368.7 $\pm$ 125.9	376.1 $\pm$ 111.3	0.54
Female	402.7 $\pm$ 74.5	380.6 $\pm$ 93.9	0.11
<i>SNR ($\pm$SD)</i>			
Male	14.1 $\pm$ 4.7	13.7 $\pm$ 3.4	0.94
Female	15.8 $\pm$ 3.3	14.9 $\pm$ 4.5	0.06
<i>CNR ($\pm$SD)</i>			
Male	17.8 $\pm$ 10.7	15.0 $\pm$ 5.1	0.45
Female	19.7 $\pm$ 7.3	16.5 $\pm$ 6.4	0.003
Distal			
<i>Mean attenuation ($\pm$SD) [HU]</i>			
Male	349.4 $\pm$ 122.1	365.4 $\pm$ 115.4	0.36
Female	390.4 $\pm$ 83.7	378.8 $\pm$ 91.9	0.41
<i>SNR ($\pm$SD)</i>			
Male	19.5 $\pm$ 7.3	19.0 $\pm$ 5.1	0.98
Female	21.5 $\pm$ 5.9	19.3 $\pm$ 5.2	0.01
<i>CNR ($\pm$SD)</i>			
Male	16.8 $\pm$ 10.6	14.4 $\pm$ 4.9	0.63
Female	19.0 $\pm$ 7.3	16.4 $\pm$ 6.4	0.02

*Mann-Whitney U test

EID, energy integrating detector; PCD, photon counting detector; HU, Hounsfield unit; SD, standard deviation; SNR, signal-to-noise ratio; CNR, contrast-to-noise ratio.

CHAPTER

5



How subjective CT image quality assessment becomes surprisingly reliable:

pairwise comparisons instead of Likert scale

E.J.I. Hoeijmakers

B. Martens

B.M.F. Hendriks

C. Muhl

R.L. Miclea

W.H. Backes

J.E. Wildberger

F.M. Zijta

H.A. Gietema

P.J. Nelemans

C.R.L.P.N. Jeukens

Published in:

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Abstract

Objectives: The aim of this study is to improve the reliability of subjective IQ assessment using a pairwise comparison (PC) method instead of a Likert scale method in abdominal CT scans.

Methods: Abdominal CT scans (single-center) were retrospectively selected between September 2019 and February 2020 in a prior study. Sample variance in IQ was obtained by adding artificial noise using dedicated reconstruction software, including reconstructions with filtered backprojection and varying iterative reconstruction strengths. Two datasets (each $n = 50$) were composed with either higher or lower IQ variation with the 25 original scans being part of both datasets. Using in-house developed software, six observers (five radiologists, one resident) rated both datasets via both the PC method (forcing observers to choose preferred scans out of pairs of scans resulting in a ranking) and a 5-point Likert scale. The PC method was optimized using a sorting algorithm to minimize necessary comparisons. The inter- and intraobserver agreement were assessed for both methods with the intraclass correlation coefficient (ICC).

Results: Twenty-five patients (mean age 61 years $\pm$ 15.5; 56% men) were evaluated. The ICC for interobserver agreement for the high-variation dataset increased from 0.665 (95%CI 0.396-0.814) to 0.785 (95%CI 0.676-0.867) when the PC method was used instead of a Likert scale. For the low-variation dataset, the ICC increased from 0.276 (95%CI 0.034-0.500) to 0.562 (95%CI 0.337-0.729). Intraobserver agreement increased for four out of six observers.

Conclusion: The PC method is more reliable for subjective IQ assessment indicated by improved inter- and intraobserver agreement.

Key words

Computed tomography (X-ray); Interobserver variability; Intraobserver variability

Introduction

Computed Tomography (CT) is one of the most important diagnostic imaging tools in daily clinical practice which requires sufficient image quality (IQ). However, increasing IQ, by decreasing image noise, implicates a higher radiation dose, which may lead to an increased lifetime attributable risk of cancer [1]. Optimization studies try to find the balance between radiation dose and IQ. However, reliable assessment of IQ remains a challenge [2; 3].

IQ is frequently assessed subjectively using a fixed (typically 5-point) Likert scale [4]. Subjective assessments can provide comprehensive information about IQ, as it takes human perception into account. However, it is prone to variability because of different interpretations of the points on the scale [5-7]. IQ is also commonly evaluated with objective parameters such as signal-to-noise ratio, and contrast-to-noise ratio in specific regions of interest. Although these parameters are quantifiable, they may not always accurately reflect the radiologist's perception of diagnostic IQ [8]. More advanced automated algorithms may be preferable for assessing IQ because of their objectivity and they do not require human observer time [9-11]. However, the development of automated algorithms requires validation using subjective IQ assessment, which could benefit from a more reliable method compared to the standard Likert scale method.

In a pairwise comparison (PC) method, the observer is repeatedly forced to choose the preferred scan out of pairs until all scans are ranked [12]. Previous studies in other disciplines suggest that the Likert scale has a low inter- and intraobserver agreement, while a PC method may be more accurate and sensitive even for small differences within the scans [5; 13-16]. When using the Likert scale method, the observers are often trained on the dataset or provided with reference scans to give an absolute score. However, studies suggest that this does not provide significant added value [13; 14]. In the PC method, the scans are rated in reference to each other avoiding the need for training or additional reference scans as well [12-17]. Additionally, observers tend to be hesitant to select the extreme Likert scores out of fear of a better or worse scan showing up afterwards [16]. This results in a bias to the middle

of the scale and provides an evaluation with a less refined classification than the PC method.

This study aims to show that the reliability of subjectively scoring IQ improves by using the PC method instead of the Likert scale method in abdominal CT examinations.

Material and methods

A waiver of written informed consent was provided by the local ethical committee and institutional review board, as retrospective data were analyzed anonymously (ref METC 2017-0250).

Study design

Data collected for a prior publication is used [18], consisting of abdominal CT scans from 30 unique patients that were retrospectively selected between September 2019 and February 2020. All scans had a reference tube voltage and tube current of respectively 120 kV_{ref} and 150 mAs_{ref}, with a slice collimation of 192 x 0.6 mm and gantry rotation time of 0.5 s. Scans were acquired at 90 kV with a dosing factor of 0.4 g I/kg contrast media and reconstructed with iterative reconstruction (IR) level 2. Movement artifacts were applied as exclusion criteria. To obtain more variation in the IQ of the scans, noise was inserted artificially using proprietary dedicated CT reconstruction software (version 13.0.0.1, prototype software, Siemens Healthineers, information on validation of the software can be found in [18]) by simulating scans at 60, 70, 80, and 90% of the original (100%) tube current. Additionally, the simulated scans were reconstructed using filtered backprojection (FBP) and IR levels 2, 3, and 4. This resulted in 510 CT scans (the original scans and 16 additional reconstructions per patient).

To investigate the reliability of subjective assessment on respectively more and less evident IQ differences, two datasets both containing 50 scans were constructed out of the total database by (1) including the original scans and (2) choosing scans obtained with a higher and lower variation in reconstruction methods and tube current values. The original scans were part of both datasets and each dataset was

complemented by randomly selecting scans (not more than one per patient) from the total database such that the desired composition was achieved.

PC method and Likert scale method

A Python-based software program was in-house developed for IQ scoring using the PC method and the Likert scale method, featuring a user interface designed to closely mimic diagnostic viewing systems and operate similarly, e.g., possibility to scroll through the slices and to adjust the window settings.

The PC method repeatedly presented the observers with pairs of CT scans and forced them to select the scan with the highest IQ, continuing until all scans could be ranked. As comparing all possible pairs of 50 scans requires 1225 comparisons, the Ford-Johnson algorithm was applied in the software which reduced the required number of comparisons to approximately 220 [13; 19]. This sorting algorithm assumes the transitivity relation of the scans: if scan A is preferred over B, and B is preferred over C, then preference of A over C is implied, thus eliminating the need to assess the latter. Depending on the previous preference the algorithm generated the next pair presented to the observer, thus, the pairs that needed to be scored were unique for every observer and assessment. The PC method resulted in a ranking of the scans for each observer (low to high IQ).

For the Likert scale method, scans were presented one by one in random order and observers were prompted to click on the corresponding Likert score for overall IQ (1, very poor; 2, poor; 3, moderate; 4, good; 5, very good) [4].

Image quality assessment

Five radiologists (C.M., F.Z., H.G., B.M., R.M.) and one radiology resident (B.H.) with respectively 12-, 12-, 10-, 7-, 13-, and 4 years of experience participated in the study. IQ was rated by the observers on diagnostic screens while being blinded to scan- and patient information. Standard window level settings (window level = 40 HU; window width = 400 HU) for abdominal CT examinations were applied, but observers were allowed to adjust them. The observers were instructed to assess the scans based on overall diagnostic IQ. All observers rated the first dataset using both the PC

method and the Likert scale method, and then, after at least a 2-week time interval, they rated the second dataset using the two methods. The original scans present in both datasets were used to evaluate the intraobserver agreement. Both assessment methods were performed without prior training or a reference image.

Data analysis

Both the inter- and intraobserver agreement were quantified using the intraclass correlation coefficient (ICC). The ICC is a reliability parameter that ranges from 0 to 1, with 1 indicating total agreement between observers [20], calculated with statistical software (SPSS, v29.0; IBM Corp). For the calculation of the ICC, the two-way random method was used for absolute agreement. The average-measures ICCs were presented for interobserver agreement, representing the reliability of the average values of multiple observers. For intraobserver agreement, the single measures ICCs were presented, representing the reliability of one single observer.

Graphical methods were used to visualize the interobserver agreement in ranking by different observers when using the PC method. Scans were sorted by the median of the ranking scores from all six observers. Per scan, the range of rankings assigned by the observers is visualized via boxplots where a smaller boxplot indicates more agreement. Colors are added to the boxplots indicating the median Likert score received by the observers. For the ICCs of the intraobserver agreement, the original scans present in both datasets were used as they were rated twice. For each observer, the assigned ranks at repeated measurements were visualized in a graph with a line between the assigned ranks. Confusion matrices were used to visualize the inter- and intraobserver agreement when using Likert scores. The Likert scores are indicated on the horizontal and vertical axes and each cell in the matrix represents the number of scans that received a particular combination of Likert scores as assigned by two observers or at repeated measurements. A diagonal matrix would indicate perfect agreement.

Results

Data from 5 patients were excluded because of substantial breathing artifacts (Figure 1). The baseline characteristics of the population are depicted in Table 1. The final compositions of the high and low IQ variation datasets are given in respectively Table 2a and b, with both datasets containing the 25 original scans and in addition 25 scans with either a high or low variety of tube currents and reconstruction techniques.

Table 1 Baseline characteristics

Parameters	N=25
Age [years]	61.1 ± 15.5
Sex [% male]	14 (56 %)
Body weight [kg]	61.1 ± 9.6
Height [m]	1.7 ± 0.1
BMI [kg/m ⁻²]	24.4 ± 2.1

BMI, body mass index;

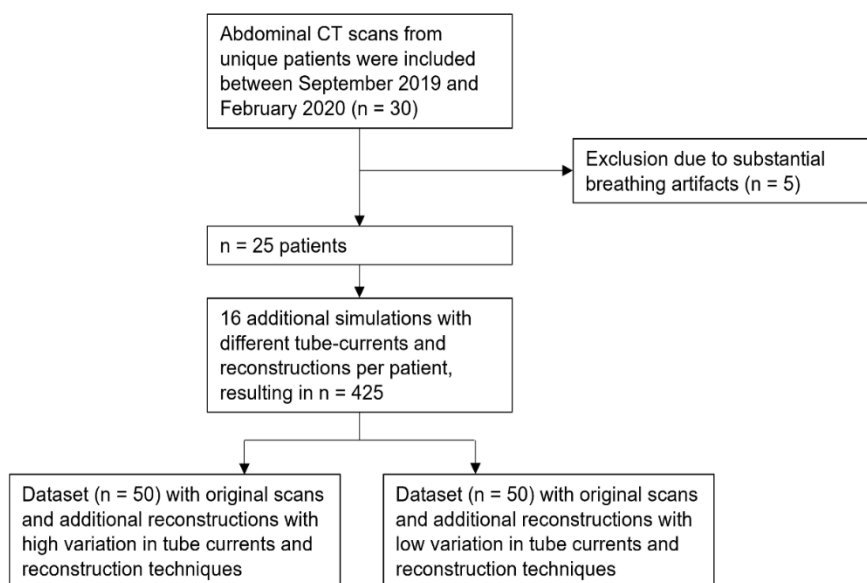


Figure 1 Flow diagram with selection of the high and low variation datasets.

Table 2 Composition of the high variation dataset (a) and the low variation dataset (b). Both datasets include the 25 original scans complemented by 25 scans, for each patient one, with either a high or low variation in reconstruction levels (IR = iterative reconstruction level; FBP = filtered backprojection) and simulated tube current (60 – 90% of the original tube current).

(a) High variation dataset	60 %	70 %	80 %	90 %	100 %
FBP	2	1	2	2	
IR2	2	2	1	2	25
IR3	2	2	2	1	
IR4	1	1	1	1	
(b) Low variation dataset	60 %	70 %	80 %	90 %	100 %
FBP					
IR2					25
IR3					
IR4			13	12	

The user interfaces of the in-house developed software are presented in Figure 2a and b for the PC method and the Likert scale method respectively. Figure 2c shows the ranking of images by observer 5, while the same observer has assigned a Likert score of 4 to all three images.

Interobserver agreement

The ICCs for both the PC method and the Likert scale method are presented in Table 3. The ICC of the high-variation dataset increased from 0.665 (95%CI 0.396-0.814) to 0.785 (95%CI 0.676-0.867) and the ICC of the low-variation dataset from 0.276 (95%CI 0.034-0.500) to 0.562 (95%CI 0.337-0.729) when using the PC method instead of the Likert scale method. Also, for both methods, the ICCs are higher for the high variation dataset than for the low variation dataset.

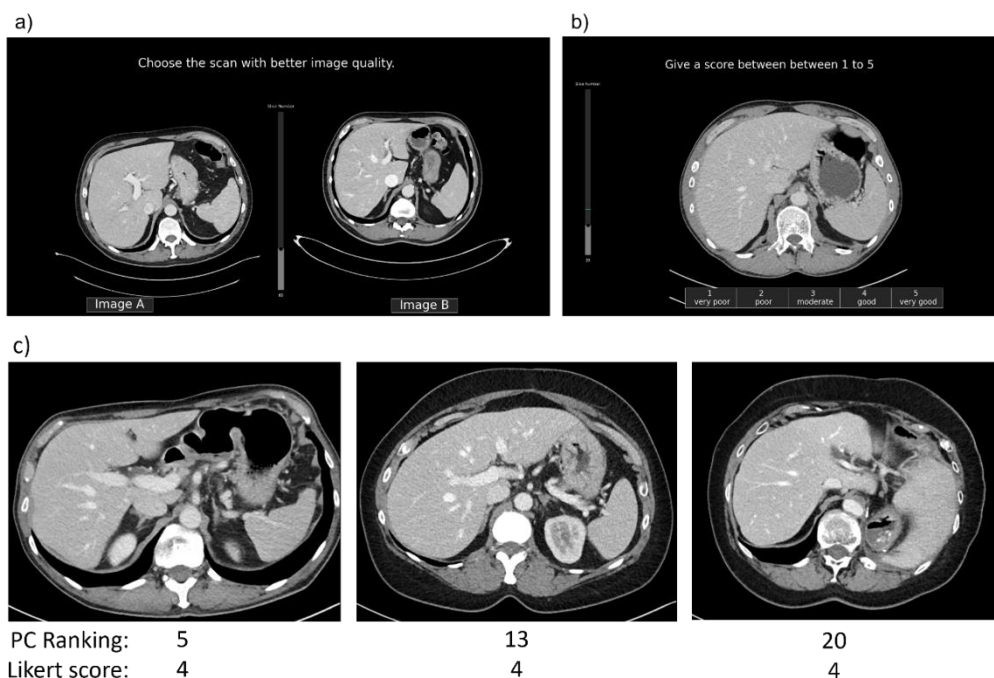


Figure 2 User interface of the in-house developed software for a) the pairwise comparison method b) the Likert scale method. c) Examples of 3 CT images ranked by an observer using the pairwise comparison method with corresponding Likert scores.

Table 3 Interobserver agreement. Intraclass correlation coefficients (ICC) for average values for both subjective scoring methods of image quality (pairwise comparison method and Likert scale) on the high and low variation dataset.

ICC	High variation dataset	Low variation dataset
Pairwise comparison method	0.785 (95%CI 0.676-0.867)	0.562 (95%CI 0.337-0.729)
Likert scale method	0.665 (95%CI 0.396-0.814)	0.276 (95%CI 0.034-0.500)

Figure 3a and b present boxplots of the PC method results for the high and low variation datasets, respectively. Via the colors of the boxplots, it can be observed that a median Likert score of 3 or 4 was assigned to the majority of the scans (more than 90%). Ranking of the IQ according to the PC method shows some correlation with

the Likert scores for IQ. For example, in Figure 3a, 15 out of 18 scans that received a median Likert score of 3 or lower were ranked in the lowest half (left side of the figure) in the PC method and in Figure 3b all six scans with 3 or lower were ranked in the lowest half.

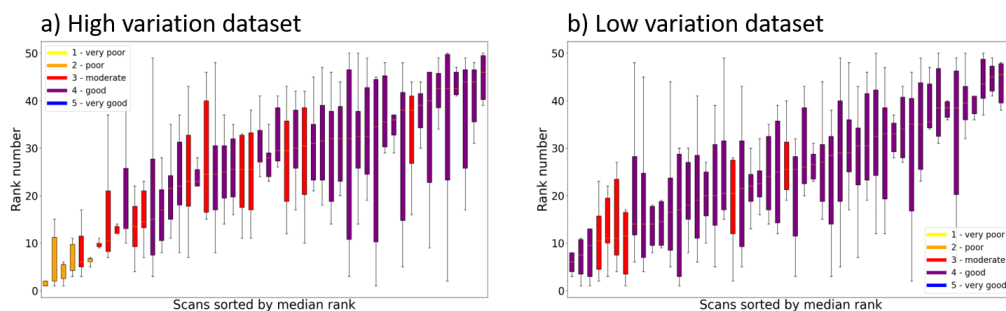


Figure 3 Boxplots representing the spread in ranks of the image quality assessment by all 6 observers using the pairwise comparison method. The boxes are sorted by the median of the 6 ranks received by the observers and colored by the median Likert score given by the observers. Results are given for the high (a) and low (b) variation dataset.

Figure 4 shows a representative selection of the confusion matrices for the Likert scale method for several pairs of observers for data from the high (Figure 4a-d) and low (Figure 4e-h) variation datasets. None of the matrices showed a clear diagonal which would have indicated high agreement. Frequently, an offset from the diagonal is observed, e.g. Figure 4a, where observer 1 most of the time rates lower than observer 2. Also, some observers use the whole range of scores 1-5, e.g., observer 5 in Figure 4c, while other observers mainly use Likert scores 3 and 4, e.g. observer 3. Sometimes large differences in scores are observed, in which one observer rates a scan with a score of 2 and another observer rates the same scan with a 5, e.g. Figure 4e. No clear differences between the high-variation data and the low-variation data can be observed from the confusion matrices.

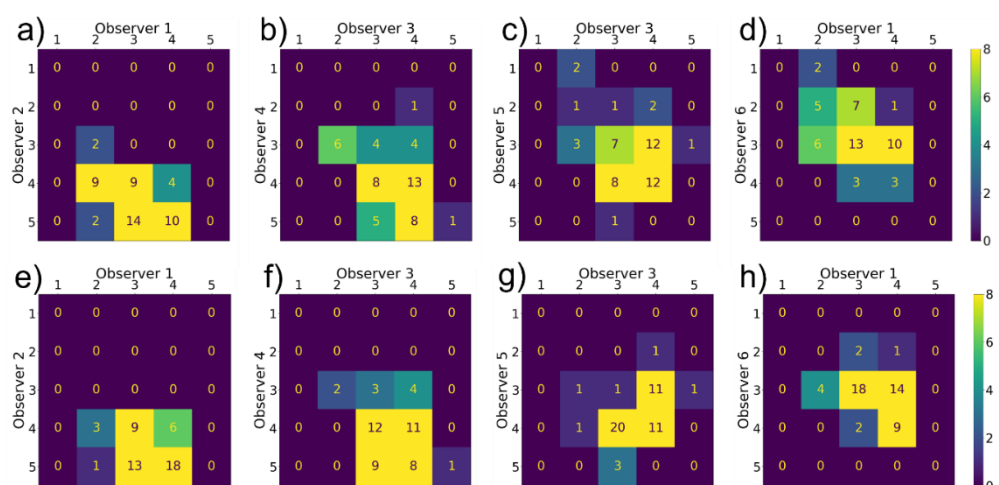


Figure 4 Confusion matrices of random combinations of two observers for the Likert scores of the high variation dataset (a-d) and low variation dataset (e-h). The horizontal and vertical axes give the Likert scores given by the observers. A diagonal matrix would indicate complete agreement.

Intraobserver agreement

Table 4 shows that the ICCs for intraobserver agreement increased substantially for four out of six observers when using the PC method instead of the Likert scale method. Observer 3 has approximately equal ICCs for both methods and for observer 4 the ICC decreased.

Table 4 Intraobserver agreement. Intraclass correlation coefficients (ICC) for both subjective scoring methods of image quality (pairwise comparison method and Likert scale) for repeated assessment by every observer.

ICC	Observer 1	Observer 2	Observer 3	Observer 4	Observer 5	Observer 6
Pairwise comparison method	0.646 (95%CI 0.331-0.831)	0.321 (95%CI 0.000-0.640)	0.596 (95%CI 0.255-0.804)	0.078 (95%CI 0.000-0.468)	0.826 (95%CI 0.638-0.921)	0.705 (95%CI 0.425-0.862)
Likert scale method	0.097 (95%CI 0.000-0.450)	0.014 (95%CI 0.000-0.225)	0.610 (95%CI 0.288-0.809)	0.337 (95%CI 0.077-0.656)	0.267 (95%CI 0.000-0.583)	0.588 (95%CI 0.238-0.800)

Figure 5 shows the ranking results using the PC method at two measurement moments for each observer. It can be observed that the scans with extreme median ranks show less spread in the repeated assessment than the scans with less extreme

median ranks, indicating more agreement on the extreme ranks. Furthermore, since the color represents the median Likert score from the two assessments for each observer, it is evident that most scans received a median Likert score of 3 or 4 independent of the observer, while demonstrating distinct rankings. The colors also show that some observers score consistently lower than other observers.

The confusion matrices of the repeated assessment of IQ with the Likert scale method are shown in Figure 6. It can be observed that observer 6 and observer 3 in particular had little variation in the Likert scores because the majority of the CT scans received the same score on IQ (64% and 72% respectively received a score of 4 during both assessments).

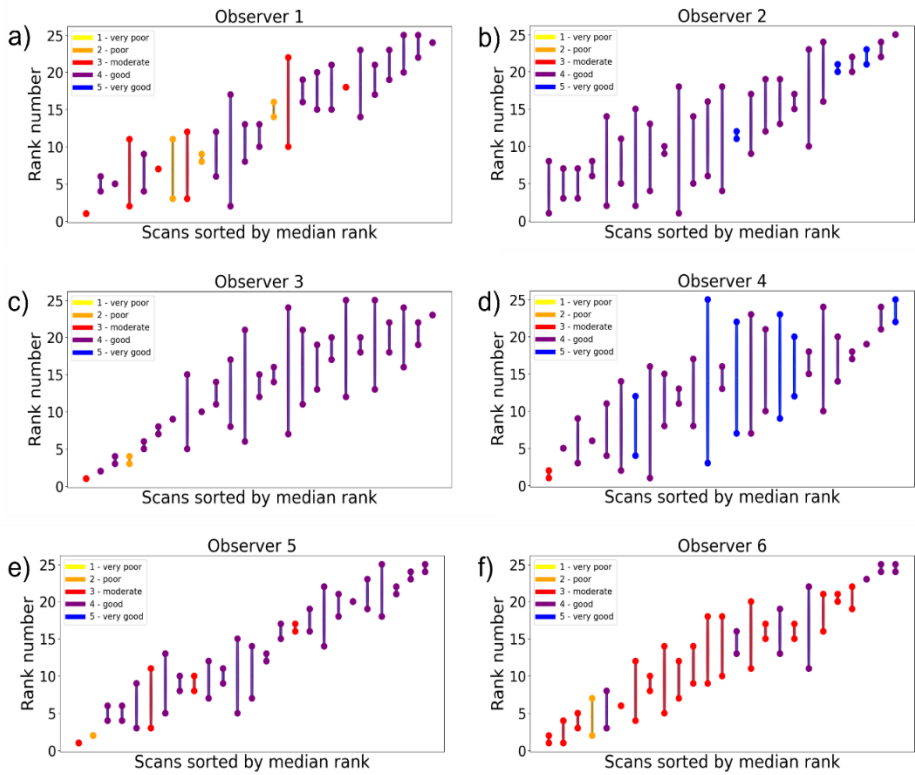


Figure 5 The agreement between two assessments using the pairwise comparison method for every single observer. For each scan, two dots are shown corresponding to the ranking in assessments 1 and 2. The color represents the mean Likert score given by the observer in the 2 assessments. The scans are sorted by the median rank. As an example, for observer 5, nearly every scan received a Likert score 4 (purple), despite the images consistently having distinct rankings in both assessments.

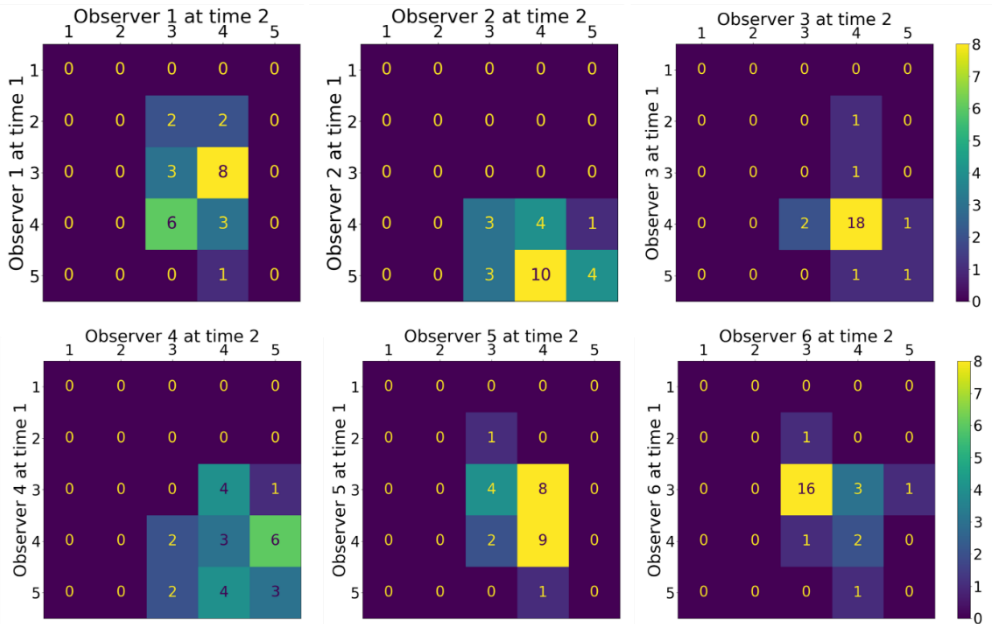


Figure 6 Confusion matrices of the repeated assessment for every single observer (intraobserver agreement). The horizontal (upper) and vertical axes give the Likert scores given by the observers. A diagonal matrix would indicate total agreement.

Discussion

This study showed that the reliability of subjective IQ assessment improved by using the PC method instead of the standard Likert scale method in the evaluation of abdominal CT scans. When using the PC method, the ICCs for interobserver agreement increased substantially irrespective of the variation in IQ between scans and for the intraobserver agreement they increased substantially for four out of six observers.

Likert scale assessments are frequently performed using a reference standard or after receiving prior training, as one might argue that this is beneficial for the accuracy. However, literature indicates that offering a reference standard or prior training does not appear to yield substantial added value. Phelps et al. [14] found that having observers repeat their Likert assessments and providing them with reference best and worst images did not noticeably improve accuracy. Similarly, Mantiuk et al. [13]

did not find any benefit from the addition of a reference scan to a Likert score method. For a fair comparison of both methods, it was decided to not use additional reference scans or prior training for either method.

Subjective evaluation of diagnostic IQ is a frequently used method to test hypotheses in diagnostic imaging and, generally, this is performed using a Likert scale method, e.g. in [6; 8; 10; 18; 21-27]. Mantiuk et al. [13] compared the four most common assessment methods, including a PC method, on photo- and video quality and found the PC method to be the most accurate and time-efficient. They also investigated the PC method with the addition of indicating how much the IQ differs on a continuous scale, but this did not improve the results. In medical IQ assessment, only a limited number of studies related to the PC method were reported. Phelps et al. [14] compared the PC method to a Likert scale method on a set of 10 clinical chest radiographs with different degrees of image sharpness (digitally blurred). They evaluated the agreement of both methods with the actual sharpness and found that the PC method performed better than the Likert scale method. Furthermore, Ellman et al. [28] used a PC method in their CT dose optimization study and found a high interobserver agreement. To the best of our knowledge, our study is the first that compares both methods for CT scans.

The current study was performed using abdominal CT scans, but we expect that results are generalizable to other anatomies and imaging modalities as improvement of IQ assessment using the PC method was shown in a non-medical setting [13] and chest radiographs [14]. Future studies could validate this observation by assessing the effectiveness of the PC method across different modalities. In the present study, the observers were given the general instruction to rate the scans based on overall diagnostic IQ, but both methods can also be used with a more specific diagnostic question, for example 'rate the image quality for discerning focal liver pathology'. This may enhance the level of agreement among the observers, particularly for the PC method. While for the Likert scale method, independent of the specificity of a diagnostic question, the same issues will remain, e.g. the lack of reference material and the tendency to refrain from extreme scores, a specific diagnostic question might simplify the choice between scans when using the PC method as demonstrated by Ellman et al. [28].

The PC method is limited by the need for a sorting algorithm that greatly reduces the number of comparisons to be evaluated because it is not feasible for observers to evaluate all possible pairs of scans. However, sorting algorithms are typically better suited to work with numerical or lexicographical data, and may not be as effective when applied to subjective data such as diagnostic IQ, in which cyclic relations can occur and there is no ground truth [5; 13]. Application of the PC method without a sorting algorithm would prevent errors due to transitivity assumptions, and would therefore lead to even more reliable results but requires a significant additional time investment. Although execution time is substantially reduced by using a sorting algorithm, it took the observers about three times longer to complete the assessment with the PC method compared to the Likert scale method. Therefore, the trade-off between execution time and reliability should be carefully considered. In many cases, the benefits of a more reliable subjective assessment and more refined classification will outweigh the additional time investment.

It is important to note that the PC method captures relative preferences and rankings; thus, it cannot be employed for the absolute evaluation of CT scans. On the other hand, the Likert scale, despite being an ordinal scale, allows for the establishment of a predefined threshold for diagnostically sufficient image quality.

In conclusion, the findings of this study indicate that the use of the PC method improves the reliability of subjective IQ assessment of clinical CT scans compared to the Likert scale method, even for small differences in IQ. This may greatly benefit future optimization studies and research on automatically determined IQ parameters for which validation with subjectively perceived IQ is of importance. Additionally, a large dataset subjectively scored on IQ in a reliable manner might be useful as training data for machine learning algorithms on automatic IQ determination. When part of a quality assurance program, automatically determined IQ can be monitored continuously, making it possible to observe small changes in IQ at an early stage.

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CHAPTER

6



Objective assessment of diagnostic image quality in CT scans:

what radiologists and researchers need to know

E.J.I. Hoeijmakers

B. Martens

J.E. Wildberger

T.G. Flohr

C.R.L.P.N. Jeukens

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Abstract

Objectives: Quantifying diagnostic image quality (IQ) is not straightforward but essential for optimizing the balance between IQ and radiation dose, and for ensuring consistent high-quality images in CT imaging. This review provides a comprehensive overview of advanced objective reference-free IQ assessment methods for CT scans, beyond standard approaches.

Methods: A literature search was performed in PubMed and Web of Science up to June 2024 to identify studies using advanced objective image quality methods on clinical CT scans. Only reference-free methods, which do not require a predefined reference image, were included. Traditional methods relying on the standard deviation of the Hounsfield Units, the signal-to-noise ratio or contrast-to-noise ratio, all within a manually selected region-of-interest, were excluded. Eligible results were categorized by IQ metric (i.e. noise, contrast, spatial resolution, and other) and assessment method (manual, automated, and artificial intelligence (AI)-based).

Results: Thirty-five studies were included that proposed or employed reference-free IQ methods, identifying 12 noise assessment methods, 4 contrast assessment methods, 14 spatial resolution assessment methods, and 7 others, based on manual, automated or AI-based approaches.

Conclusion: This review emphasizes the transition from manual to fully automated approaches for IQ assessment, including the potential of AI-based methods and it provides a reference tool for researchers and radiologists who need to make a well-considered choice in how to evaluate IQ in CT imaging.

Keywords

Tomography (X-ray computed); Diagnostic imaging; Image quality; Automation

Background

Computed Tomography (CT) has revolutionized conventional X-ray-based medical imaging by enabling 3D visualization of internal structures. However, this advancement comes with an increase in radiation exposure, which raises potential health risks to patients. A key challenge in CT imaging remains in concurrently optimizing radiation dose and diagnostic image quality (IQ) [1]. While there are well-defined metrics for straightforward quantification of radiation dose, accurate quantification of diagnostic IQ is far more complex and requires a comprehensive understanding of how various IQ metrics (e.g. noise, spatial resolution, contrast) influence IQ and how to evaluate them effectively [2-4].

Diagnostic IQ can be evaluated subjectively and objectively. Subjective methods, such as radiologist assessments using a Likert scale, remain up to this day the gold standard. While this method is widely used, it has several drawbacks, such as high inter- and intraobserver variability, reader fatigue and a tendency to avoid extreme ratings [5-7]. To address some of these issues, a new subjective assessment method using pairwise comparisons has been proposed, which shows improved consistency between observers [8]. However, this approach cannot completely eliminate observer variability.

To objectively evaluate diagnostic IQ of CT scans, various standard physics based techniques, such as region-of-interest (ROI)-based noise measurements, contrast-to-noise ratio (CNR), and signal-to-noise ratio (SNR) calculations, have traditionally been employed [2; 9]. While these techniques provide quantitative insights into certain aspects of IQ, they often lack the ability to capture the nuanced perceptual preferences of radiologists.

Objective assessment methods are generally categorized into two approaches: reference-free IQ assessment and full-reference IQ assessment [10]. Full-reference methods compare IQ to a predefined reference image. However, obtaining a reference image for every scenario is challenging and might even be impossible due to variability in patient anatomy, pathology, and imaging. In contrast, reference-free IQ assessment methods evaluate IQ based solely on intrinsic image properties, such as noise, contrast, and spatial resolution, without the need for a reference.

Ideally, objective IQ assessment methods should be consistent with human observers and allow automated IQ assessment [11]. A fully automated IQ assessment method, implemented directly after CT scan acquisition would allow for instant feedback if IQ is below a certain threshold. Such a system could also support continuous quality control for CT scanners, ensuring consistently high-quality images and enabling the optimization of the dose-IQ trade-off across large datasets [10; 11].

This review focuses on existing reference-free advanced objective IQ assessment methods, including (artificial intelligence [AI]) algorithms, beyond the standard objective IQ methods (i.e. ROI-based noise, CNR and SNR analysis). We provide a comprehensive overview based on our own experience in developing, analyzing and applying various IQ assessment methods [8; 12-14]. It is intended to serve as a reference tool for researchers and radiologists when designing studies involving CT scans that require IQ assessment.

Methods

Data sources and study selection

For this systematic review, a literature search was performed in PubMed and Web of Science to identify all papers published in peer-reviewed literature until June 2024. The search was conducted using combinations of the terms: *CT, contrast, noise, spatial resolution, image quality, diagnostic, clinical, patient, algorithm, computer-assisted, automated, quantification, model observer, measurement, assessment, machine learning, deep learning, development, proposes, subjective and likert*, defined in consensus by all authors.

Studies were included if they met the following criteria:

- Written in English.
- Explicitly discussed, proposed or applied objective quantitative reference-free methods for IQ assessment of CT, including AI-based methods.
- Applied to or applicable to clinical data.

Exclusion criteria were:

- Studies that used only standard quantitative IQ assessment methods, i.e., an ROI-based analysis extracting mean Hounsfield Unit (HU) $\pm$ standard deviation (SD), SNR, and CNR.
- Studies that lack detailed descriptions of their methods, rendering them non-reproducible.
- Studies focusing solely on artifacts as an IQ measure were also excluded, as artifacts are not directly related to the performance of the scanner self or are only partially influenced by the scanner.
- Studies with full-reference IQ assessment methods (e.g., the Structural Similarity Index Measure)

After removing duplicates, the initial screening was conducted by one author (E.H.) based on the titles to exclude studies that were clearly unrelated to CT imaging. Subsequently, studies that did not meet the inclusion and exclusion criteria were filtered out by reviewing the abstracts and, when necessary, full texts. A second author (C.J.) also reviewed the full texts independently to verify eligibility. In case of disagreement, consensus was reached through discussion.

Both authors independently performed data extraction from the included studies. The final selection of studies was categorized into four main groups based on the aspect of IQ they addressed: noise, contrast, spatial resolution, and other factors. Within each category, studies were further subcategorized by the type of method used:

- Manual: Techniques that require human involvement for measurement or evaluation, such as manual ROI placement
- Automatic: Fully deterministic algorithms that operate without human involvement once configured
- AI-based: (automated) methods using machine or deep learning techniques that automatically learn patterns from data.

Finally, the following information was extracted for each study: the IQ assessment methodology, the body region for which the method is applicable, the key principle, whether the method is ROI-based, key advantages and disadvantages of the method

and if and how the method was validated. Based on this information a decision tree was constructed as a tool to help deciding on a relevant method corresponding to the research goal.

Results

The Preferred Reporting Items for Systematic reviews and Meta-Analysis (PRISMA) flowchart of the screening process is given in Figure 1. The primary literature search yielded 2559 studies of which 304 were duplicates, resulting in 2254 studies. From these, 78 studies were selected for full text review, after which 35 studies were included in this systematic review.

The subdivision of the studies in the four categories resulted in: $n = 12$ IQ assessment methods focusing on noise, $n = 4$ on contrast, $n = 14$ on spatial resolution and $n = 7$ on other aspects, where two studies were included in more than one category.

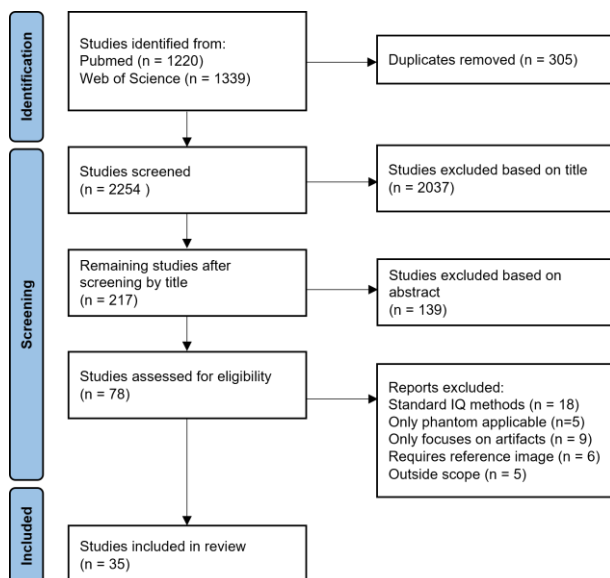


Figure 1 Preferred Reporting Items for Systematic reviews and Meta-Analysis (PRISMA) flowchart of study selection

Figure 2 gives a decision tree providing a structured overview of the selected objective IQ assessment techniques.

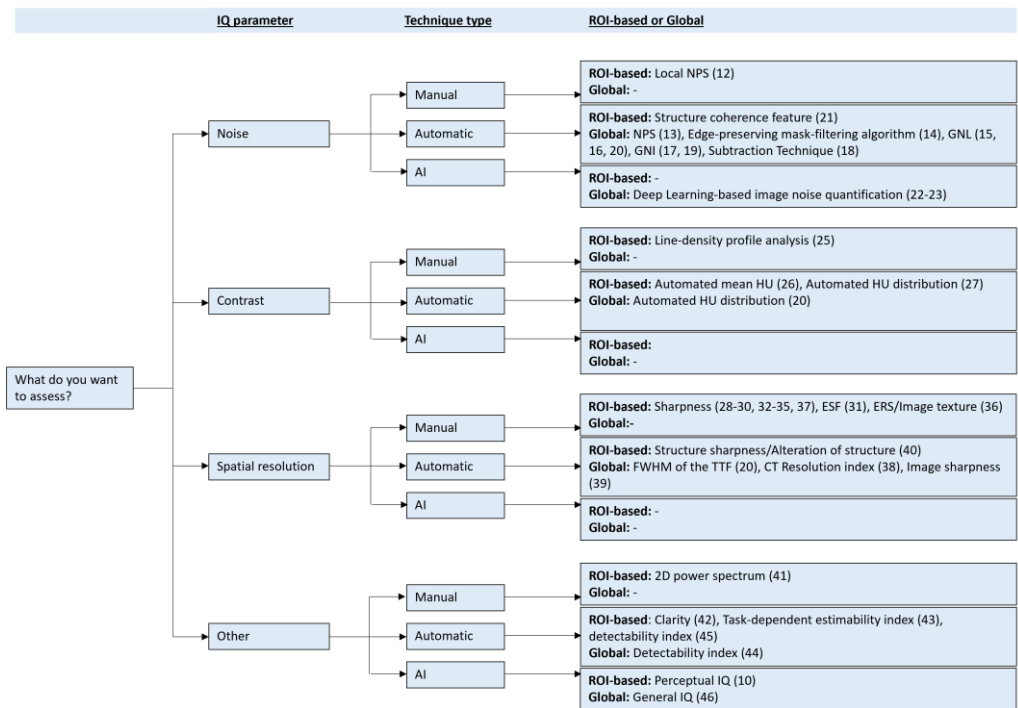


Figure 2 Decision tree providing a structured overview of advanced image quality (IQ) assessment techniques for CT. ROI, region-of-interest; IQ, image quality; AI, artificial intelligence; NPS, noise power spectrum; GNL, global noise level; GNI, global noise index; HU, Hounsfield unit; ESF, edge spread function; ERS, edge rise slope; FWHM, full width half maximum; TTF, task transfer function

Noise

Twelve studies reported on noise assessment techniques (Table 1 and Table S1 for more details). One study required manual ROI placement [15], nine studies employed automatic noise assessment methods [12; 16-23] and two used AI-based approaches [24; 25].

Manual methods

Zhang et al reported a manual method determining the local Noise Power Spectrum (NPS) for photon-counting detector CT, which measured noise in localized ROIs

using raw detector counts [15]. It eliminates the need for repeated scans (and thus repeated exposure) as is necessary for standard NPS measurements that are applied on phantoms [4]. The method is however not applicable to conventional energy-integrating detector CT data.

Automatic methods

Smith et al suggested an automatic method to do NPS measurements in the liver without repeated exposure, assuming the HU variation in the uniform liver as noise after automatic segmentation [16].

Additionally, automatic global noise evaluations were addressed in seven studies [12; 17-22], some of which are related. An early approach by Kortensniemi et al [17] developed an edge-preserving mask-filtering algorithm to create noise maps by calculating the SD in pixel neighborhoods while excluding sharp transitions. This method provides a comprehensive image-wide noise assessment and has the potential for a series-wide application, i.e., including all slices of a series as opposed to a single slice analysis. Subsequently, Christianson et al introduced the Global Noise Level (GNL) method, based on segmentation of tissue types using threshold values, and using the mode of the histogram of pixel-wise SDs as a robust measure of noise [18] applicable in abdominal CT scans. Anam et al extended the GNL to be able to use it anatomy independent by using a threshold-based segmentation of the whole body and selecting homogeneous regions for noise evaluation via the histogram [19]. Further refinement of the GNL was executed by Alsaihati et al, averaging it over slices to obtain the global noise index (GNI) [20]. However, both studies rely on successful threshold-based segmentation of tissue types.

An alternative to the GNL approach was presented by Tian and Samei [21] with a method that involves subtracting adjacent CT slices to eliminate anatomical variations, and filtering out edges, with noise being measured in the resulting uniform regions. However, a key limitation of this approach is the requirement for sufficiently large uniform areas for accurate noise measurement. Building on Tian and Samei's work, Malkus et al [22] enhances the method's applicability by showing that noise can be measured in air regions. Subsequently, Jeukens et al combined this result with the results from Christianson et al to obtain a GNL measured in air [12].

Their study showed significant correlation with subjective IQ scores in abdominal CT scans, demonstrating feasibility of algorithm-determined objective IQ.

Another alternative approach was proposed by Chun et al, who developed a method based on the structure coherence feature [23]. This technique 'filters out' small anatomical structures, preventing them from influencing noise measurements, and calculates noise by averaging the SDs within automatically selected ROIs instead of globally. This method has the advantage of reducing the impact of anatomical detail on noise assessment, making it potentially more accurate in complex clinical images.

AI methods

The two AI-based approaches for measuring noise in CT images utilized convolutional neural networks (CNNs) trained on anthropomorphic phantom data. These methods are designed to automate the traditionally manual process of standard deviation (SD) measurements for noise by generating pixel-wise noise maps automatically. Ketola et al developed a CNN trained with true noise (standard deviation, SD) maps obtained from two consecutive scans, focusing exclusively on chest CT [24]. In contrast, Huber et al trained their CNN using true noise maps derived from per-pixel SDs calculated across 100 replicate scans, covering chest, head, and pelvis CT [25]. Both studies validated their methods on clinical data.

Contrast

Four studies reported on contrast assessment methods (Table 2 and Table S2 for more details), of which one required manual placement of an ROI [26] and three were fully automatic [12; 27; 28].

Manual methods

One manual measurement method was included that evaluated the contrast with a line-density profile analysis, by considering minimum and maximum HU values within a 10-mm line crossing tumorous and healthy tissue [26]. While such a method is very effective for targeted evaluations, it has similar disadvantages as the standard CNR method: it offers no information on the overall contrast distribution and merely focuses on contrast between predefined regions.

Automatic methods

Three studies used automated approaches for contrast assessment. Pallenberg et al automated contrast measurement in aortic CT angiography (CTA) volumes by using a template-matching technique to automatically place ROIs in the aorta [27]. The reliance on reference templates may restrict the method's applicability across different patient cases and anatomical variations. In contrast, Abadi et al and Jeukens et al developed broader, automated contrast measurement techniques that analyze the entire CT image rather than relying on specific regions [12; 28]. Abadi et al introduced an automatic method to measure the distribution of Hounsfield Units (HUs) across various organs in chest CT scans, starting with body contour segmentation followed by automated ROI placement in the lungs, liver, aorta, and spine. The median HU value within each ROI was used as a contrast measure, allowing for application across both contrast-enhanced and non-contrast-enhanced scans. This approach benefits from standardized ROI placement, which provides consistent contrast measurements within predefined anatomical structures. Although this method does not require manual ROI placement, it still only gives a measure for contrast in specific areas. Jeukens et al offered a different approach by basing contrast measurement on HU histogram analysis of the entire scan volume. Following a threshold-based segmentation to remove background air and bone pixels, a histogram of HU values is generated where the peaks correspond to different soft tissues. Contrast metrics are derived from the histogram's shape and peaks, providing insights into the distribution and prominence of different tissue contrasts.

Spatial Resolution

Fourteen studies reported on methods to determine spatial resolution (Table 3 and Table S3 for more details). The majority (n=10) of the studies were manual methods [29-38], and four studies contained automatic methods [12; 39-41].

Manual methods

Manual methods for assessing spatial resolution predominantly focused on edge sharpness or slope measurements of a line profile across specific anatomical structures, i.e., edges in the head/brain region [29-33], in vascular structures such as

coronary arteries and the portal vein [34-37] and in the intestines [38]. These methods typically involved drawing a single or multiple lines perpendicular to the edge of a structure to obtain a profile of the CT density values along the line, which shows a change in values at the edge. The spatial resolution or sharpness is subsequently calculated by determining the maximum slope of the line profile or differences between CT values at either side of the edge. These methods are effective for consistent comparisons of spatial resolution within defined regions and for specific protocols. However, they require radiological expertise for accurate line/ROI placement, making them time-intensive and prone to observer variability. Additionally, the localized nature limits the ability to represent global spatial resolution. Anatomical variability, such as differences in vessel diameter or tissue composition, can further affect results even with standardized scanning parameters. Partial volume effects and complex anatomical structures may also distort sharpness measurements, reducing the generalizability of these methods [30]. Additionally, to determine the maximal slope, Cao et al also considered the skewness of the CT number histogram in the liver parenchyma as a measure for image texture [37].

Automatic methods

Sanders et al [39] introduced the CT Resolution Index, which evaluates spatial resolution by measuring the edge spread function (ESF) across the air-skin interface after segmenting the patient's skin and generating a tetrahedral mesh. While this method allows for quick, objective assessments across different anatomical regions, it is sensitive to noise and depends heavily on accurate segmentation. Jeukens et al [12] enhanced Sanders' method by adding normalization and averaging steps but maintaining the dependence on accurate segmentation. Both methods offer the benefit of independence from specific body areas, as they can be consistently applied to the skin-air interface.

Other approaches, by Salimova et al [40] and Chun et al [41], focused on specific anatomical areas or tissue structures. Salimova's method assessed image sharpness in the lower extremities by first segmenting and excluding air and bone edges, then analyzing the 2D image gradient of automatically detected significant edges per slice. Chun et al developed a structure coherence feature (SCF) technique to

distinguish between homogeneous and edge containing regions within the liver. In the edge containing regions the SCF is determined based on the presence of edges and the directionality of the edges. Similar to other automatic techniques, they remain constrained by dependencies on high-quality segmentation.

Others

In addition to conventional metrics like noise, spatial resolution, and contrast, various other methods have been developed to objectively assess IQ in CT scans (Table 4 and Table S4 for more details). These approaches often combined multiple image attributes and computational models to provide a more nuanced evaluation of IQ. The seven studies encompassed one manual method [42], four automatic methods [43-46] and two AI methods [10; 47].

Manual methods

Svalkest et al analyzed the frequency components of clinical CT scans by computing the power spectrum (PS) from reconstructed image slices. By averaging the PS from non-overlapping pixel regions, mirroring to reduce edge artifacts and radially averaging the PS of each image slice, the study quantified the spatial frequency distribution [42].

Automatic methods

One significant aspect of IQ is how clearly a lesion or abnormality is rendered in a scan for which Cheng et al introduced the term 'clarity' [43]. This metric combines the spatial resolution via the modulation transfer function (MTF) of the clinical image at hand with the noise texture derived from phantom images, matching the clinical scan protocol. It is assumed that the noise in the phantom images matches the clinical noise. However, its dependence on pre-acquired phantom images, limits its applicability across different imaging settings.

Task-specific methods were a key focus among the identified automatic methods, valuable in contexts where the primary task is not to quantify the IQ in general, but how well the system can detect and characterize specific features required for a

specific task within the image. Samei et al introduced a task-dependent estimability index for quantifying stenosis using cardiac CT angiography (CTA). This method integrated multiple objective factors, such as motion blur, noise, and spatial resolution, to quantify how accurate the imaging system can assess stenosis in cardiac vessels [44]. Similarly, Smith et al developed a detectability index with a model observer function for evaluating the system's ability to detect liver lesions. This approach combined spatial resolution and noise measurements from previous validated techniques [18; 39] extracted from the clinical image with a reference task function (detecting a disk lesion with -15 HU contrast) to provide a metric for lesion detectability [45; 46]. While highly relevant for their respective clinical applications, these task-focused metrics are often limited in their generalizability to other anatomical regions or imaging tasks.

AI methods

Lee et al developed an AI-based approach that used a self-supervised training strategy to assess the perceptual IQ of chest and abdominal CT scans [10]. This method aimed to provide a quantitative score that reflects the overall quality of the image as perceived by radiologists, offering a more standardized and potentially more reliable measure of IQ.

Nakanishi et al addressed the challenge of generalizability in AI methods by training a model for diagnostic versus non-diagnostic image classification of coronary CTA scans on multi-center and multi-vendor datasets. This approach increased the robustness of the AI model across various imaging systems [47].

Table 1 Objective CT image quality assessment methods focusing on noise

Study	Methodology	Technique type	Body region	ROI or Global	Key principle
Zhang et al (2024) [15]	NPS	Manual	All	ROI	Novel mathematical method that uses the raw counts data of PCD-CT scan to obtain a local NPS
Smith et al (2021) [16]	NPS	Automatic	Abdomen	Global	Segmenting the liver, selecting a uniform area within it and measuring noise patterns within the data via noise auto covariance
Kortesniemi et al (2008) [17]	Edge-preserving mask-filtering algorithm	Automatic	All	Global	Creating a map of filtered local minimum SD's using a moving square mask, forming an IQ score
Christianson et al (2015) [18]	GNI	Automatic	Abdomen	Global	Finding the most frequent noise level in areas of homogeneous tissue
Anam et al (2021) [19]	GNI	Automatic	All	Global	Finding minimum standard deviation in any region of the body as a measure for noise
Alsaïhati et al (2023) [20]	GNI	Automatic	Soft tissue	Global	Enhanced version of Christianson et al (2015) but averaging it over slices to obtain the GNI of a complete series
Tian et al (2016) [21]	Subtraction technique	Automatic	Abdomen, lung	Global	Subtracting sequential slices and filtering out edges to measure SD in resultant uniform area
Malkus et al (2017) [22]	GNI	Automatic	All	Global	Determination of the mode/mean of the SD in air, by subtracting adjacent slices and with an edge-finding algorithm to remove morphological variations
Jeukens et al (2023) [12]	GNI	Automatic	All	Global	Combines Malkus et al and Christianson et al by measuring GNI in soft tissue and air
Chun et al (2015) [23]	Automated noise measurement with a novel structure coherence feature	Automatic	Regions with fat	ROI	Calculation of the structure coherence feature to determine homogeneous ROIs and using the average SD as noise level
Ketola et al (2024) [24]	Deep learning-based image noise quantification framework	AI	Chest	Global	Convolutional neural network trained on anthropomorphic phantom data labeled with true noise maps via 2 consecutive scans
Huber et al (2023) [25]	Deep learning-based image noise quantification framework	AI	Chest, head and pelvis	Global	Convolutional neural network trained on anthropomorphic phantom data labeled with true noise maps with per-pixel SDs from 100 replicate scans

ROI, region-of-interest; NPS, noise power spectrum; PCD-CT, photon-counting detector CT; SD, standard deviation; GNI, global noise level; GNI_g, global noise index; AI, artificial intelligence;

Table 2 Objective CT image quality assessment methods focusing on contrast

Study	Methodology	Technique type	Body region	ROI or Global	Key principle
Beer et al (2019) [26]	Line-density profile analysis	Manual	Abdomen	ROI	Difference between maximum and minimum HU values on the line
Pallenberg et al (2020) [27]	Automated mean HU measurement of aortic CTA volumes	Automatic	Aorta	ROI	Classification of insufficient, optimal or excessive contrast based on the mean HU values in ROIs that were automatically positioned in the CTA volume (CTA image slices were automatically localized as well)
Abadi et al (2017) [28]	Automated HU distribution measurement	Automatic	Chest (lung, liver, aorta, spine)	ROI	Median of the distribution of HUs by segmentation of organs and automated ROI placement
Jeukens et al (2023) [12]	Automated HU distribution measurement	Automatic	Abdomen	Global	Generating an histogram of HUs after removing air and bone with threshold based segmentation. Various contrast metrics are defined with the histogram analysis

ROI, region-of-interest; HU, Hounsfield Unit; CNR, contrast-to-noise ratio; SNR, signal-to-noise ratio; IQ, image quality; CTA, computed tomography angiography;

Table 3 Objective CT image quality assessment methods focusing on spatial resolution

Study	Methodology	Technique type	Body region	ROI or Global	Key principle
Korn et al (2013) [29]	Sharpness	Manual	Head	ROI	Sharpness was quantified in terms of the maximal slope (change in HU per pixel) across a line profile perpendicular to the skull circumference
Brodoefel et al (2014) [30]	Sharpness	Manual	Head	ROI	Sharpness was quantified in terms of the gradient or maximal slope (change in HU per pixel) across a line profile perpendicular to the skull circumference
Ernst et al (2016) [31]	Sharpness	Manual	Head	ROI	Sharpness was quantified in terms of the maximal slope (change in HU per pixel) across a line profile perpendicular to the skull circumference
Wu et al (2020) [32]	Edge spread function	Manual	Head	ROI	Spatial resolution was quantified in terms of the width of the edge spread function (ESF) across line profiles perpendicular to the boundary between the lateral ventricle and surrounding brain parenchyma
Altmann et al (2023) [33]	Sharpness	Manual	Head/Neck	ROI	Sharpness was quantified in terms of the maximum slope (change in HU per pixel) across line profiles perpendicular to a border between fat and muscle tissue
Mergen et al (2022) [34]	Sharpness	Manual	Coronary arteries	ROI	Sharpness was quantified in terms of the maximum slope (change in HU per pixel) across line profiles perpendicular to the left anterior descending artery
Feger et al (2015) [35]	Sharpness	Manual	Coronary arteries	ROI	Sharpness was quantified in terms of the difference between 25% and 75% of the maximal HU value and in terms of the maximal slope (change in HU per pixel) across a line profile perpendicular to the edge of a vessel
Heinrich et al (2023) [36]	Sharpness	Manual	Coronary arteries	ROI	Sharpness was quantified in terms of the maximum slope (change in HU per pixel) across a line profile perpendicular to the edge of the vessel lumen

ROI, region-of-interest; HU, Hounsfield Unit; IQ, image quality; ESF, edge spread function; CTA, CT-angiography; ERS, edge rise slope; MTF, modulation transfer function; FWHM, full width half maximum; TTF, task transfer function; SCF, structure coherence feature; SD, standard deviation

Table 3 continued Objective CT image quality assessment methods focusing on spatial resolution

Study	Methodology	Technique type	Body region	ROI or Global	Key principle
Cao et al (2023) [37]	Edge rise slope (ERS)	Manual	Abdomen	ROI	Sharpness was quantified in terms of the ERS (the ratio between the difference between the first peak and last dip in HU values, divided by their distance) across a line profile perpendicular to the edge of a portal vein
Johnson et al (2014) [38]	Image texture	Manual	Abdomen	ROI	Skewness of the CT number histogram on the relatively uniform area of the liver parenchyma
Sanders et al (2016) [39]	Sharpness	Manual	Intestines	ROI	Sharpness was quantified in terms of the edge-width across an ROI crossing the psoas-fat edge
	CT Resolution index based on the edge spread function	Automatic	Skin	Global	Spatial resolution was quantified by segmentation of the patient's body and measuring the edge spread function (ESF) across line profiles perpendicular to the air-skin interface
Jeukens et al (2023) [12]	FWHM of the Task transfer function (TTF)	Automatic	Skin	Global	Enhanced version of the method of Sanders et al (2016) by calculating the FWHM from the ESF
Salimova et al (2022) [40]	Image sharpness	Automatic	Lower extremities	Global	Sharpness was quantified in terms of the 2D image gradient on a per-slice bases after automated segmentation of air and bone to exclude sharp edges
Chun et al (2022) [41]	Structure sharpness/ Alteration of structure	Automatic	Abdomen	ROI	Using a structure coherence feature (SCF) homogeneous and structure edge regions were defined. Structure sharpness and alteration were evaluated using the SCF and SD's between homogeneous and structured regions

ROI, region-of-interest; HU, Hounsfield Unit; IQ, image quality; ESF, edge spread function; CTA, CT-angiography; ERS, edge rise slope; MTF, modulation transfer function; FWHM, full width half maximum; TTF, task transfer function; SCF, structure coherence feature; SD, standard deviation

Table 4 Objective CT image quality assessment methods not solely focusing on noise, contrast or spatial resolution

Study	Methodology	Technique type	Body region	ROI or Global	Key principle
Svalkest et al (2022) [42]	2D power spectrum (PS)	Manual	Chest	ROI	IQ was quantified in terms of the spatial frequency distribution with a 2D PS per slice, by averaging the PS of non-overlapping ROIs on each slice (ROIs were mirrored for artefact reduction)
Cheng et al (2019) [43]	Clarity	Automatic	Lesions	ROI	Combination of the modulation transfer function on the clinical CT images and the NPS calculated on previously made phantom images
Samei et al (2021) [44]	Task-dependent estimability index	Automatic	Stenosis quantification in CTA	ROI	Combination of motion point spread functions, CT image motion blur, noise and an automated maximum-likelihood estimator, to form an estimability index e' as a task-based measure of IQ for stenosis quantification in cardiac CTA
Smith et al (2018) [45]	Detectability index	Automatic	Abdomen	Global	Combination of noise, spatial resolution and a reference task function to form an detectability index for the detection of lesions in the liver
Smith et al (2021) [46]	Detectability index	Automatic	Abdomen	ROI	Combination of image derived noise, spatial resolution and a reference task function to calculate a detectability index for the detection of lesions in the liver
Lee et al (2022) [10]	Perceptual IQ	AI	Chest and abdomen	ROI	Self-supervised training strategy for object detection that results in a quantitative quality of CT images
Nakanishi et al (2018) [47]	General IQ	AI	CCTA	Global	Machine learning algorithm that predicts an IQ score, with features as noise, contrast, misregistration scores and uninterpretable index as input, meant to predict diagnostic / non-diagnostic

ROI, region-of-interest; PS, power spectrum; IQ, image quality; NPS, noise power spectrum; CTA, CT angiography; AI, artificial intelligence; CCTA, coronary CT angiography;

Discussion

A systematic review of the literature up to June 2024 was performed on objective diagnostic IQ assessment methods. An overview was provided by categorizing the methods in 'noise', 'contrast', 'spatial resolution' and 'others' and further dividing them into manual, automatic and AI-based methods. Thirty-five studies were identified that proposed or employed reference-free IQ methods beyond traditional methods. The traditional methods encompassed IQ assessment performed within a manually selected ROI by determining the SD of the HU values, which can be used to determine the SNR or CNR. As these methods are very well-known, but limited in their ability to capture more complex aspects of IQ, this review focuses on more advanced IQ assessment methods.

There is an ongoing transition towards more advanced, automated and AI-driven IQ assessment techniques, which offer opportunities for large-scale, real-time, and reproducible assessments. ROI-based methods are regarded still valuable for localized and task-specific evaluation. However, they are time-consuming and less reproducible and inherently limited in generalizability due to observer dependence leading to variability in shape (square, circular, line) and size. The American Association of Physicists in Medicine recommends an ROI size of at least 1 cm in diameter for reliable measurements [1; 2]. Large ROI sizes are not always practical in clinical settings, especially when dealing with small structures, in images that have a complex anatomy or in regions with limited homogeneous tissue. These manual placement challenges underscore the need for automatic and AI-based methods, offering advantages in standardization and scalability. This would eliminate the need for user input, improve reproducibility and they align with the goal to achieve real-time integration of IQ assessment into scanner workflows and quality control programs.

AI-based approaches, particularly CNNs, are promising for IQ assessment. When trained on subjectively assessed datasets, CNNs can simulate subjective IQ assessment while providing greater consistency. These models have demonstrated accuracy in automating noise and general IQ measurements, trained on anthropomorphic phantom data and validated with clinical datasets [1; 3-5].

However, AI methods face challenges, such as the need for large, diverse, annotated datasets, their limited generalizability across different body regions and examination protocols. They are difficult to validate to the lack of a clinical universal IQ “gold standard”. Additionally, many automated and AI-based approaches also rely on accurate image segmentation, which are often prone to failure due to unexpected anatomy or artifacts.

Two noticeable gaps in the existing literature were identified: First a lack of comprehensive metrics that integrate multiple IQ characteristics (e.g., noise, contrast, spatial resolution) into a single IQ score. Second, absence of robust, objective validation frameworks of IQ assessment methods, as there is no reference or “gold standard” to compare results.

Looking forward, CT IQ assessment should move toward a fully objective approach to overcome human variability. Rather than evaluating isolated aspects of IQ, IQ assessment should comprise a single IQ score that accounts for all relevant parameters, within the context of the intended diagnostic task. In parallel, validation frameworks should shift toward objective, task-based reference standards to replace commonly used subjective assessments. Relying on subjective methods as ultimate truth and “gold standard” to validate new objective approaches can introduce circular reasoning, particularly when the new methods aim to surpass the limitations of subjective ones. Consequently, claims of accuracy or robustness of the IQ assessment methods may be overstated. Ideally, IQ assessment should be embedded into the scanner’s acquisition or reconstruction pipeline, offering real-time feedback. This would support proactive quality assurance, can help with scan and reconstruction protocol optimization, and even lead to a reduction of unnecessary repeat scans.

As researchers are constantly striving to optimize CT IQ for each individual patient and examination [6], and in particular to balance IQ with radiation dose, it would be beneficial to keep improving existing IQ evaluation methods. Future research should prioritize the development of automated methods to prevent observer variability due to manual intervention. There should be focus on robust segmentation-free approaches, or improvements in segmentation reliability. Additionally, expanding training datasets for AI models, including diverse imaging conditions, patient

demographics, and scanner technologies, will largely improve the potential of AI in IQ assessment, but this requires collaboration and data sharing between institutions. Finally, this review highlights the need for standardized metrics and benchmarks in objective IQ assessment, as well as the need for reliable validation methods. Establishing common frameworks would improve the reproducibility of IQ research and further foster the development of clinically meaningful IQ evaluation methods.

Conclusion

By giving an overview of existing reference-free advanced objective IQ assessment methods, this review shows the progression from manual ROI-based techniques to fully automated and/or AI-based approaches. Further development of objective IQ assessment methods is required, including methods integrating multiple IQ parameters, expanding training datasets for AI models, and developing robust segmentation-free methods. While no single method can yet be considered universally optimal, this review provides a structured overview to guide in selecting currently existing IQ assessment methods appropriate to their specific goals.

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Supplementary material

Table S1 Objective CT image quality assessment methods focusing on noise ROI, region-of-interest; NPS, noise power spectrum; PCD-CT, photon-counting detector CT; SD, standard deviation; GNL, global noise level, GNI, global noise index, AI, artificial intelligence.

Study	Methodology	Technique type	Body region	Key principle	ROI or Global	Key advantages / disadvantages	Method of validation
Zhang et al. (2024) [15]	NPS	Manual	All	Novel mathematical method that uses the raw counts data of PCD-CT scan to obtain a local NPS	ROI	(+) Enables NPS measurement without exposing a patient to repeated radiation (-) Only applicable to PCD-CT scans (-) Depends on correct placement of ROI by expert	Validated with a conventional local NPS measurement method using repeated scans on phantom data
Smith et al. (2021) [16]	NPS	Automatic	Abdomen	Segmenting the liver, selecting a uniform area within it and measuring noise patterns within the data via noise auto covariance	Global	(+) Also measurement over complete CT-series is described (-) Method cannot fully rule out possible anatomical "contamination"	Validated with anthropomorphic phantom data with ground-truth NPS measurements
Kortesniemi et al. (2008) [17]	Edge-preserving mask-filtering algorithm	Automatic	All	Creating a map of filtered local minimum SD's using a moving square mask, forming an IQ score	Global	(+) Works on slices but method can easily be extended to complete CT-series (+) Relatively independent of high-detail content (+) Not anatomy dependent	Validated with technical and anthropomorphic phantom data with noise measured in manually placed ROIs in the background and by testing the method on a clinical head CT scan Also validated in Jeukens et al. (2023) and Franck et al. (2017) [12; 51]
Christianson et al. (2015) [18]	GNL	Automatic	Abdomen	Finding the most frequent noise level in areas of homogeneous tissue	Global	(-) Small-scale anatomic texture may contribute to the global noise level (-) Depends on successful threshold based segmentation (-) Requires sufficiently large region of soft tissue	Validated with phantom data with noise measured by subtracting repeated scans and validated with an observer study via the SD in manually placed ROIs on clinical abdominal CT scans of 6 patients
Anam et al. (2021) [19]	GNL	Automatic	All	Finding minimum standard deviation in any region of the body as a measure for noise	Global	(+) Applicable to all body regions (+) Not dependent on segmentation (-) Only validated on abdominal data	Validated with results of noise measurements from Christianson et al. (2015) [18] on a homogeneous phantom and tested on clinical abdominal CT scans of 27 patients

Continued Table S1 Objective CT image quality assessment methods focusing on noise
ROI, region-of-interest; NPS, noise power spectrum; PCD-CT, photon-counting detector; CT, SD, standard deviation; GNL, global noise level; GNI, global noise index; AI, artificial intelligence;

Study	Methodology	Technique type	Body region	Key principle	ROI or Global	Key advantages / disadvantages	Method of validation
Alsalhati et al. (2023) [20]	GNI	Automatic	Soft tissue	Enhanced version of Christianson et al. (2015) but averaging it over slices to obtain the GNI of a complete series	Global	(+) Gives noise of a complete CT-series (-) Depends on successful segmentation (-) Requires large region of soft tissue (-) Small-scale anatomic texture may contribute to the global noise level	Compared to results of Christianson et al. (2015) [18]
Tian et al. (2016) [21]	Subtraction technique	Automatic	Abdomen, lung	Subtracting sequential slices and filtering out edges to measure SD in resultant uniform area	Global	(+) GNI cannot be influenced by anatomical variability	Tian et al. (2016)
Malikus et al. (2017) [22]	GNI	Automatic	All	Determination of the mode/mean of the SD in air, by subtracting adjacent slices and with an edge-finding algorithm to remove morphological variations	Global	(-) Sufficient amount of air is needed around the patient	Validated with results of the noise measurement method from Tian and Samei et al. (2016) [21] measured inside a phantom and with the SD in manually placed ROIs on clinical abdominal CT scans of 40 patients
Jeukens et al. (2023) [12]	GNL	Automatic	All	Combines Malkus et al. and Christianson et al. by measuring GNL in soft tissue and air	Global	(+) GNL in air cannot be influenced by anatomical variability (+) GNL in air is not limited to body region. (-) Depends on successful threshold based segmentation	Validated with an observer study on anthropomorphic phantom data and clinical abdominal CT scans of 30 patients via subjective IQ assessment (overall diagnostic IQ)
Chun et al. (2015) [23]	Automated noise measurement with a novel structure coherence feature	Automatic	Regions with fat	Calculation of the structure coherence feature to determine homogeneous ROIs and using the average SD as noise level	ROI	(+) Small anatomical structures do not influence the noise results (+) Not limited to body region except for head (not enough fat) (-) Depends on successful segmentation	Validated with the SD in manually placed ROIs on clinical chest and abdominal CT scans of 100 patients

Continued Table S1 Objective CT image quality assessment methods focusing on noise
ROI, region-of-interest; NPS, noise power spectrum; PCD-CT, photon-counting detector CT; SD, standard deviation; GNL, global noise level; GNI, global noise index; AI, artificial intelligence;

Study	Methodology	Technique type	Body region	Key principle	ROI or Global	Key advantages / disadvantages	Method of validation
Ketola et al. (2024) [24]	Deep learning-based image noise quantification framework	AI	Chest	Convolutional neural network trained on anthropomorphic phantom data labeled with true noise maps via 2 consecutive scans	Global	(-) AI is only trained on phantom data and generalizability to human anatomy is dependent on the ability to include all relevant features in the phantom training data	Validated with true noise maps of phantom data and with noise maps (subtracting 5-slice running average) of clinical chest CT scans of 5 patients
Huber et al. (2023) [25]	Deep learning-based image noise quantification framework	AI	Chest, head and pelvis	Convolutional neural network trained on anthropomorphic phantom data labeled with true noise maps with per-pixel SDs from 100 replicate scans	Global	(-) AI is only trained on phantom data and generalizability to human anatomy is dependent on the ability to include all relevant features in the phantom training data	Validated with true noise maps of phantom data and with SD in manually placed ROIS on clinical CT scans of 10 patients

Table S2 Objective CT image quality assessment methods focusing on contrast

<i>ROI, region-of-interest; HU, Hounsfield Unit; CNR, contrast-to-noise ratio; IQ, image quality; CTA, computed tomography angiography;</i>					
Study	Methodology	Technique type	Body region	Key principle	ROI or Global Key advantages / disadvantages Method of validation
Beer et al. (2019) [26]	Line-density profile analysis	Manual	Abdomen	Difference between maximum and minimum HU values on the line	ROI (-) Local contrast assessment, not completely representative for full image (-) Depends on correct placement of the line profile by expert Compared to CNR/SNR measurements and to results of subjective IQ (vessel contrast) assessment via a Likert score on clinical abdominal CT scans of 45 patients
Pallenberg et al. (2020) [27]	Automated mean HU measurement of aortic CTA volumes	Automatic	Aorta	Classification of insufficient, optimal or excessive contrast based on the mean HU values in ROIs that were automatically positioned in the CTA volume (CTA image slices were automatically localized as well)	ROI (-) Automatic ROI placement shows varying results (-) Slice detection works with a template matching approach which requires sufficient reference templates Validated with manual contrast measurements on clinical CTA scans of 73 patients
Abadi et al. (2017) [28]	Automated HU distribution measurement	Automatic	Chest (lung, liver, aorta, spine)	Median of the distribution of HUs by segmentation of organs and automated ROI placement	ROI (+) Shown to work on both contrast and non-contrast-enhanced exams (-) Depends on successful segmentation Validated with manual contrast measurements on clinical chest CT scans of 732 patients
Jeukens et al. (2023) [12]	Automated HU distribution measurement	Automatic	Abdomen	Generating an histogram of HUs after removing air and bone with threshold based segmentation. Various contrast metrics are defined with the histogram analysis	Global (+) Works on entire scan area instead of specific ROI locations (-) Shape of histogram is dependent on contrast protocol and anatomical area Validated with an observer study on anthropomorphic phantom data and clinical abdominal CT scans of 30 patients via subjective IQ assessment (overall diagnostic IQ)

Table S3 Objective CT image quality assessment methods focusing on spatial resolution ROI, region-of-interest; HU, Hounsfield Unit; IQ, image quality; ESF, edge spread function; CTA, CT-angiography; ERS, edge rise slope; MTF, modulation transfer function; FWHM, full width half maximum; TTF, task transfer function; SCF, structure coherence feature; SD, standard deviation

Study	Methodology	Technique type	Body region	Key principle	ROI or Global	Key advantages / disadvantages	Method of validation
Korn et al. (2013) [29]	Sharpness	Manual	Head	Sharpness was quantified in terms of the maximal slope (change in HU per pixel) across a line profile perpendicular to the skull circumference	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Compared to results of subjective IQ (sharpness) assessment via a Likert score on clinical head CT scans of 15 patients
Brodoefel et al. (2014) [30]	Sharpness	Manual	Head	Sharpness was quantified in terms of the gradient or maximal slope (change in HU per pixel) across a line profile perpendicular to the skull circumference	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Compared to results of subjective IQ (sharpness) assessment via a Likert score on clinical head CT scans of 25 patients
Ernst et al. (2016) [31]	Sharpness	Manual	Head	Sharpness was quantified in terms of the maximal slope (change in HU per pixel) across a line profile perpendicular to the skull circumference	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Compared to results of subjective IQ (sharpness) assessment via a Likert score on clinical head CT scans of 24 patients
Wu et al. (2020) [32]	Edge spread function	Manual	Head	Spatial resolution was quantified in terms of the width of the edge spread function (ESF) across line profiles perpendicular to the boundary between the lateral ventricle and surrounding brain parenchyma	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Observer study is subject of ongoing work

Continued Table S3 Objective CT image quality assessment methods focusing on spatial resolution ROI, region-of-interest; HU, Hounsfield Unit; IQ, image quality; ESF, edge spread function; CTA, CT-angiography; ERS, edge rise slope; MTF, modulation transfer function; FWHM, full width half maximum; TTF, task transfer function; SCF, structure coherence feature; SD, standard deviation

Study	Methodology	Technique type	Body region	Key principle	ROI or Global	Key advantages / disadvantages	Method of validation
Altmann et al. (2022) [33]	Sharpness	Manual	Head/Neck	Sharpness was quantified in terms of the maximum slope (change in HU per pixel) across line profiles perpendicular to a border between fat and muscle tissue	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Compared to results of subjective IQ –(sharpness) assessment via a Likert score on clinical head and neck CT scans of 40 patients
Mergen et al. (2022) [34]	Sharpness	Manual	Coronary arteries	Sharpness was quantified in terms of the maximum slope (change in HU per pixel) across line profiles perpendicular to the left anterior descending artery	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Compared to results of subjective IQ (vessel and plaque delineation) assessment via a Likert score on clinical CTA scans of 20 patients
Feger et al. (2015) [35]	Sharpness	Manual	Coronary arteries	Sharpness was quantified in terms of the difference between 25% and 75% of the maximal HU value and in terms of the maximal slope (change in HU per pixel) across a line profile perpendicular to the edge of a vessel	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Not compared to or validated with other IQ methods

Continued Table S3 Objective CT image quality assessment methods focusing on spatial resolution ROI, region-of-interest; HU, Hounsfield Unit; IQ, image quality; ESF, edge spread function; CTA, CT-angiography; ERS, edge rise slope; MTF, modulation transfer function; FWHM, full width half maximum; TTF, task transfer function; SCF, structure coherence feature; SD, standard deviation

Study	Methodology	Technique type	Body region	Key principle	ROI or Global	Key advantages / disadvantages	Method of validation
Heinrich et al. (2023) [36]	Sharpness	Manual	Coronary arteries	Sharpness was quantified in terms of the maximum slope (change in HU per pixel) across a line profile perpendicular to the edge of the vessel lumen	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) No direct correlation was visible with subjective assessment (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Compared to results of subjective IQ (how well images were suited for performing specific tasks) assessment via a Likert score on clinical CTA scans of 50 patients, but no direct correlation was visible
Cao et al. (2023) [37]	Edge rise slope (ERS)	Manual	Abdomen	Sharpness was quantified in terms of the ERS (the ratio between the difference between the first peak and last dip in HU values, divided by their distance) across a line profile perpendicular to the edge of a portal vein	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Compared to results of subjective IQ (sharpness) assessment via a Likert score on clinical head CT scans of 40 patients
	Image texture	Manual	Abdomen	Skewness of the CT number histogram on the relatively uniform area of the liver parenchyma	ROI		
Johnson et al. (2014) [38]	Sharpness	Manual	Intestines	Sharpness was quantified in terms of the edge-width across an ROI crossing the psoas-fat edge	ROI	(+) Practical for easy comparison of spatial resolution between CT scans made with the same protocol (-) Requires expert in radiology for manual ROI drawing (-) Local spatial resolution assessment, not completely representative for full image	Validated by testing the method on clinical data of 5 patients with added blur

Continued Table S3 Objective CT image quality assessment methods focusing on spatial resolution ROI, region-of-interest; HU, Hounsfield Unit; IQ, image quality; ESF, edge spread function; CTA, CT-angiography; ERS, edge rise slope; MTF, modulation transfer function; FWHM, full width half maximum; TTF, task transfer function; SCF, structure coherence feature; SD, standard deviation

Study	Methodology	Technique type	Body region	Key principle	ROI or Global	Key advantages / disadvantages	Method of validation
Sanders et al. (2016) [39]	CT Resolution index based on the edge spread function	Automatic	Skin	Spatial resolution was quantified by segmentation of the patient's body and measuring the edge spread function (ESF) across line profiles perpendicular to the air-skin interface	Global	(+) Independent of scanned body area (-) Higher noise will make it harder to extract clean ESFs (-) Highly dependent on successful segmentation	Validated with MTF measurements on phantom data and compared to results of subjective IQ (sharpness) assessment via a two-alternative-forced-choice methodology on clinical abdominal and chest CT scans of 72 pairs of edges extracted from different images
Jeukens et al. (2023) [12]	FWHM of the Task transfer function (TTF)	Automatic	Skin	Enhanced version of the method of Sanders et al. (2016) by calculating the FWHM from the ESF	Global	(+) Independent of scanned body area (-) Highly dependent on successful segmentation	Validated with an observer study on anthropomorphic phantom data and clinical abdominal CT scans of 30 patients via subjective IQ assessment (overall diagnostic IQ)
Salimova et al. (2022) [40]	Image sharpness	Automatic	Lower extremities	Sharpness was quantified in terms of the 2D image gradient on a per-slice bases after automated segmentation of air and bone to exclude sharp edges	Global	(-) Depends on successful segmentation of bone structures	Compared to results of subjective IQ assessment (overall diagnostic IQ) via a Likert score on clinical CTA scans of 100 patients
Chun et al. (2022) [41]	Structure sharpness/ Alteration of structure	Automatic	Abdomen	Using a structure coherence feature (SCF) homogeneous and structure edge regions were defined. Structure sharpness and alteration were evaluated using the SCF and SD's between homogeneous and structured regions	ROI	(+) Method can differentiate between homogeneous area and structural transitions (+) Tested on scans from different vendors (-) Depends on successful liver segmentation via deep learning segmentation model	Validated with manually measured edge slopes on clinical abdominal CT scans of 120 patients and good agreement is found

Table S4 Objective CT image quality assessment methods not solely focusing on noise, contrast or spatial resolution
ROI, region-of-interest; PS, power spectrum; IQ, image quality; NPS, noise power spectrum; CTA, CT angiography; AI, artificial intelligence; CCTA, coronary CT angiography;

Study	Methodology	Technique type	Body region	Key principle	ROI or Global	Key advantages / disadvantages	Method of validation
Svalkest et al. (2022) [42]	2D power spectrum (PS)	Manual	Chest	IQ was quantified in terms of the spatial frequency distribution with a 2D PS per slice, by averaging the PS of non-overlapping ROIs on each slice (ROIs were mirrored for artefact reduction)	ROI	(+) Can be extended to 3D to be applicable to a complete series (+) Captures both noise and spatial resolution characteristics (-) No clear correlation was observed with subjectively assessed IQ	Compared to subjective IQ (overall diagnostic IQ) assessment via a Likert score on clinical chest CT scans of 25 patients
Cheng et al. (2019) [43]	Clarity	Automatic	Lesions	Combination of the modulation transfer function on the clinical CT images and the NPS calculated on previously made phantom images	ROI	(-) Requires the NPS of previously made phantom CT-images with the same scanning protocol, potentially hampering generalizability	Validated with an observer study based on ranking 23 sets of up to 6 clinical abdominal CT scans with respect to their IQ
Samei et al. (2021) [44]	Task-dependent estimability index	Automatic	Stenosis quantification in CTA	Combination of motion point spread functions, CT image motion blur, noise and an automated maximum-likelihood estimator, to form an estimability index e' as a task-based measure of IQ for stenosis quantification in cardiac CTA	ROI	(+) Framework accounts for a wide range of influential factors, including cardiac motion, spatial resolution, noise and others (-) Framework is only usable to quantify stenosis estimation task performance, not generalizable	Validated with subjective IQ assessment using a 4-point Likert scale on clinical CTA scans of 132 patients
Smith et al. (2018) [45]	Detectability index	Automatic	Abdomen	Combination of noise, spatial resolution and a reference task function to form an detectability index for the detection of lesions in the liver	Global	(+) Method is not limited to one specific reconstruction algorithm but can compare images made with different reconstruction algorithms (-) IQ parameter is specifically designed for lesion detection in liver scans and not easily generalizable (-) Method relies on phantom images made at forehead	Validated with an observer detection study on clinical abdominal CT scans of 21 patients

Continued Table S4 Objective CT image quality assessment methods not solely focusing on noise, contrast or spatial resolution
ROI, region-of-interest; PS, power spectrum; IQ, image quality; NPS, noise power spectrum; CTA, CT angiography; AI, artificial intelligence; CCTA, coronary CT angiography;

Study	Methodology	Technique type	Body region	Key principle	ROI or Global	Key advantages / disadvantages	Method of validation
Smith et al. (2021) [46]	Detectability index	Automatic	Abdomen	Combination of image derived noise, spatial resolution and a reference task function to calculate a detectability index for the detection of lesions in the liver	ROI	(+) This method does not need phantom scans, as opposed to Smith et al. (2018).	Validated on clinical abdomen scans against the detection accuracy in an observer study of clinical abdominal CT scans of 21 patients
Lee et al. (2022) [10]	Perceptual IQ	AI	Chest and abdomen	Self-supervised training strategy for object detection that results in a quantitative quality of CT images	ROI	(+) Study shows potential generalizability to other anatomies	Tested on clinical and anthropomorphic phantom CT images among different dose levels and validated with subjective IQ assessment via a Likert score on clinical CT scans of 30 patients
Nakanishi et al. (2018) [47]	General IQ	AI	CCTA	Machine learning algorithm that predicts an IQ score, with features as noise, contrast, misregistration scores and un-interpretability index as input, meant to predict diagnostic / non-diagnostic	Global	(+) Trained on multicenter data (different CT scanners) (-) Algorithm might be biased because training data contained more good than poor IQ	Validated with subjective IQ assessment via a Likert score on clinical CCTA studies of 172 patients

CHAPTER

7



Automated CT image quality metrics: replacing manual ROI-based assessment

EMBARGOED

E.J.I. Hoeijmakers

B. Martens

J.E. Wildberger

J.J.A. de Jong

C.R.L.P.N. Jeukens

Submitted

CHAPTER

8



General Discussion

While recent years have seen rapid advances in CT detector technology and image reconstruction possibilities, their true clinical value depends on whether they change clinical practice, by making imaging better, faster, and safer, and addressing unmet diagnostic needs. This requires careful translation of innovation into optimized imaging protocols. This thesis explored this challenge in part I by exploiting the potential of photon-counting detector CT (PCD-CT) for contrast media optimization. In part II, the focus shifted from acquisition to image quality evaluation, highlighting the challenges in image quality assessment and proposing alternative approaches that improve current image quality assessment methods. Overall, this thesis underscores that technological progress in CT should at all times be accompanied by image quality optimization and quantification.

This thesis firstly addressed the fundamental differences between polyenergetic images acquired with conventional energy-integrating detector CT (EID-CT) and virtual monoenergetic images (VMIs) derived from PCD-CT (**Chapter 2**). Although tube potential (kV) and monoenergetic energy levels (keV) are often intuitively interpreted as comparable measures of X-ray beam energy, this work highlighted that they represent fundamentally different representations of the energy spectrum. Whereas kV defines the polychromatic X-ray spectrum emitted by the tube, keV values in VMIs correspond to reconstructed energy levels derived from spectral information after data acquisition. The ability to generate VMIs is not unique to PCD-CT and has previously been available with dual-energy CT (DE-CT) systems [1-4]. However, with the first clinically approved PCD-CT system, the vendor provides VMIs as part of the default image output, rather than as optional or advanced post-processing tools [5; 6]. Therefore, the integration of VMIs into routine clinical workflows has become far more widespread. Spectral imaging techniques are of clinical relevance and offer additional degrees of (diagnostic) freedom, as the ability to retrospectively select optimal keV levels enables flexible optimization of image quality without additional data acquisition. **Chapter 2** highlighted the potential benefits of VMIs for diagnostic confidence and radiation dose optimization by increased image contrast and decreased image noise.

Building on the knowledge presented in **Chapter 2**, **Chapter 3** and **Chapter 4** used the spectral imaging capabilities of PCD-CT to optimize contrast media protocols in

CT imaging. **Chapter 3**, which focused on routine abdominal CT imaging, demonstrated that PCD-CT enables approximately 20% reduction in total iodine load while maintaining sufficient diagnostic image quality, compared with already optimized and highly individualized scan and contrast protocols on EID-CT. Similarly, **Chapter 4** showed that a reduction in iodine load is also feasible for CT pulmonary angiography (CTPA) applications for pulmonary embolism detection, of approximately 27%.

Furthermore, the latter was accompanied by an approximately 24% reduction in radiation dose index volume (CTDIvol), while preserving diagnostic image quality, also compared with optimized and highly individualized EID-CT protocols. These findings support that the inherently improved contrast performance and reduced noise inherent to PCD-CT can be used to tailor protocols according to the ALARA principle for dose and contrast, if applicable.

Across the literature, reported percentages of contrast media volume reduction with PCD-CT vary widely [7-17]. The achievable degree of contrast media reduction is inherently task-dependent and depends directly on the EID-CT protocols used in these centers. The studies in this thesis compared PCD-CT to highly individualized EID-CT protocols, while other studies compare PCD-CT to fixed- or non-individualized EID-CT protocol emphasizing that the reference standard against which contrast reduction is evaluated plays a crucial role. Tasks with a (too) high intrinsic contrast, such as fixed- or non-individualized CTPA, may tolerate larger reductions in this respect, compared to applications requiring subtle only soft-tissue contrast differentiation.

Taken together, this emphasizes that no single percentage of contrast media reduction can be applied universally. Instead, contrast optimization should be viewed as an iterative, task-specific and protocol-dependent process. The percentages in **Chapter 3 and 4** therefore illustrate the achievable potential of contrast media reduction with PCD-CT in our dedicated setting, rather than proposing absolute reduction percentage applicable across different clinical indications and imaging protocols. They will contribute and built upon the steps that have been taken in the last 10-20 years in our center, including protocol individualization and the introduction of the 10-to-10 rule (Figure 1) [18-21].

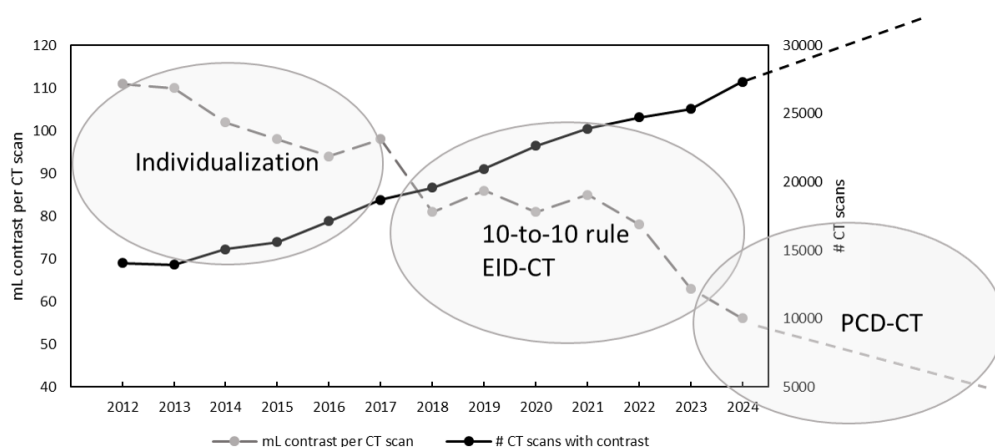


Figure 1 While the number of CT scans (with contrast media) increases over time, the contrast media volume has decreased due to protocol individualization and the 10-to-10 rule. PCD-CT offers the opportunity for further reduction of contrast media volume. (Numbers from Maastricht University Medical Centre+)

At the same time, the widespread availability of VMIs with PCD-CT introduces new complexity for radiologists in daily practice. Different monoenergetic reconstruction levels influence contrast and noise in varying ways, making it challenging to determine which VMI is optimal for a specific diagnostic task and with what contrast media dose. In clinical practice, radiologists may subjectively prefer certain VMIs over others, without being able to precisely specify the underlying reasons. Quantitative image quality metrics can therefore play a crucial role in supporting rational selection of VMI levels and in defining appropriate trade-offs between contrast enhancement and noise. In this context, the findings presented in **Chapter 5, 6, and 7** on image quality assessment are highly relevant.

Chapter 5 focused on the reliability and interpretation of subjective image quality assessment in CT. The findings on the pairwise comparison method represent an important methodological contribution to image quality assessment. The results demonstrate that subjective image quality, traditionally regarded as inherently variable and difficult to reproduce [22; 23], can be assessed with substantially greater reliability when relative rather than absolute evaluation methods (e.g. Likert scale assessment) are used. Even in datasets with only subtle differences in image quality, which is often the case in protocol optimization and dose-reduction studies,

interobserver agreement was substantially higher for the pairwise comparison method than for the conventional Likert scale method. This suggests that part of the variability previously attributed to “subjectivity” may in fact originate from limitations of the image quality assessment method that was used.

Although these advantages, the pairwise comparison method requires a considerable number of individual comparisons per observer, even when optimized using efficient sorting algorithms. The time investment per observer limits its feasibility in large-scale studies or routine clinical use. This underscores the need for objective image quality assessment methods that are reproducible, scalable, and even less observer-dependent. Pairwise comparisons assessment can serve as a valuable reference standard for validation of such objective approaches.

During the research for **Chapter 5**, we observed that one radiologist showed substantial deviations in image quality assessment compared with the other radiologists. Excluding the particular radiologist from the analysis increased the interobserver agreement from an intraclass correlation coefficient (ICC) of 0.562 (95%CI 0.337-0.729) to 0.857 (95%CI 0.782-0.912). Further analysis revealed that this radiologist consistently preferred images reconstructed with iterative reconstruction level 2. At the time of the initial measurements, the radiologist had been working at our hospital for only a short period and was not yet accustomed to the local imaging protocols and corresponding image quality. Previously, he had worked for ten years at a different hospital, where different scanner vendors, reconstruction settings and image characteristics were standard. Therefore, we speculate that differences in prior experience and familiarity with image appearance contributed to the observed discrepancy in image quality assessment. Repeated measurements after 3 years supported this hypothesis. The ICC with the other five radiologists that were included in the study increased from the initially 0.562 to an ICC of 0.689 (95%CI 0.529-0.807). The radiologist’s evaluations now more closely align with those of the other radiologists at our institution.

Radiologists become accustomed to a particular level of image quality, which represents a balance among factors such as image noise, contrast, and spatial resolution. When new protocols are introduced or existing protocols are modified, a period of adaptation is required before radiologists fully adjust to the new image

appearance. This phenomenon was also observed during the clinical implementation of iterative reconstruction (IR) in CT imaging [24; 25]. Radiologists initially disliked iterative reconstruction in CT because early versions were computationally slow, created unfamiliar “plastic” or “oversmoothed” image appearances (different from filtered back-projection that had more noise in the image), and produced unique artifacts. This raised concern that subtle pathology might be obscured. Acceptance grew once studies demonstrated non-inferiority in diagnostic performance at a favorable radiation dose level. Over time, improvements in reconstruction algorithms, combined with growing experience and training, normalized the new image appearance, and IR became widely accepted as a standard tool.

This required adaptation period highlights a fundamental challenge in image quality research: how can image quality be quantified when assessments are inherently subjective and strongly influenced by experience, familiarity, and clinical context? Although the software developed in **Chapter 5** improved the reliability of subjective assessment, radiologist evaluations remain inherently influenced by experience and familiarity with local protocols. These limitations motivated the use of advanced objective image quality methods, which were explored in Chapter 6.

The systematic review presented in **Chapter 6** provides a comprehensive overview of advanced objective image quality assessment methods in CT beyond traditional physics-based metrics such as region-of-interest-based (ROI-based) noise, signal-to-noise ratio, and contrast-to-noise ratio. It highlights that while simple ROI-based measurements have long served as convenient parameters for certain image quality characteristics, they suffer to comprehend the total image quality.

Crucially, this review documents a clear transition within the field from manual, ROI-based techniques toward fully automated and, increasingly, artificial intelligence (AI)-driven approaches. The review also underscores two fundamental gaps that are highly relevant. First, although numerous techniques assess noise, contrast, or spatial resolution individually, there remains an absence of objective methods that integrate multiple parameters into a single, task-relevant score, that can replace the opinion of the radiologist if an image is of sufficient diagnostic image quality. Second, there is a lack of robust validation frameworks in objective image quality research, in part

because no universally accepted reference standard for diagnostic image quality currently exist.

Chapter 7 presents a fully automated, segmentation-based approach for objective image quality assessment in abdominal CT. By automatically segmenting the liver and paraspinal muscles, contrast- and noise metrics were determined which effectively replaces manual ROI-based measurements and provides reproducible, observer-independent results. Comparable performance across EID-CT and PCD-CT demonstrates that this approach is not tied to a specific scanner technology.

However, objective image quality metrics such as contrast and noise measurement represent only one component of overall image quality. Diagnostic image quality ultimately depends on the clinical context, including the diagnostic task, and the experience of the radiologist interpreting the images. Consequently, objective metrics such as noise and contrast can still not fully capture the diagnostic usability of an image, while subjective assessments, although clinically meaningful, are more difficult to standardize and automate.. The framework presented in **Chapter 7** contributed by enabling scalable and standardized objective image quality assessment across large datasets.

Future directions

Building upon this foundation, an important future direction in CT image quality research is the development of fully automated objective assessment methods that closely reflect the subjective judgement of radiologists. The optimal combination of objective metrics (Figure 2) likely depends on the diagnostic task, and identifying these relationships will be essential for developing fully automated assessment methods that accurately capture clinical IQ. Ideally, such methods would be performed without human intervention, while maintaining a strong correlation with subjective IQ assessment. This would enable consistent, reproducible, and scalable IQ assessment across scanners, institutions, and patient populations.

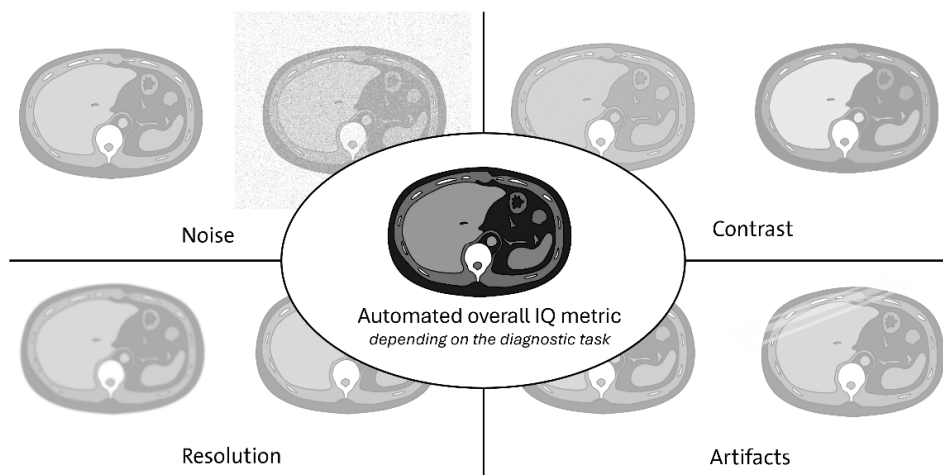


Figure 2 Illustration of the main components of CT image quality and their conceptual integration into a single, automated, task-dependent image quality (IQ) score.

Post-acquisition, near real-time image quality assessment directly on the CT scanner would represent a substantial clinical benefit. If insufficient IQ could be detected immediately after image reconstruction, patients could be rescanned during the same visit when necessary, avoiding delayed recalls and improving workflow efficiency. An even more ambitious goal would be the ability to predict diagnostic IQ a priori, based solely on acquisition and reconstruction parameters, and patient information. Accurate pre-scan quality prediction could prevent non-diagnostic examinations by allowing protocol adjustments before image acquisition, thereby reducing the need for repeat scans. Although, timing-related errors in contrast-enhanced arterial phase imaging cannot be fully eliminated [26].

A critical requirement for such automated decision-making is the definition of clinically meaningful IQ threshold. While research is currently devoted to developing quantitative IQ metrics, their clinical use ultimately depends on clearly defined acceptability criteria. Without established thresholds that distinguish diagnostically sufficient from insufficient IQ, objective metrics remain descriptive rather than actionable.

Artificial intelligence (AI) offers powerful tools to facilitate the transition toward automated and task-specific IQ assessment. It has the potential to integrate multiple

image features simultaneously, adapt to complex, task-specific definitions of IQ, and operate near real-time on modern CT systems [27]. In addition, such systems could potentially support the immediate detection of acute pathology, further enhancing clinical decision-making [28]. However, use of AI also introduces important limitations and ethical considerations. One major concern relates to responsibility and accountability. If an AI system makes an incorrect decision, this may directly affect patient care. For example, if an AI algorithm determines that a scan is of insufficient quality and requests a repeat scan, while in reality the original scan was adequate, the patient may be unnecessarily exposed to additional radiation and contrast media, respectively. This raises the question of accountability: who is responsible for the consequences of such errors, the radiologist, the institution, or the vendor of the AI system?

Another fundamental limitation is the reliance of most AI models on large, high-quality annotated datasets [27]. Although unsupervised or self-supervised approaches may reduce the need for labeled data, many current methods still depend on expert annotations that serve as a reliable ground truth. Collecting these datasets can be time-consuming and costly. In addition, if the annotations are based on a subjective judgment (e.g. radiologists' opinions), these are inherently affected by inter- and intraobserver variation. Moreover, the collection, storage, curation, and governance of such data pose substantial technical, logistical, and regulatory challenges.

Finally, AI models are sensitive to changes in their environment. If new imaging equipment is introduced, such as a scanner that produces images with different characteristics, the AI system may no longer perform optimally and may need to be retrained. This limits long-term robustness and increases maintenance efforts. In addition, AI models are not inherently generalizable. Their performance depends strongly on the data they were trained on. If certain patient populations, imaging conditions, or pathologies are underrepresented in the training data, the model may perform poorly in real-world clinical settings.

Conclusion

In summary, this thesis demonstrates that advances in CT technology, particularly the introduction of photon-counting detectors and the routine availability of VMIs, create new opportunities for optimizing diagnostic imaging. PCD-CT enables substantial reduction of contrast media and, in specific cases, radiation dose, without compromising diagnostic performance, highlighting the potential for safer and more patient-tailored imaging protocols. At the same time, the increasing complexity of image reconstruction and task-dependent nature of image quality assessment underscores the importance of reliable and objective measures. Subjective evaluation can be made more reproducible through pairwise comparison approaches, while fully automated, segmentation-based metrics provide scalable, observer-independent alternatives. Nevertheless, the optimal combination of objective parameters depends on the specific diagnostic task and remains an important area for future research.

Taken together, these findings emphasize that technological progress in CT is most valuable when going together with systematic, task-specific image quality optimization and assessment.

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CHAPTER

9



**Summary
&
Nederlandse Samenvatting**

Summary

Computed tomography (CT) plays a central role in modern diagnostic imaging. However, its increasing use poses important challenges with regard to radiation exposure, contrast media administration, and the assurance of sufficient diagnostic image quality. Recent technological developments, particularly the introduction of photon-counting detector CT (PCD-CT), have created new opportunities to improve contrast efficiency, dose efficiency, and spectral imaging capabilities, while simultaneously increasing the complexity of protocol optimization. In addition, reliable and scalable methods for assessing diagnostic image quality remain limited, as image quality is inherently subjective and task-specific. This thesis addresses these challenges by investigating how PCD-CT can be used for further protocol optimization, particularly aimed at contrast media reduction, and by developing improved subjective and objective methods for image quality assessment required in protocol optimization studies.

Chapter 2 provides an educational and physics-based overview of the fundamental differences between conventional energy-integrating detector CT (EID-CT) and photon- PCD-CT, with particular attention to the distinction between polyenergetic kV imaging and virtual monoenergetic images (VMIs, expressed in keV). Using an anthropomorphic phantom with varying iodine concentrations, it illustrates how energy settings influence the balance between contrast and noise. In conventional CT, tube potential (kV) determines the entire polychromatic X-ray spectrum and therefore jointly affects contrast and noise. In contrast, PCD-CT systems enable routine spectral acquisition, allowing VMIs to be reconstructed at specific keV levels. This shifts optimization from prospective kV selection to a retrospective choice of the optimal keV level tailored to the clinical question.

The knowledge presented in Chapter 2 can be applied to optimize patient protocols, particularly with respect to radiation dose and contrast media usage. Chapters 3 and 4 evaluate contrast media reduction using retrospective clinical studies. **Chapter 3** assesses the potential of PCD-CT to reduce contrast media usage in routine contrast-enhanced abdominal CT compared with conventional EID-CT using individualized kV- and total body weight-adapted contrast protocols. In a cohort of 102 PCD-CT

and 91 EID-CT examinations, a 20.1% reduction in total iodine load was achieved with PCD-CT, while objective image quality parameters were higher compared to EID-CT. Subjective assessment by experienced radiologists confirmed that all scans were of sufficient diagnostic image quality.

Chapter 4 presents a comparable study in CT pulmonary angiography (CTPA) for the diagnosis of pulmonary embolism. In a retrospective cohort of 118 PCD-CT and 140 EID-CT scans, PCD-CT achieved a 26.7% lower total iodine load and a 24.4% lower radiation dose (CT Dose Index volume), without clinically relevant deterioration in objective or subjective image quality. Together, these findings demonstrate that PCD-CT with individualized protocols can maintain diagnostic image quality while substantially reducing contrast media usage, potentially improving patient safety and sustainability.

Chapter 5 addresses a key methodological challenge in CT protocol optimization: the assessment of diagnostic image quality. In studies such as those presented in Chapters 3 and 4, accurate quantification of image quality is essential to determine whether modifications in acquisition or contrast protocols affect diagnostic performance. Traditionally, image quality is assessed subjectively by radiologists using Likert scales (e.g., a score from 1 to 5). However, this approach is prone to considerable inter- and intraobserver variability, and extreme values are often avoided. Chapter 5 introduces an alternative evaluation method: a pairwise comparison method. Using in-house developed software, observers were repeatedly asked to select the scan with superior diagnostic image quality from pairs of images. Based on these repeated comparisons, an absolute ranking of scans was generated. In an experimental setup, six radiologists evaluated two datasets (each containing 50 CT scans) using both Likert scale assessment and pairwise comparison. One dataset contained high variation in image quality, while the other contained low variation, with differences artificially created by adding noise. Part of the data was present in both datasets to assess intraobserver variability. Interobserver variability, measured using the intraclass correlation coefficient (ICC), improved with pairwise comparison from 0.665 (95% CI 0.396–0.814) to 0.785 (95% CI 0.676–0.867) in the high-variation dataset and from 0.276 (95% CI 0.034–0.500) to 0.562 (95% CI 0.337–0.729) in the

low-variation dataset. Intraobserver agreement also improved substantially for four of the six observers.

Although subjective assessment remains essential, diagnostic image quality should ideally be determined in an objective, reproducible, and observer-independent manner. In the literature, region-of-interest (ROI)-based methods are commonly used, in which noise is expressed as the standard deviation of CT values (in Hounsfield Units), and additional parameters such as signal-to-noise ratio (SNR) and contrast-to-noise ratio (CNR) are calculated. However, these methods are sensitive to variation in ROI size and placement and generally provide only a local representation of signal and noise within an entire CT series.

Therefore, **Chapter 6** presents a systematic review of existing objective image quality assessment methods that go beyond standard ROI-based approaches. Only methods that do not require a reference image were included, in order to focus on approaches that are applicable in routine clinical practice where ground-truth reference images are typically unavailable. Ultimately, 35 studies were included, identifying 12 different noise assessment methods, 4 contrast assessment methods, 14 spatial resolution methods, and 7 additional approaches. The results demonstrate a gradual shift toward automated methods, with increasing attention to AI-based techniques. This review therefore provides a structured reference framework for radiologists and researchers when selecting appropriate objective image quality assessment methods in both clinical and research settings.

Finally, in **Chapter 7**, a fully automated, segmentation-based framework is proposed for objective noise and contrast assessment, which overcomes the limitations of manual ROI-based measurements. Liver and paraspinal muscle segmentations were used to quantify contrast as the difference between attenuation modes, and noise was assessed using a global noise level metric both before, and after segmentation of the liver. Both slice-based (with automatic slice selection) and volume-based measurements were considered. The method was evaluated on 91 EID-CT scans and 101 PCD-CT scans. Automated contrast measurements showed strong agreement with manual measurements ($ICC = 0.85\text{--}0.95$). Noise measurements showed good agreement after segmentation ($ICC = 0.67\text{--}0.81$), whereas agreement was lower without segmentation ($ICC = 0.38\text{--}0.44$). Volumetric and slice-based analyses

demonstrated comparable performance. The framework enables standardized, observer-independent image quality assessment suitable for large-scale implementation.

Overall, this thesis demonstrates that technological advances, such as PCD CT and virtual monoenergetic imaging, provide important opportunities for contrast media and radiation dose optimization without compromising diagnostic image quality. At the same time, the development of reliable and reproducible image quality assessment methods is essential to support reliable protocol evaluation and optimal implementation of new technology. This thesis contributes to this goal by proposing improved subjective and automated objective image quality assessment methods.

Nederlandse Samenvatting

Computed tomography (CT) speelt een centrale rol binnen de moderne diagnostische beeldvorming, maar het toenemende gebruik ervan brengt belangrijke uitdagingen met zich mee op het gebied van stralingsbelasting, toediening van contrastmiddel en het waarborgen van adequate diagnostische beeldkwaliteit. Recente technologische ontwikkelingen, in het bijzonder de introductie van *photon-counting detector CT* (PCD-CT), hebben nieuwe mogelijkheden gecreëerd om contrastmiddelgebruik en ioniserende straling te verminderen en spectrale beeldvorming te verbeteren, maar maken protocloptimalisatie tegelijkertijd complexer. Daarnaast ontbreken nog steeds betrouwbare en schaalbare methoden voor het beoordelen van diagnostische beeldkwaliteit, mede omdat beeldkwaliteit intrinsiek subjectief en taakspecifiek is. Dit proefschrift richt zich op deze uitdagingen door te onderzoeken hoe PCD-CT kan worden ingezet voor verdere protocloptimalisatie, met name gericht op reductie van contrastmiddel, en door verbeterde subjectieve en objectieve methoden voor beeldkwaliteitsbeoordeling te ontwikkelen die nodig zijn in protocloptimalisatie.

Hoofdstuk 2 biedt een educatief en fysisch onderbouwd overzicht van de fundamentele verschillen tussen conventionele *energy-integrating detector CT* (EID-CT) en PCD-CT, met specifieke aandacht voor het onderscheid tussen polyenergetische beeldvorming (bepaald door de buisspanning kV) en virtuele monoenergetische beelden (VMIs, uitgedrukt in keV). Met behulp van een antropomorf fantoom met verschillende jodiumconcentraties wordt geïllustreerd hoe energie-instellingen de balans tussen contrast en ruis beïnvloeden. Waar bij conventionele CT de buisspanning (kV) het volledige polyenergetische spectrum bepaalt en daarmee contrast en ruis gezamenlijk beïnvloedt, maken PCD-CT-systemen standaard spectrale acquisitie mogelijk en kunnen VMIs op specifieke keV-niveaus worden gereconstrueerd. Dit verschuift de optimalisatie van een prospectieve kV-keuze naar een retrospectieve selectie van het optimale keV-niveau, afgestemd op de klinische vraagstelling.

De kennis uit Hoofdstuk 2 kan gebruikt worden om protocollen voor patiënten te optimaliseren, met name op het gebied van stralingsdosis en contrastmiddel. In Hoofdstuk 3 en 4 wordt door middel van retrospectieve studies gekeken naar reductie van jodiumhoudend contrastmiddel reductie. **Hoofdstuk 3** evalueert de potentie van PCD-CT om het gebruik van contrastmiddel in routine *contrast-enhanced* abdomen scans te verminderen ten opzichte conventionele EID-CT scans, met een geïndividualiseerd kV- en lichaamsgewicht-gebaseerd contrastprotocol. In een cohort van respectievelijk 101 PCD-CT en 91 EID-CT onderzoeken werd een 20,1% reductie in totaal jodiumgehalte gevonden met PCD-CT, terwijl objectieve beeldkwaliteitsparameters hoger waren dan bij EID-CT. Subjectieve beoordeling door ervaren radiologen toonde dat alle scans voldoende diagnostische beeldkwaliteit hadden.

Hoofdstuk 4 bestaat uit een vergelijkbare studie, echter dan uitgevoerd voor *CT pulmonary angiography* (CTPA) scans voor de diagnose van pulmonale embolie. In een retrospectieve studie van 118 PCD-CT en 140 EID-CT scans bleek dat PCD-CT een 26,7% lagere totale jodiumgehalte én 24,4% lagere stralingsdosis (CT Dose Index volume) gebruikte, zonder klinisch relevante achteruitgang in objectieve of subjectieve beeldkwaliteit.

Beide hoofdstukken tonen aan dat PCD-CT met een geïndividualiseerd protocol diagnostische beeldkwaliteit kan behouden terwijl contrastmiddel substantieel verminderd kan worden. Dit kan potentieel patiëntveiligheid en duurzaamheid bevorderen.

Hoofdstuk 5 richt zich op een belangrijk methodologisch knelpunt binnen CT-protocol optimalisatie: beoordeling van diagnostische beeldkwaliteit. In studies zoals beschreven in Hoofdstuk 3 en 4 is het essentieel om beeldkwaliteit nauwkeurig te kwantificeren om te bepalen of wijzigingen in acquisitie- of contrastprotocol de diagnostische nauwkeurigheid beïnvloeden. Traditioneel wordt beeldkwaliteit subjectief beoordeeld door radiologen met behulp van Likert score-schalen (bv. een score van 1 tot 5). Deze methode is echter gevoelig voor aanzienlijke inter- en intraobserver variabiliteit, en tevens worden extreme waarden (de hoogste en laagste score) vaak vermeden. Hoofdstuk 5 introduceert een alternatief

beoordelingssysteem: de paarsgewijze vergelijkingsmethode. Met behulp van een in-huis ontwikkeld algoritme werden beoordelaars herhaaldelijk gevraagd om uit twee scans de scan met de beste diagnostische beeldkwaliteit te selecteren. Op basis van deze herhaalde vergelijkingen werd een absolute ranking van de scans bepaald. In een experimentele opzet hebben zes radiologen twee datasets (elk 50 CT scans) beoordeeld met beide methoden (Likert score-schaal en paarsgewijze vergelijking). Één dataset bevatte hoge variatie in beeldkwaliteit en de andere een lage variatie, waarbij verschillen kunstmatig werden gecreëerd door toevoeging van ruis. Een deel van de data was aanwezig in beide datasets om intraobserver variabiliteit te onderzoeken. De interobserver variabiliteit, gemeten met de intraclass correlatie coefficient (ICC), verbeterde bij gebruik van de paarsgewijze vergelijkingsmethode van 0.665 (95%CI 0.396–0.814) naar 0.785 (95%CI 0.676–0.867) in de hoge variatie dataset en van 0.276 (95%CI 0.034–0.500) naar 0.562 (95%CI 0.337–0.729) in de lage variatie dataset. Ook de intraobserver variabiliteit steeg substantieel voor 4 van de 6 observers.

Hoewel subjectieve beoordeling essentieel blijft, zou diagnostische beeldkwaliteit idealiter objectief, reproduceerbaar en onafhankelijk van de beoordelaar moeten worden vastgesteld. In de literatuur worden hiervoor vaak *region-of-interest* (ROI)-gebaseerde methoden gebruikt, waarbij ruis wordt uitgerukt als de standaarddeviatie van CT-waardes (in Hounsfield Units) en aanvullende parameters zoals *signal-to-noise* ratio (SNR) en *contrast-to-noise* ratio (CNR) worden berekend. Deze methoden zijn echter gevoelig voor variatie in ROI-grootte en -locatie en geven doorgaans slechts een lokale representatie van signaal en ruis binnen een volledige CT-serie.

In **Hoofdstuk 6** is daarom een systematische review gedaan naar bestaande objectieve beoordelingsmethodes die verder gaan dan de standaard ROI-gebaseerde methodes. Hierbij is gekozen om methoden te includeren die geen referentiebeeld vereisen, omdat beeldkwaliteit subjectief is en er dus geen gouden standaard is voor goede of slechte beeldkwaliteit. Uiteindelijk werden 35 studies geïnccludeerd, waarin 12 verschillende ruisbepalingsmethodes, 4 contrastmethoden,

14 spatiële resolutie methoden en 7 overige methodes werden geïdentificeerd. De resultaten tonen een verschuiving richting het ontwikkelen van automatische methodes, met toenemende aandacht voor AI-gebaseerde methodes. Dit overzicht biedt daarmee een gestructureerd referentiekader voor radiologen en onderzoekers bij de keuze voor een geschikte objectieve methode voor beeldkwaliteitsbeoordeling in klinische en onderzoekscontext.

Ten slotte is in **Hoofdstuk 7** een volledige geautomatiseerd, segmentatie-gebaseerde methode ontwikkeld en gevalideerd voor de bepaling van ruis en contrast in abdominale CT, die de limitaties van handmatige ROI-gebaseerde metingen aanpakt. Lever- en paraspinale spiersegmentatie werd gebruikt om contrast te berekenen als het absolute verschil tussen de modale CT-waardes, terwijl ruis werd bepaald met een *global noise level* meting die voor én na segmentatie van de lever is bepaald. De metingen zijn op één enkele automatisch gekozen CT-*slice* uitgevoerd, en ook op het gehele CT volume. De methode is geëvalueerd op 91 EID-CT scans en 101 PCD-CT scans. Geautomatiseerde contrastmetingen toonden een sterke overeenkomst met handmatige metingen (ICC = 0.85–0.95). Ruismetingen vertoonden goede overeenkomst na segmentatie (ICC = 0.67–0.81), maar lagere overeenkomst zonder segmentatie (ICC = 0.38–0.44). Volumetrische en slice-gebaseerde analyses gaven vergelijkbare resultaten. De methode maakt gestandaardiseerde, observer-onafhankelijke beeldkwaliteitsanalyse mogelijk voor grootschalige toepassing.

Samenvattend toont dit proefschrift aan dat technologische vooruitgang, zoals PCD-CT en het gebruik van VMIs, belangrijke kansen biedt voor contrast- en dosioptimalisatie zonder verlies van diagnostische beeldkwaliteit. Tegelijkertijd is de ontwikkeling van betrouwbare en reproduceerbare methoden voor beeldkwaliteitsbeoordeling essentieel om een zorgvuldige protocolevaluatie en optimale implementatie van nieuwe technologie te ondersteunen. Dit proefschrift draagt bij aan dit doel door verbeterde subjectieve en geautomatiseerde objectieve methoden voor beeldkwaliteitbeoordeling voor te stellen.

CHAPTER

10



Scientific Impact

Research

Computed tomography (CT) plays an essential role in modern medical diagnosis. Technological developments have made CT scanners more advanced than ever. However, improving technology alone is not enough. To fully exploit these advances, we must ensure that the clinical protocols are optimized, contrast media use is safe and sustainable, and image quality can be evaluated in a reliable and reproducible way.

One of the most important recent developments in CT imaging is the introduction of photon-counting detector CT. This technology enables improved spectral imaging and introduces routine clinical use of monoenergetic reconstructions instead of conventional polyenergetic images. As monoenergetic reconstructions become more available for radiologists, it is essential that they understand how energy selection influences image contrast, noise, and overall diagnostic image quality.

This thesis therefore firstly aimed to provide a clear educational explanation on the physical and practical differences between polyenergetic images and monoenergetic reconstructions. It demonstrates quantitatively how energy level selection affects image quality parameters. A thorough understanding of these mechanisms allows radiologists and researchers to fully exploit the advantages of spectral imaging with photon-counting detector CT and to make conscious, task-based choices in image reconstruction. This thereby enables more effective protocol optimization.

The improved contrast capabilities of photon-counting detector CT create opportunities to reduce iodinated contrast media dose. This thesis showed that substantial contrast media reduction is possible while maintaining diagnostic image quality. These findings underline that hospitals should critically re-evaluate and optimize their contrast protocols rather than continue to apply fixed or traditional dosing strategies.

Technological progress in CT is often incremental, consisting of small improvements in detector efficiency, reconstruction speed, or dose optimization. Occasionally, however, developments are disruptive, such as the introduction of spiral CT, dual-source CT, and PCD-CT. In such transitions, radiologists are

confronted with unfamiliar image appearances. In these situations, robust and objective image quality assessment methods are essential to provide a consistent framework for safe clinical implementation and protocol optimization.

However, diagnostic image quality is inherently subjective and may vary between radiologists and diagnostic tasks. Quantifying image quality in a consistent manner is therefore challenging. This thesis addresses both subjective and objective image quality assessment.

For subjective evaluation, a software tool based on pairwise comparison was developed and validated. In contrast to the commonly used Likert scale (e.g., scoring image quality from 1 to 5), pairwise comparison leads to higher inter- and intraobserver agreement. The developed software can be widely applied and has already been used across multiple hospitals and imaging modalities.

In addition, this thesis provides a structured overview of advanced objective image quality assessment methods beyond traditional manual region-of-interest (ROI) measurements. Building on this overview, a fully automated method was developed to assess contrast and noise. This automated approach can fully replace standard ROI-based measurements, reducing workload and eliminating observer dependency.

Together, the present thesis makes a step in moving CT imaging to a future in which optimization of image protocols and evaluation of image quality are integrated into one continuous and automated workflow.

Relevance

This thesis demonstrates that CT developments and image quality assessment are closely connected and should be addressed together. New CT techniques offer opportunities to improve diagnostic image quality while reducing radiation and contrast doses, but these benefits can only be realized through systematic protocol optimization, a core principle of radiation protection. This thesis illustrates how spectral imaging can maintain or improve diagnostic quality while reducing iodine contrast dose.

Beyond patient safety, also the environmental impact of medical imaging is gaining increasing attention. Iodinated contrast agents are only partially removed by

wastewater purification systems and can accumulate in surface and drinking water. By demonstrating the feasibility of contrast reduction with photon-counting CT, this thesis contributes to more sustainable imaging practice.

Efficiency and workflow are also highly relevant. Modern CT systems generate increasing amounts of spectral and reconstruction data. Especially with photon-counting CT, radiologists are confronted with multiple possible image reconstructions and energy levels. Choosing the most appropriate reconstruction for a specific diagnostic task is crucial to maintain efficiency. Structured image quality evaluation and automated quality control can support these decisions. From a scientific and clinical perspective, reliable and standardized image quality assessment is essential. Without well-established consistent assessment methods, it is difficult to compare studies across hospitals, scanners, and time points. The introduction of pairwise comparison and automated assessment methods contributes to improved reproducibility, scalability, and methodological robustness in imaging research.

As imaging datasets continue to grow and artificial intelligence becomes more integrated into radiology, fast and reliable quality control will become increasingly important. Automated image quality assessment may support clinical workflow, research efficiency, and future scanner-based quality monitoring systems.

Target groups

The results of this thesis are relevant to multiple stakeholder groups:

- Radiologists benefit from improved understanding of spectral imaging principles and practical guidance on selecting monoenergetic reconstructions for specific diagnostic tasks. They also gain access to more reliable image quality assessment methods.
- Radiographers may in the future be supported by fully automated quality monitoring that simplify workflow and may reduce the need for repeat scans.
- Medical physicists and imaging researchers benefit from the quantitative frameworks, structured methodological overview, and validated tools

developed in this work. These tools can be applied to evaluate emerging CT technologies and support innovation in image quality research.

- Healthcare institutions benefit from potential cost reduction through optimized contrast usage and improved efficiency.
- Patients ultimately benefit from safer imaging protocols, maintained diagnostic accuracy, fewer repeat scans, and reduced environmental impact of medical imaging practices.

Future perspectives and long-term impact

CT technology will continue to evolve, and careful evaluation of image quality will remain essential to ensure that technical innovations truly translate into better patient care. Ideally, image quality assessment would become fully automated and integrated directly into the scanner, possibly even in real time. Such systems could immediately detect when image quality is insufficient, allowing scans to be repeated right away if necessary. This would prevent patients from having to return at a later stage for repeat examinations.

In addition, automated image quality assessment will make consistent quality control easier to achieve. By applying the same objective criteria to every scan, hospitals could monitor and maintain imaging standards reliably over time. This will help ensure that future CT imaging is not only more advanced, but also stays consistently safe, efficient, sustainable, and centered around high-quality patient care.

CHAPTER

A



Appendices

Dankwoord

Curriculum Vitae

Publications

Dankwoord

Soms heb ik het idee dat een PhD doen me een beetje overkomen is. Wat begon als een tof wetenschapsproject tijdens mijn klinische fysica opleiding is enkele jaren later ineens geëindigd in dit proefschrift. Maar wat ben ik dankbaar dat dit gebeurd is.

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How wonderful life is, while you're in the world.

Curriculum Vitae

Evie Hoeijmakers was born on 15 October 1996 in Venlo, the Netherlands. She completed her secondary education (Gymnasium) cum laude at Dendron College in Horst in 2014, after which she began a propaedeutic programme in Applied Mathematics at the Eindhoven University of Technology. She completed this programme and continued with a Bachelor's degree in Applied Physics, graduating in 2018. She subsequently pursued a Master's degree in Applied physics.

During her Master's program, she completed an internship at Sioux Lime in Eindhoven, where she worked on the accurate real-time prediction of wafer thermal expansion in photolithography processes. She completed her Master's thesis at TU/e in collaboration with the Máxima Medical Centre in Veldhoven, focusing on the continuous monitoring of procalcitonin levels in neonates for the early diagnosis of sepsis.

After obtaining her Master's degree in 2021, Evie started her residency in Medical Physics at Maastricht University Medical Centre+, under the supervision of Prof. Dr. W.H. Backes and Dr. C.R.L.P.N. Jeukens. During her residency, she started researching subjective diagnostic image quality assessment of CT scans. In 2023, she started as a PhD candidate under the supervision of Prof. dr. J.E. Wildberger, Dr. C.R.L.P.N. Jeukens and Dr. B. Martens. Her research focused on optimization and quantification of image quality in computed tomography. She worked part-time on her PhD research while continuing her work as a medical physics resident. In 2026, she completed her medical physics residency and started working as a medical physicist in VieCuri Medical Centre in Venlo.

Publications

This thesis:

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